Safe prompts, safe fields:
the order-1 Wasserstein conformal region for linear flow matching

Distilled from the proof-of-concept note of this folder
companion to *Pipeline-Agnostic Meta-Learning: an Optimal Transport Approach*

Abstract

A prompt-conditioned flow-matching model turns each prompt into a time-dependent velocity field. We restrict the field of a new prompt to the order-1 Wasserstein conformal region built from the fields of approved prompts. For velocity fields that are linear in their parameters, a class that contains the first-order linearisations of conditional networks, we define every object involved. The region contains the field of a new approved prompt with probability at least $1-\beta$. Projecting a field onto the region is a second-order cone program; in the region's own metric, its solution is a mixture of the prompt's field and the approved fields, and for a model affine in the prompt it is the field of a mixed prompt. When fields are compared through their noise-to-output maps, every field of the region generates outputs within order-1 Wasserstein distance $(M+1)\varepsilon$ of the approved outputs, with no Grönwall factor. This map metric is a Mahalanobis distance for one-step samplers, and multi-step samplers pay their straightness defects. We summarise the toy evidence of the proof of concept, then test the method on a two-dimensional prompt-conditioned toy. Coverage holds and time behaves as predicted, but on a diverse family of approved prompts the region as written is too wide to exclude small unsafe modes, and its certificate is vacuous. Two changes keep the guarantees and repair both: scoring against the nearest approved field, and a truncated map metric whose certificate needs no Markov step. Under a white-box attack on the prompt embedding, field-space regions cap the unsafe mass at a few percent; prompt-space regions do not.

1 Problem and summary of the proof of concept

A prompt-conditioned flow-matching model maps a prompt $\mathbf{e}$ (a text embedding, a class, a goal) to a velocity field $v : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$; integrating the field from Gaussian noise generates the outputs of the prompt. The fields of approved prompts $\mathbf{e}_1, \dots, \mathbf{e}_M$ generate safe outputs. For a new prompt, which may be unsafe or adversarial, we want to (i) decide whether its field is compatible with the safe fields, (ii) replace it by a compatible field otherwise, and (iii) know what compatibility implies for the generated outputs. On a space of weights, the manuscript answers (i) and (ii) with the order-1 score $s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) = \frac{1}{M(M+1)} \sum_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$, the order-1 Wasserstein distance between the empirical law of the historical weights before and after the candidate is added, and with a split-conformal radius ε such that $\bar{\Theta} = \{\boldsymbol{\theta} : s_{d,1}(\boldsymbol{\theta}; \mathcal{D}) \le \varepsilon\}$ contains the weights of a new task with probability at least $1-\beta$. This note builds the region on a space of fields and answers (iii).

What the proof of concept established. The note `flow_matching_safe_set_POC.tex` (the POC below) contains the following facts, which are kept here in a reorganised form.

1. The closed form of $s_{d,1}$ only uses the triangle inequality, so it holds on any pseudometric space of fields, and split-conformal coverage holds for any score (Lemma 5.3, Proposition 5.4). The substantive choice is the distance between fields.
2. Where in space and time two fields are compared decides what the region sees. Costs evaluated along the safe fields' trajectories (*off-policy*: an L^2 distance with a fixed Gram matrix, the forward path divergence) are convex in the parameters but nearly blind to a small unsafe mode: a mode of mass m costs at most $\log \frac{1}{1-m} \approx m$ in forward divergence. Costs evaluated along the candidate's own trajectories (*on-policy*: the reverse divergence) charge about $m \log(m/q)$, q the safe probability of the unsafe region, but are not convex (Appendix A).
3. For a velocity linear in its parameters, L^2 distances between fields are Mahalanobis distances between parameters, the region is the manuscript's Euclidean region after whitening, and constrained training is a second-order cone program.
4. Region membership certifies a total-variation bound to the safe mixture (from path divergences; valid but loose), a rare-event bound (from the reverse divergence; informative), and closeness of ODE trajectories through Grönwall's lemma (valid but vacuous: the factor is about 10^{15} in the toy; Remark 2.14).

5. On a two-dimensional toy, coverage is nominal; on-policy costs exclude contaminated fields that off-policy costs admit; the certificates hold on every split; and constrained few-shot fitting beats least squares but not tuned shrinkage (Section 8).
6. The prompt-conditioned results (POC Section 8) and the experiment plan (POC Section 10) are untested there; Section 9 tests them.

What this note adds. Section 2 collects the mathematical background: measures and L^2 spaces of fields, projections in Hilbert spaces, conditional expectations, flows, couplings, convexity and exchangeability. Section 3 defines linear flow matching, reflow (which straightens its fields by re-pairing noise and outputs), and the prompt-conditioned version. Sections 4–7 build the theory: distances between fields and the role of time (Section 4); the order-1 region of the safe fields and its coverage (Section 5); the constrained field of a new prompt, which mixes its field with the safe fields and, for a model affine in the prompt, is the field of a mixed prompt (Section 6); and a Wasserstein certificate on the outputs of every field of the region (Section 7). Section 8 summarises the evidence of the POC, Section 9 implements the method on a prompt-conditioned toy and follows one request through it step by step, with figures of its fields, of the regions and of the certificates, then criticises the method and improves it; Section 10 compares regions built on the prompts and on the fields, with static and adaptive weights, and tests reflow, pull-back metrics and denser anchors as remedies; and Section 11 lists what remains to be tested.

2 Mathematical preliminaries

This section fixes the objects that the note manipulates and states every fact that a later proof uses. Textbook results are quoted with a reference; the others are proved, and later proofs cite the items they use.

Notation. $[\![M]\!] = \{1, \dots, M\}$, $\Delta^{M-1} = \{w \in \mathbb{R}^M : w_i \geq 0, \sum_i w_i = 1\}$ is the probability simplex, and $\mathbf{1}_b$ is the b-th unit vector. On $\mathbb{R}^{n'}$, $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b}$ and $\|\cdot\|$ are the Euclidean inner product and norm. For a matrix A, $\|A\|_F = (\sum_{j,l} A_{jl}^2)^{1/2}$ is the Frobenius norm, $\|A\mathbf{a}\| \leq \|A\|_F \|\mathbf{a}\|$ by the Cauchy–Schwarz inequality applied to each row, and $A \otimes A'$ is the Kronecker product. $H \succeq 0$ ($H \succ 0$) means that H is symmetric positive semidefinite (definite). For $B \subseteq \mathbb{R}^{n'}$ and $r > 0$, $\operatorname{dist}(\mathbf{y}, B) = \inf_{\mathbf{y}' \in B} \|\mathbf{y} - \mathbf{y}'\|$ and $B^{+r} = \{\mathbf{y} : \operatorname{dist}(\mathbf{y}, B) < r\}$. $\arg\min$ is the set of minimisers, identified with its element when it has exactly one. $\mathcal{U}[a, b]$, $\mathcal{N}(\mathbf{c}, \Sigma)$ and $\delta_{\mathbf{x}}$ are the uniform, Gaussian and Dirac laws, $\mathbb{1}_E$ is the indicator of E, and $\mu_0 = \mathcal{N}(0, I_n)$ throughout.

Measures and integrals.

Definition 2.1 (Borel maps and laws)**.** *The spaces below are Borel subsets of Euclidean spaces, such as $\mathbb{R}^{n'}$, $\mathbb{R}^n \times [0, 1]$ or $\mathbb{R}^{n \times D}$, with their Borel σ-algebras. A map between two of them is* Borel *if the preimage of every Borel set is Borel, and $\mathcal{P}(\mathcal{X})$ is the set of Borel probability measures on $\mathcal{X}$. Random variables are measurable maps from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ into these spaces, and the* law *of X is $\mathbb{P} \circ X^{-1}$. The* push-forward *of $\mu \in \mathcal{P}(\mathcal{X})$ by a Borel map T is $T_{\#}\mu = \mu \circ T^{-1}$, the law of $T(X)$ when $X \sim \mu$. The* product *$\mu \otimes \nu$ is the law of (X, X') for independent $X \sim \mu$ and $X' \sim \nu$. For $w \in \Delta^{M-1}$, the* mixture *$\sum_i w_i \mu_i$ is the measure $B \mapsto \sum_i w_i \mu_i(B)$, which is the law of X_I when $X_i \sim \mu_i$ and $I \sim w$ is independent of $(X_1, \dots, X_M)$. A law μ on $\mathbb{R}^{n'}$ has a* finite *p-th moment if $\int \|\mathbf{x}\|^p d\mu < \infty$. A property holds μ-*almost everywhere *(a.e.), or* almost surely *(a.s.) under $\mathbb{P}$, if it holds outside a set of measure zero.*

Remark 2.2 (Standard tools). (i) Continuous maps are Borel, and so are sums, products and compositions of Borel maps. (ii) *Change of variables*: $\int f d(T_{\#}\mu) = \int f \circ T d\mu$ for every Borel $f \geq 0$ and every f integrable under $T_{\#}\mu$; in particular $\mathbb{E}\, f(X) = \int f d\mu$ for $X \sim \mu$ (Williams 1991, Sec. 6.12). Also $(S \circ T)_{\#}\mu = S_{\#}(T_{\#}\mu)$. (iii) *Fubini–Tonelli*: the integral against $\mu \otimes \nu$ of a nonnegative or integrable function is the iterated integral, in either order (Brezis 2011, Thms. 4.4–4.5). For instance $\int f d(\mu \otimes \delta_t) = \int f(\mathbf{x}, t) \mu(d\mathbf{x})$, and $\mathbb{E}\, g(I, X) = \sum_i w_i \mathbb{E}\, g(i, X)$ when $I \sim w$ and X are independent. (iv) *Dominated convergence* (Brezis 2011, Thm. 4.2); the *Cauchy–Schwarz inequality* $\mathbb{E}\,|\langle X, X' \rangle| \leq (\mathbb{E}\,\|X\|^2)^{1/2} (\mathbb{E}\,\|X'\|^2)^{1/2}$; and *Markov's inequality* $\mathbb{P}(Z \geq r) \leq \mathbb{E}\, Z / r$ for $Z \geq 0$ and $r > 0$ (Williams 1991, Secs. 6.4 and 6.8). (v) For every $B \subseteq \mathbb{R}^{n'}$, $|\operatorname{dist}(\mathbf{y}, B) - \operatorname{dist}(\mathbf{y}', B)| \leq \|\mathbf{y} - \mathbf{y}'\|$, so B^{+r} is open, hence Borel.

Pseudometrics, Hilbert spaces and projections.

Definition 2.3 (Pseudometrics, seminorms, inner products)**.** *An* extended pseudometric *on a set V is a map $d : V \times V \to [0, +\infty]$ with $d(x, x) = 0$, $d(x, y) = d(y, x)$ and $d(x, z) \leq d(x, y) + d(y, z)$; it is a* pseudometric *if it is finite, and a* metric *if moreover $d(x, y) = 0$ only when $x = y$. A map $f : V \to \mathbb{R}$ is L-*Lipschitz *if $|f(x) - f(y)| \leq L d(x, y)$; for a pseudometric d and $y \in V$, $d(\cdot, y)$ is 1-Lipschitz by the triangle inequality. A* seminorm *on a real vector space V is a map $\|\cdot\| : V \to [0, \infty)$ with $\|cx\| = |c| \|x\|$ for $c \in \mathbb{R}$ and $\|x + y\| \leq \|x\| + \|y\|$; its* kernel *$\{x : \|x\| = 0\}$ is a subspace, and it is a* norm *if its kernel is $\{0\}$. An* inner product *is a symmetric bilinear form with $\langle x, x \rangle > 0$ for $x \neq 0$; with the norm $\|x\| = \langle x, x \rangle^{1/2}$, V is a* Hilbert space *if every Cauchy sequence converges.*

Lemma 2.4 (Elementary properties)**.** *(i) A seminorm induces the pseudometric $(x, y) \mapsto \|x - y\|$, which is a metric if and only if the kernel is $\{0\}$. (ii) If d is an extended pseudometric on V and $g : U \to V$, then $(u, u') \mapsto d(g(u), g(u'))$ is*

an extended pseudometric on U. (iii) If d is a metric and $r > 0$, then $\min(d, r)$ is a metric with values in $[0, r]$. (iv) If (d_ω) are pseudometrics on V, indexed by the points ω of a probability space, with $\omega \mapsto d_\omega(x, y)$ measurable for all x, y, then $(x, y) \mapsto \int d_\omega(x, y)\mathbb{P}(d\omega)$ is an extended pseudometric. (v) In an inner-product space, $|\langle x, y\rangle| \le \|x\|\|y\|$ (Cauchy–Schwarz), and $\|x+y\|^2 = \|x\|^2+\|y\|^2$ when $\langle x, y\rangle = 0$ (Pythagoras). A finite-dimensional inner-product space is a Hilbert space, and a finite-dimensional subspace of a Hilbert space is closed.

Proof: (i)–(ii) The axioms transfer directly, and $\|x-y\| = 0$ means that $x-y$ is in the kernel. (iii) $\min(\cdot, r)$ is nondecreasing and $\min(a + b, r) \le \min(a, r) + \min(b, r)$ for $a, b \ge 0$. (iv) The three axioms hold for every ω and are preserved by integration. (v) Both follow from bilinearity (Brezis 2011, Sec. 5.1); finite-dimensional normed spaces are complete, and a complete subset of a metric space is closed. □

Lemma 2.5 (Mahalanobis seminorms and Gram matrices)**.** *For $H \in \mathbb{R}^{D\times D}$ with $H \succeq 0$, let $\|\mathbf{a}\|_H = (\mathbf{a}^\top H\mathbf{a})^{1/2}$ and $\langle \mathbf{a}, \mathbf{b}\rangle_H = \mathbf{a}^\top H\mathbf{b}$. (i) If $H = CC^\top$ (for instance $C = H^{1/2}$, or the Cholesky factor when $H \succ 0$), then $\|\mathbf{a}\|_H = \|C^\top\mathbf{a}\|$. Hence $\|\cdot\|_H$ is a seminorm with kernel $\ker H$; it is a norm, and $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_H)$ is a Hilbert space, if and only if $H \succ 0$. (ii) If P is the orthogonal projector onto the range of H, then $\|\mathbf{a}\|_H = \|P\mathbf{a}\|_H$ for all $\mathbf{a}$, and $\|\cdot\|_H$ is a norm on that range. (iii) Let $\mathsf{L} : \mathbb{R}^D \to \mathcal{H}$ be linear, $\mathcal{H}$ a Hilbert space. Its* Gram matrix *$G = (\langle \mathsf{L}\mathbf{1}_j, \mathsf{L}\mathbf{1}_l\rangle)_{j,l}$ satisfies $G \succeq 0$, $\langle \mathsf{L}\mathbf{a}, \mathsf{L}\mathbf{b}\rangle = \mathbf{a}^\top G\mathbf{b}$, $\|\mathsf{L}\mathbf{a}\| = \|\mathbf{a}\|_G$ and $\ker G = \ker \mathsf{L}$: a linear map pulls the geometry of $\mathcal{H}$ back to a Mahalanobis seminorm on $\mathbb{R}^D$.*

Proof: (i) $\mathbf{a}^\top CC^\top\mathbf{a} = \|C^\top\mathbf{a}\|^2$, so $\|\cdot\|_H$ is the Euclidean norm composed with a linear map, a seminorm. With $C = H^{1/2}$, $\|\mathbf{a}\|_H = 0$ if and only if $H^{1/2}\mathbf{a} = 0$, that is, $H\mathbf{a} = 0$. The form $\langle\cdot,\cdot\rangle_H$ is symmetric and bilinear, and it is an inner product exactly when this kernel is trivial; completeness is Lemma 2.4(v). (ii) H is symmetric with range $(\ker H)^\perp$, so $H = PHP$ and $\mathbf{a}^\top H\mathbf{a} = (P\mathbf{a})^\top H(P\mathbf{a})$; and $\ker H \cap \mathrm{range}(H) = \{0\}$. (iii) By bilinearity, $\langle \mathsf{L}\mathbf{a}, \mathsf{L}\mathbf{b}\rangle = \sum_{j,l} a_j b_l \langle \mathsf{L}\mathbf{1}_j, \mathsf{L}\mathbf{1}_l\rangle = \mathbf{a}^\top G\mathbf{b}$. Hence $\mathbf{a}^\top G\mathbf{a} = \|\mathsf{L}\mathbf{a}\|^2 \ge 0$, and by (i) $\mathbf{a}^\top G\mathbf{a} = 0$ if and only if $G\mathbf{a} = 0$. □

Theorem 2.6 (Projection onto a closed convex set)**.** *Let $\mathcal{C}$ be a non-empty closed convex subset of a Hilbert space $\mathcal{H}$. Every $x \in \mathcal{H}$ has a unique nearest point $P_\mathcal{C}x$ in $\mathcal{C}$, characterised by $P_\mathcal{C}x \in \mathcal{C}$ and $\langle x - P_\mathcal{C}x, y - P_\mathcal{C}x\rangle \le 0$ for all $y \in \mathcal{C}$ (Brezis 2011, Thm. 5.2). Consequently: (i) $\|x-y\|^2 \ge \|x-P_\mathcal{C}x\|^2+\|P_\mathcal{C}x-y\|^2$ for all $y \in \mathcal{C}$; (ii) $\|P_\mathcal{C}x - P_\mathcal{C}x'\|^2 \le \langle x-x', P_\mathcal{C}x-P_\mathcal{C}x'\rangle$, hence $\|P_\mathcal{C}x - P_\mathcal{C}x'\| \le \|x - x'\|$ (Brezis 2011, Prop. 5.3); (iii) if $\mathcal{C}$ is a closed subspace, $P_\mathcal{C}$ is linear and characterised by $P_\mathcal{C}x \in \mathcal{C}$ and $x - P_\mathcal{C}x \perp \mathcal{C}$ (the* orthogonal projection*) (Brezis 2011, Cor. 5.4), and $\|x\|^2 = \|P_\mathcal{C}x\|^2 + \|x - P_\mathcal{C}x\|^2$.*

Proof of (i)–(iii): (i) $\|x - y\|^2 = \|x - P_\mathcal{C}x\|^2 + \|P_\mathcal{C}x - y\|^2 - 2\langle x - P_\mathcal{C}x, y - P_\mathcal{C}x\rangle$. (ii) Add the characterisation at x with $y = P_\mathcal{C}x'$ and at x' with $y = P_\mathcal{C}x$, then apply Cauchy–Schwarz. (iii) Pythagoras, since $P_\mathcal{C}x \perp x - P_\mathcal{C}x$. □

Definition 2.7 (L^p spaces, linear growth)**.** *Let $\mu \in \mathcal{P}(\mathcal{X})$ and $p \in \{1, 2\}$. For a Borel map $F : \mathcal{X} \to \mathbb{R}^{n'}$, $\|F\|_{L^p(\mu)} = (\int \|F\|^p d\mu)^{1/p} \in [0, +\infty]$, and $L^p(\mu; \mathbb{R}^{n'})$ is the set of Borel F with $\|F\|_{L^p(\mu)} < \infty$, two of which are identified when they agree μ-a.e.; on $L^2(\mu; \mathbb{R}^{n'})$, $\langle F, F'\rangle_{L^2(\mu)} = \int\langle F, F'\rangle d\mu$. Matrix-valued maps are treated alike with $\|\cdot\|_F$, and the same definitions apply to random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. A Borel map F on $\mathbb{R}^n$ or on $\mathbb{R}^n \times [0, 1]$ has* linear growth *if $\|F(\mathbf{x})\| \le C(1 + \|\mathbf{x}\|)$, respectively $\|F(\mathbf{x}, t)\| \le C(1 + \|\mathbf{x}\|)$, for some C and all arguments.*

Proposition 2.8 (Facts on L^p spaces)**.** *(i) $\|F + F'\|_{L^p(\mu)} \le \|F\|_{L^p(\mu)} + \|F'\|_{L^p(\mu)}$ for all Borel F, F' (Minkowski). Hence $(F, F') \mapsto \|F - F'\|_{L^p(\mu)}$ is an extended pseudometric on Borel maps, which vanishes exactly when $F = F'$ μ-a.e., and $\|\cdot\|_{L^p(\mu)}$ is a norm on $L^p(\mu; \mathbb{R}^{n'})$. (ii) $\|F\|_{L^1(\mu)} \le \|F\|_{L^2(\mu)}$. (iii) $L^2(\mu; \mathbb{R}^{n'})$ is a Hilbert space. (iv) $\|F \circ T\|_{L^p(\mu)} = \|F\|_{L^p(T_\#\mu)}$ for every Borel T. (v) If F has linear growth with constant C and $m_2 = \int \|\mathbf{x}\|^2 d\mu < \infty$, where $\mathbf{x}$ is the $\mathbb{R}^n$-coordinate, then $\|F\|^2_{L^2(\mu)} \le 2C^2(1 + m_2)$. For μ_0, $m_2 = n$.*

Proof: (i) Apply Minkowski's inequality (Brezis 2011, Thm. 4.7) to the nonnegative functions $\|F\|$ and $\|F'\|$, whose sum dominates $\|F + F'\|$; the inequality is trivial when its right side is infinite. (ii) Cauchy–Schwarz with the constant function 1, since μ is a probability. (iii) A Cauchy sequence of $L^2(\mu; \mathbb{R}^{n'})$ has Cauchy coordinates, which converge in $L^2(\mu; \mathbb{R})$ by the Fischer–Riesz theorem (Brezis 2011, Thm. 4.8). (iv) Remark 2.2(ii) with $f = \|F\|^p$. (v) $(1 + a)^2 \le 2(1 + a^2)$. □

Conditional expectation.

Definition 2.9 (Conditional expectation)**.** *Let $U \in L^2(\mathbb{P}; \mathbb{R}^{n'})$ and let Z be a random variable with values in $\mathcal{X}$. A* version *of $\mathbb{E}[U \mid Z]$ is a Borel map $g : \mathcal{X} \to \mathbb{R}^{n'}$ such that $g(Z) \in L^2(\mathbb{P}; \mathbb{R}^{n'})$ and $\mathbb{E}\langle U - g(Z), h(Z)\rangle = 0$ for every Borel h with $h(Z) \in L^2(\mathbb{P}; \mathbb{R}^{n'})$; we then write $\mathbb{E}[U \mid Z] = g(Z)$. For square-integrable U this is the usual conditional expectation given Z (Williams 1991, Ch. 9), written as a function of Z.*

Lemma 2.10 (Existence, uniqueness and Bayes' formula)**.** *(i) A version exists and is unique* $\mathrm{Law}(Z)$*-a.e. The variable $g(Z)$ is the orthogonal projection of U onto the closed subspace $\mathcal{H}_Z = \{h(Z) : h \text{ Borel}, h(Z) \in L^2(\mathbb{P}; \mathbb{R}^{n'})\}$ of $L^2(\mathbb{P}; \mathbb{R}^{n'})$. Hence $\mathbb{E}\,\|U\|^2 = \mathbb{E}\,\|g(Z)\|^2 + \mathbb{E}\,\|U - g(Z)\|^2$, and $\mathbb{E}[\Psi(Z)^\top U] = \mathbb{E}[\Psi(Z)^\top g(Z)]$ for every Borel Ψ with values in $\mathbb{R}^{n'\times D}$ and $\Psi(Z) \in L^2(\mathbb{P}; \mathbb{R}^{n'\times D})$. (ii) (Bayes' formula.) Let Y be a random variable with law p on $\mathcal{Y}$, ℓ a σ-finite Borel measure on $\mathcal{X}$ (in our use, the Lebesgue measure), and $\varphi : \mathcal{X} \times \mathcal{Y} \to [0, \infty)$ Borel with $\mathbb{P}(Y \in A, Z \in B) = \int_A \int_B \varphi(z \mid y)\ell(dz)p(dy)$ for all Borel A, B (a* conditional density *of Z given Y). If $U = F(Y, Z)$ for a Borel F, then*

$$g(z) = \frac{\int F(y, z)\varphi(z \mid y)p(dy)}{\int \varphi(z \mid y)p(dy)},$$

set to 0 *where the ratio is not defined, is a version of* $\mathbb{E}[U \mid Z]$.

Proof: (i) By Proposition 2.8(iv), $h \mapsto h(Z)$ is a linear isometry from $L^2(\mathrm{Law}(Z);\mathbb{R}^{n'})$ onto $\mathcal{H}_Z$, so $\mathcal{H}_Z$ is complete (Proposition 2.8(iii)), hence closed. A version is exactly a g with $g(Z) \in \mathcal{H}_Z$ and $U - g(Z) \perp \mathcal{H}_Z$, so Theorem 2.6(iii) gives the existence and uniqueness of $g(Z)$, and the isometry gives the uniqueness of g up to a $\mathrm{Law}(Z)$-null set. The identity is Pythagoras, and the last claim is orthogonality applied to each column of $\Psi(Z)$. (ii) The law of (Y,Z) and the measure with density $\varphi(z \mid y)$ with respect to $p \otimes \ell$ agree on products $A \times B$, hence coincide; by Tonelli's theorem, Z has the density $\bar{\varphi}(z) = \int \varphi(z \mid y)p(dy)$ with respect to ℓ, so $0 < \bar{\varphi}(Z) < \infty$ a.s., and $\int \|F(y,z)\|^2 \varphi(z \mid y)p(dy) < \infty$ for $\mathrm{Law}(Z)$-a.e. z, because its integral against ℓ is $\mathbb{E}\,\|U\|^2$. For such z, the Cauchy–Schwarz inequality under the probability $\varphi(z \mid \cdot)p/\bar{\varphi}(z)$ gives $\|g(z)\|^2\bar{\varphi}(z) \le \int \|F(y,z)\|^2\varphi(z \mid y)p(dy)$, so $\mathbb{E}\,\|g(Z)\|^2 \le \mathbb{E}\,\|U\|^2$. Finally, for $h(Z) \in L^2(\mathbb{P};\mathbb{R}^{n'})$, $\langle U, h(Z)\rangle$ is integrable, and Fubini's theorem gives $\mathbb{E}\langle U, h(Z)\rangle = \int\langle \int F(y,z)\varphi(z \mid y)p(dy), h(z)\rangle\ell(dz) = \int\langle g(z), h(z)\rangle\bar{\varphi}(z)\ell(dz) = \mathbb{E}\langle g(Z), h(Z)\rangle$. □

Time-dependent vector fields.

Definition 2.11 (Fields and flows)**.** *A (time-dependent vector)* field *is a Borel map* $v : \mathbb{R}^n \times [0,1] \to \mathbb{R}^n$. *It is* L-Lipschitz *if* $\|v(\mathbf{x},t) - v(\mathbf{x}',t)\| \le L\|\mathbf{x} - \mathbf{x}'\|$ *for all* $\mathbf{x}, \mathbf{x}', t$, *and* locally Lipschitz *if its restriction to* $B \times [0,1]$ *is* L_B*-Lipschitz for every compact* $B \subset \mathbb{R}^n$. *Fields form a vector space under pointwise operations, and a* mixture *of fields is a convex combination* $\sum_a \gamma_a v_a$, γ *in a simplex. A* solution *of* $\dot{X}_t = v(X_t,t)$ *from* $\mathbf{x} \in \mathbb{R}^n$ *is a differentiable curve* $X : [0,1] \to \mathbb{R}^n$ *(with one-sided derivatives at the endpoints) such that* $X_0 = \mathbf{x}$ *and* $\dot{X}_t = v(X_t,t)$ *for every* $t \in [0,1]$. *The equation is* well posed *if every* $\mathbf{x}$ *has exactly one solution, and its* flow *is then the family of maps* $\mathbf{x} \mapsto X_t$, $t \in [0,1]$.

Remark 2.12 (Fields as vectors of a Hilbert space). For $\rho \in \mathcal{P}(\mathbb{R}^n \times [0,1])$, the fields with $\|v\|_{L^2(\rho)} < \infty$ form the Hilbert space $L^2(\rho;\mathbb{R}^n)$, with $\langle v, v'\rangle_{L^2(\rho)} = \int\langle v(\mathbf{x},t), v'(\mathbf{x},t)\rangle\rho(d\mathbf{x},dt)$ (Proposition 2.8). Its distance compares two fields only where ρ puts mass: fields that differ on a ρ-null set are at distance 0. Samplers, in contrast, evaluate a field at points that depend on the field itself (Definition 3.4), so two fields at $L^2(\rho)$-distance 0 may generate different outputs. Fields are therefore Borel maps in this note, to which L^2 norms are applied, rather than equivalence classes. Two choices of ρ recur: $\mu_0 \otimes \delta_t$, for which $\|v\|_{L^2(\mu_0\otimes\delta_t)} = \|v(\cdot,t)\|_{L^2(\mu_0)}$ compares the slices at time t on the noise (Remark 2.2(iii)), and the interpolation path of Definition 3.1, on which flow matching is trained.

Lemma 2.13 (Linear models in $L^2(\rho;\mathbb{R}^n)$)**.** *Let* $\rho \in \mathcal{P}(\mathbb{R}^n \times [0,1])$ *and* $\Phi : \mathbb{R}^n \times [0,1] \to \mathbb{R}^{n\times D}$ *Borel with* $\int \|\Phi\|_F^2 d\rho < \infty$, *and let* $\Phi_\rho : \mathbb{R}^D \to L^2(\rho;\mathbb{R}^n)$, $\boldsymbol{\theta} \mapsto \Phi\boldsymbol{\theta}$. *(i)* Φ_ρ *is well defined and linear, and its range* $\{\Phi\boldsymbol{\theta} : \boldsymbol{\theta} \in \mathbb{R}^D\}$ *is a closed subspace of* $L^2(\rho;\mathbb{R}^n)$. *(ii)* $G_\rho = \int \Phi^\top\Phi d\rho$ *(entrywise) is finite and is the Gram matrix of* Φ_ρ: $\langle \Phi\boldsymbol{\theta}, \Phi\boldsymbol{\theta}'\rangle_{L^2(\rho)} = \boldsymbol{\theta}^\top G_\rho \boldsymbol{\theta}'$ *and* $\|\Phi\boldsymbol{\theta}\|_{L^2(\rho)} = \|\boldsymbol{\theta}\|_{G_\rho}$. *(iii) For* $f \in L^2(\rho;\mathbb{R}^n)$, $\Phi_\rho^* f = \int \Phi^\top f d\rho \in \mathbb{R}^D$ *is finite and* $\langle \Phi\boldsymbol{\theta}, f\rangle_{L^2(\rho)} = \boldsymbol{\theta}^\top\Phi_\rho^* f$: Φ_ρ^* *is the adjoint of* Φ_ρ. *(iv) For* $f \in L^2(\rho;\mathbb{R}^n)$ *and* $\lambda > 0$, $\|\Phi\boldsymbol{\theta} - f\|^2_{L^2(\rho)} + \lambda\|\boldsymbol{\theta}\|^2 = \boldsymbol{\theta}^\top(G_\rho + \lambda I)\boldsymbol{\theta} - 2\boldsymbol{\theta}^\top\Phi_\rho^* f + \|f\|^2_{L^2(\rho)}$, *and its unique minimiser over* $\mathbb{R}^D$ *is* $(G_\rho + \lambda I)^{-1}\Phi_\rho^* f$.

Proof: Pointwise, $\|\Phi\boldsymbol{\theta}\| \le \|\Phi\|_F\|\boldsymbol{\theta}\|$, $|(\Phi^\top\Phi)_{jl}| \le \|\Phi\|_F^2$ and $\|\Phi^\top f\| \le \|\Phi\|_F\|f\|$, so all the integrals are finite, the last one by the Cauchy–Schwarz inequality. (i) A finite-dimensional subspace is closed (Lemma 2.4(v)). (ii)–(iii) Integrate $\langle \Phi\boldsymbol{\theta}, \Phi\boldsymbol{\theta}'\rangle = \boldsymbol{\theta}^\top\Phi^\top\Phi\boldsymbol{\theta}'$ and $\langle \Phi\boldsymbol{\theta}, f\rangle = \boldsymbol{\theta}^\top\Phi^\top f$, and apply Lemma 2.5(iii). (iv) Expand the square with (ii)–(iii); since $G_\rho + \lambda I \succ 0$, Lemma 2.19(iv) gives the minimiser. □

Remark 2.14 (Well-posedness and Grönwall's lemma). If v is continuous, locally Lipschitz and of linear growth, the equation $\dot{X}_t = v(X_t,t)$ is well posed (Teschl 2012, Ch. 2). *Grönwall's lemma*: a continuous $f : [0,1] \to [0,\infty)$ with $f(t) \le a + \int_0^t (Lf(s) + b)ds$ for all t satisfies $f(t) \le (a + bt)e^{Lt}$ (differentiate $e^{-Lt}(a + \int_0^t (Lf + b)ds)$). If X and X' solve the equations of a continuous L-Lipschitz field v and of a continuous field v' with $\|v - v'\| \le b$ everywhere, then $f = \|X - X'\|$ satisfies this inequality with $a = \|X_0 - X_0'\|$, so $\|X_1 - X_1'\| \le (\|X_0 - X_0'\| + b)e^L$. Bounds on outputs derived from distances between fields along this route carry the factor e^L, with L the Lipschitz constant of the candidate field; it is about 10^{15} in the toy of Section 8. The certificates of Section 7 compare sampler maps directly and avoid it.

Couplings and distances between laws.

Definition 2.15 (Couplings and Wasserstein distances)**.** *A* coupling *of* $\mu \in \mathcal{P}(\mathcal{X})$ *and* $\nu \in \mathcal{P}(\mathcal{Y})$ *is a law on* $\mathcal{X} \times \mathcal{Y}$ *with marginals* μ *and* ν, *and* $\Pi(\mu,\nu)$ *is the set of couplings. For* $\mu, \nu \in \mathcal{P}(\mathbb{R}^{n'})$, $W_1(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \int \|\mathbf{y} - \mathbf{y}'\| d\pi(\mathbf{y},\mathbf{y}') \in [0,+\infty]$. *For laws* $\mu = \sum_x a_x\delta_x$ *and* $\nu = \sum_y b_y\delta_y$ *with finite supports* S_μ *and* S_ν *in a pseudometric space* (V,d), *a coupling is a matrix* $\pi \ge 0$ *on* $S_\mu \times S_\nu$ *with row sums* a_x *and column sums* b_y, *and* $W_{d,1}(\mu,\nu) = \min_{\pi\in\Pi(\mu,\nu)} \sum_{x,y} \pi(x,y)d(x,y)$, *the minimum of a linear function over a non-empty compact polytope.*

On laws with finite first moments, W_1 is a metric, and the Kantorovich–Rubinstein duality $W_1(\mu,\nu) = \sup\{\int f d\mu - \int f d\nu : f \text{ is 1-Lipschitz}\}$ holds (Villani 2009, Chs. 5–6). The proofs below use only the two inequalities of the next lemma.

Lemma 2.16 (Shared noise and Lipschitz test functions)**.** *(i) Let* $X \sim \mu$, *let* $F, F_1, \dots, F_M$ *be Borel maps into* $\mathbb{R}^{n'}$, *and let* $I \sim w \in \Delta^{M-1}$ *be independent of* X. *Then* $Y = F(X)$ *and* $Y' = F_I(X)$ *have the laws* $F_\#\mu$ *and* $\sum_i w_i(F_i)_\#\mu$, *and* $W_1(F_\#\mu, \sum_i w_i(F_i)_\#\mu) \le \mathbb{E}\,\|Y - Y'\| = \sum_i w_i\|F - F_i\|_{L^1(\mu)}$. *(ii) If* $\mu = \sum_x a_x\delta_x$ *and* $\nu = \sum_y b_y\delta_y$ *are finitely supported in* (V,d) *and* $f : V \to \mathbb{R}$ *is* 1*-Lipschitz, every coupling of* μ *and* ν *costs at least* $\sum_x a_x f(x) - \sum_y b_y f(y)$.

Proof: (i) By independence (Remark 2.2(iii)), $\mathbb{P}(Y' \in B) = \sum_i w_i \mathbb{P}(F_i(X) \in B)$ and $\mathbb{E}\,\|Y - Y'\| = \sum_i w_i\, \mathbb{E}\,\|F(X) - F_i(X)\|$; the law of (Y, Y') is a coupling of the two laws. (ii) $d(x, y) \geq f(x) - f(y)$; sum against the coupling, whose marginals are μ and ν. □

Definition 2.17 (Divergences)**.** *For laws P, Q on the same space,* $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$ *over measurable A. If P is absolutely continuous with respect to Q ($P \ll Q$), with Radon–Nikodym density $\frac{dP}{dQ}$, then* $\mathrm{KL}(P\|Q) = \int \log \frac{dP}{dQ} dP \in [0, +\infty]$*; otherwise* $\mathrm{KL}(P\|Q) = +\infty$*. For $x, q \in [0, 1]$,* $\mathrm{kl}(x\|q) = x \log \frac{x}{q} + (1 - x) \log \frac{1-x}{1-q}$ *(with $0 \log 0 = 0$) is the divergence between Bernoulli laws.*

Appendix A uses the chain rule (Cover and Thomas 2006, Thm. 2.5.3) and the joint convexity (Cover and Thomas 2006, Thm. 2.7.2) of KL; the inequality $\mathrm{kl}(P(A)\|Q(A)) \leq \mathrm{KL}(P\|Q)$ for every event A (Jensen's inequality for $x \log x$ under Q conditioned on A and on its complement); Pinsker's inequality $\mathrm{TV} \leq (\mathrm{KL}/2)^{1/2}$ (Tsybakov 2009, Lemma 2.5) and the Bretagnolle–Huber inequality $\mathrm{TV} \leq (1 - e^{-\mathrm{KL}})^{1/2}$ (Bretagnolle and Huber 1979, Canonne 2022); and, by direct computation, $\mathrm{KL}(\mathcal{N}(\mathbf{c}, \sigma^2 I)\|\mathcal{N}(\mathbf{c}', \sigma^2 I)) = \|\mathbf{c} - \mathbf{c}'\|^2/(2\sigma^2)$.

Remark 2.18 (Monte Carlo bounds and the evaluation metric). *Hoeffding's inequality* (Hoeffding 1963): if $Z_1, \ldots, Z_L$ are i.i.d. with values in $[0, 1]$ and mean $\bar{z}$, then $\mathbb{P}(\bar{z} \geq \frac{1}{L}\sum_l Z_l + s) \leq e^{-2Ls^2}$, so $\bar{z} \leq \frac{1}{L}\sum_l Z_l + (\log(1/\delta)/(2L))^{1/2}$ with probability at least $1 - \delta$. The bound needs bounded variables, which is why Section 9 asks for a bounded distance. The *Clopper–Pearson* bound (Clopper and Pearson 1934) is the exact binomial upper confidence bound on a probability estimated from i.i.d. indicators. The *maximum mean discrepancy* with the kernel $k(\mathbf{y}, \mathbf{y}') = \sum_{s \in \{0.1, 0.3, 1\}} e^{-\|\mathbf{y} - \mathbf{y}'\|^2/(2s^2)}$ is $\mathrm{MMD}^2(\mu, \nu) = \mathbb{E}\,k(Y, Y') + \mathbb{E}\,k(Z, Z') - 2\,\mathbb{E}\,k(Y, Z)$ for independent $Y, Y' \sim \mu$ and $Z, Z' \sim \nu$. It is the squared distance between the mean embeddings of μ and ν in the reproducing-kernel Hilbert space of k (Gretton et al. 2012, Lemmas 4 and 6). It is additive in the kernel, and Gaussian kernels are characteristic on $\mathbb{R}^{n'}$ (Gretton et al. 2012, Sec. 2.2), so it vanishes only when $\mu = \nu$. Sections 8 and 9 report its unbiased estimate (Gretton et al. 2012, Lemma 6).

Convexity and cone programs.

Lemma 2.19 (Convexity)**.** *(i) Seminorms and nonnegative combinations of convex functions are convex, and so is a convex function composed with an affine map. Affine maps preserve convex combinations, and the preimage of a convex set by an affine map is convex. (ii) The sublevel set $\{f \leq c\}$ of a convex function f is convex, and closed if f is continuous. The convex hull* $\mathrm{conv}\{x_0, \ldots, x_M\} = \{\sum_a \gamma_a x_a : \gamma \in \Delta^M\}$ *of finitely many points of $\mathbb{R}^D$ is compact. (iii) A continuous function that is* coercive *on a non-empty closed set $\mathcal{C} \subseteq \mathbb{R}^D$ ($f(\boldsymbol{\theta}) \to \infty$ as $\|\boldsymbol{\theta}\| \to \infty$ in $\mathcal{C}$) attains its minimum on $\mathcal{C}$, and a strictly convex function has at most one minimiser on a convex set. (iv) If $Q \succ 0$, then $\boldsymbol{\theta}^\top Q\boldsymbol{\theta} - 2\mathbf{b}^\top\boldsymbol{\theta} + c = \|\boldsymbol{\theta} - Q^{-1}\mathbf{b}\|_Q^2 + c - \mathbf{b}^\top Q^{-1}\mathbf{b}$, so $Q^{-1}\mathbf{b}$ is the unique minimiser.*

Proof: (i)–(ii) Direct from the definitions; the hull is the image of the compact simplex by a continuous map. (iii) A minimising sequence is bounded by coercivity, so a subsequence converges, to a point of the set, which is a minimiser by continuity; two minimisers $x \neq y$ would give $f(\frac{x+y}{2}) < f(x)$. (iv) Expand the square. □

Definition 2.20 (Second-order cone program)**.** *A* second-order cone program *minimises a linear function of $\mathbf{x} \in \mathbb{R}^{n'}$ subject to affine equalities and to constraints $\|A_j\mathbf{x} + \mathbf{c}_j\| \leq \mathbf{f}_j^\top\mathbf{x} + g_j$, $j \in [\![J']\!]$ (Boyd and Vandenberghe 2004, Sec. 4.4.2). It is a convex problem, solved by interior-point methods (Clarabel, Goulart and Chen 2024, in our experiments). A norm objective becomes a constraint through an epigraph variable: minimising $\|C^\top(\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}})\|$ is minimising r_0 subject to $\|C^\top(\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}})\| \leq r_0$.*

Exchangeability.

Definition 2.21 (Exchangeability)**.** *Random variables $Z_1, \ldots, Z_N$ are* exchangeable *if $(Z_{\sigma(1)}, \ldots, Z_{\sigma(N)})$ has the law of $(Z_1, \ldots, Z_N)$ for every permutation σ of $[\![N]\!]$, as i.i.d. variables are. They are* exchangeable given *a random variable $\mathcal{D}$ if $(\mathcal{D}, Z_{\sigma(1)}, \ldots, Z_{\sigma(N)})$ has the law of $(\mathcal{D}, Z_1, \ldots, Z_N)$ for every σ; then so are $h(\mathcal{D}, Z_1), \ldots, h(\mathcal{D}, Z_N)$, for every Borel h. For an event E and $c \in [0, 1]$, $\mathbb{P}(E \mid \mathcal{D}) \geq c$ means that $\mathbb{P}(E \cap \{\mathcal{D} \in B\}) \geq c\mathbb{P}(\mathcal{D} \in B)$ for every Borel B, that is, every version of $\mathbb{E}[\mathbb{1}_E \mid \mathcal{D}]$ is at least c almost surely.*

3 Linear flow matching

Definition 3.1 (Flow matching with the linear interpolant)**.** *Let p be a Borel probability measure on $\mathbb{R}^n$ with finite second moment (the* target*; Definition 2.1), and let $\mathbf{x}_0 \sim \mu_0$, $\mathbf{x}_1 \sim p$ and $t \sim \mathcal{U}[0, 1]$ be independent. Set $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$ and $\mathbf{u} = \mathbf{x}_1 - \mathbf{x}_0$, and let ρ_p be the law of $(\mathbf{x}_t, t)$ on $\mathbb{R}^n \times [0, 1]$ (the* interpolation path *of p). The* marginal velocity $v_p^\star$ *is a version of $\mathbb{E}[\mathbf{u} \mid \mathbf{x}_t, t]$ (Definition 2.9); it exists and is unique ρ_p-a.e. because $\mathbf{u}$ is square integrable (Lemma 2.10(i)). The* flow-matching loss *of a field $v \in L^2(\rho_p; \mathbb{R}^n)$ is $\mathcal{L}_p(v) = \mathbb{E}\,\|v(\mathbf{x}_t, t) - \mathbf{u}\|^2$.*

Remark 3.2 (Why the marginal velocity). For $s \in [0, 1]$, let p_s be the law of $(1 - s)\mathbf{x}_0 + s\mathbf{x}_1$, so that $p_0 = \mu_0$ and $p_1 = p$. For $f \in C_c^1(\mathbb{R}^n)$, differentiating $\mathbb{E}\,f((1 - s)\mathbf{x}_0 + s\mathbf{x}_1)$ under the expectation, and applying Definition 2.9 with $h(\mathbf{x}, t) = c(t)\nabla f(\mathbf{x})$ for bounded Borel c, gives $\frac{d}{ds}\int f\,dp_s = \int \langle \nabla f, v_p^\star(\cdot, s)\rangle\, dp_s$ for almost every s: the laws p_s solve the continuity equation $\partial_s p_s + \nabla \cdot (p_s v_p^\star(\cdot, s)) = 0$ in the weak sense. Under local boundedness, Lipschitz and integrability

conditions on $v_p^\star$, the solutions of $\dot{X}_s = v_p^\star(X_s, s)$ then exist from μ_0-almost every starting point and push μ_0 forward to p_s (Ambrosio et al. 2008, Prop. 8.1.8); in particular, the flow transports μ_0 to p (Lipman et al. 2023, Liu et al. 2023, Albergo and Vanden-Eijnden 2023). This motivates the construction; no result below uses it.

Lemma 3.3 (Loss decomposition)**.** *For every $v \in L^2(\rho_p; \mathbb{R}^n)$, $\mathcal{L}_p(v) = \|v - v_p^\star\|^2_{L^2(\rho_p)} + \mathcal{L}_p(v_p^\star)$.*

Proof: $\mathbf{u}$ is square integrable because p and μ_0 have finite second moments. By Lemma 2.10(i), $\mathbf{u} - v_p^\star(\mathbf{x}_t, t)$ is orthogonal to $\mathcal{H}_{(\mathbf{x}_t,t)}$, which contains $(v - v_p^\star)(\mathbf{x}_t, t)$: its $L^2(\mathbb{P})$ norm is $\|v - v_p^\star\|_{L^2(\rho_p)} < \infty$ (Proposition 2.8(iv)). Pythagoras (Lemma 2.4(v)) applied to $v(\mathbf{x}_t, t) - \mathbf{u} = (v - v_p^\star)(\mathbf{x}_t, t) + (v_p^\star(\mathbf{x}_t, t) - \mathbf{u})$ gives the claim. □

Definition 3.4 (Samplers, outputs, straightness defect)**.** *Fix a grid $0 = t_0 < t_1 < \cdots < t_J = 1$ and let $h_k = t_{k+1} - t_k$. For a field v (Definition 2.11), the* sampler map *$T_v : \mathbb{R}^n \to \mathbb{R}^n$ is $T_v(\mathbf{x}) = \mathbf{y}_J$, where $\mathbf{y}_0 = \mathbf{x}$ and $\mathbf{y}_{k+1} = \mathbf{y}_k + h_k v(\mathbf{y}_k, t_k)$ (explicit Euler). The* one-step map *is $T_v^1(\mathbf{x}) = \mathbf{x} + v(\mathbf{x}, 0)$, the case $J = 1$. The* outputs *of v have law $\nu_v = (T_v)_\# \mu_0$, and its* straightness defect *is $\zeta_v = \|T_v - T_v^1\|_{L^2(\mu_0)} \in [0, +\infty]$. Both maps are Borel (Remark 2.2(i)).*

Theorem 7.1 and Proposition 7.3 below only use that T_v is a Borel map, so they hold for any other deterministic sampler (higher-order schemes, or the exact flow when it exists).

Definition 3.5 (Linear flow-matching model)**.** *A* feature map *is a Borel map $\Phi : \mathbb{R}^n \times [0,1] \to \mathbb{R}^{n \times D}$ of linear growth (Definition 2.7): $\|\Phi(\mathbf{x}, t)\|_F \le C_\Phi(1 + \|\mathbf{x}\|)$ for all $(\mathbf{x}, t)$. The model is $v_{\boldsymbol{\theta}} = \Phi\boldsymbol{\theta}$, $\boldsymbol{\theta} \in \mathbb{R}^D$, and we write $T_{\boldsymbol{\theta}}, T^1_{\boldsymbol{\theta}}, \nu_{\boldsymbol{\theta}}, \zeta_{\boldsymbol{\theta}}$ for $T_{v_{\boldsymbol{\theta}}}, T^1_{v_{\boldsymbol{\theta}}}, \nu_{v_{\boldsymbol{\theta}}}, \zeta_{v_{\boldsymbol{\theta}}}$. For a target p, the* moments *are $G_p = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top \Phi(\mathbf{x}_t, t)]$ and $\mathbf{b}_p = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top \mathbf{u}]$, and for $\lambda > 0$ the* ridge fit *is $\boldsymbol{\theta}_p = \arg\min_{\boldsymbol{\theta}} \mathcal{L}_p(v_{\boldsymbol{\theta}}) + \lambda\|\boldsymbol{\theta}\|^2$. Given samples $\mathbf{y}_1, \dots, \mathbf{y}_N$, the* few-shot fit *is $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_{\hat{p}}$ with $\hat{p} = \frac{1}{N}\sum_j \delta_{\mathbf{y}_j}$, and its moments are $\hat{G} = G_{\hat{p}}$ and $\hat{\mathbf{b}} = \mathbf{b}_{\hat{p}}$.*

Lemma 3.6 (Linear flow matching)**.** *Let p have finite second moment and $\lambda > 0$. (i) G_p and $\mathbf{b}_p$ are finite, $\mathcal{L}_p(v_{\boldsymbol{\theta}}) = \boldsymbol{\theta}^\top G_p \boldsymbol{\theta} - 2\mathbf{b}_p^\top \boldsymbol{\theta} + \mathbb{E}\,\|\mathbf{u}\|^2$, and $\boldsymbol{\theta}_p = (G_p + \lambda I)^{-1}\mathbf{b}_p$. (ii) $\boldsymbol{\theta}_p = \arg\min_{\boldsymbol{\theta}} \|v_{\boldsymbol{\theta}} - v_p^\star\|^2_{L^2(\rho_p)} + \lambda\|\boldsymbol{\theta}\|^2$: the ridge fit is the regularised $L^2(\rho_p)$-projection of the marginal velocity onto the model. (iii) For every $\boldsymbol{\theta}$, $T_{\boldsymbol{\theta}}$ is Borel and $\|T_{\boldsymbol{\theta}}(\mathbf{x})\| \le e^{C_\Phi\|\boldsymbol{\theta}\|}(1 + \|\mathbf{x}\|)$, so $T_{\boldsymbol{\theta}} \in L^2(\mu_0)$. The one-step map $T^1_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{x} + \Phi(\mathbf{x}, 0)\boldsymbol{\theta}$ is affine in $\boldsymbol{\theta}$; for $J \ge 2$, $T_{\boldsymbol{\theta}}$ is in general not, because Φ is evaluated at points that depend on $\boldsymbol{\theta}$.*

Proof: (i)–(ii) $\mathbf{x}_t$ has a finite second moment, so $\mathbb{E}\,\|\Phi(\mathbf{x}_t, t)\|_F^2 < \infty$ (Proposition 2.8(v)), and Lemma 2.13 applies with $\rho = \rho_p$ and $G_{\rho_p} = G_p$; in particular $v_{\boldsymbol{\theta}} \in L^2(\rho_p; \mathbb{R}^n)$. The columns of $\Phi(\mathbf{x}_t, t)$ lie in $\mathcal{H}_{(\mathbf{x}_t,t)}$, so $\mathbf{b}_p = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top \mathbf{u}] = \mathbb{E}[\Phi(\mathbf{x}_t, t)^\top v_p^\star(\mathbf{x}_t, t)] = \Phi^*_{\rho_p} v_p^\star$ is finite (Lemma 2.10(i)). By Lemmas 3.3 and 2.13(iv), $\mathcal{L}_p(v_{\boldsymbol{\theta}}) = \|\Phi\boldsymbol{\theta} - v_p^\star\|^2_{L^2(\rho_p)} + \mathcal{L}_p(v_p^\star) = \boldsymbol{\theta}^\top G_p \boldsymbol{\theta} - 2\mathbf{b}_p^\top \boldsymbol{\theta} + \|v_p^\star\|^2_{L^2(\rho_p)} + \mathcal{L}_p(v_p^\star)$, and the last two terms add up to $\mathbb{E}\,\|\mathbf{u}\|^2$ (Lemma 2.10(i)). The objectives of (i) and (ii) therefore both differ from $\|\Phi\boldsymbol{\theta} - v_p^\star\|^2_{L^2(\rho_p)} + \lambda\|\boldsymbol{\theta}\|^2$ by constants, and Lemma 2.13(iv) gives their unique minimiser $(G_p + \lambda I)^{-1}\Phi^*_{\rho_p} v_p^\star = (G_p + \lambda I)^{-1}\mathbf{b}_p$. (iii) $T_{\boldsymbol{\theta}}$ is a composition of Borel maps (Remark 2.2(i)). Since $\|\Phi\boldsymbol{\theta}\| \le \|\Phi\|_F\|\boldsymbol{\theta}\|$, $1 + \|\mathbf{y}_{k+1}\| \le (1 + h_k C_\Phi\|\boldsymbol{\theta}\|)(1 + \|\mathbf{y}_k\|)$, and $\prod_k (1 + h_k C_\Phi\|\boldsymbol{\theta}\|) \le e^{C_\Phi\|\boldsymbol{\theta}\|}$ because $1 + x \le e^x$ and $\sum_k h_k = 1$; Proposition 2.8(v) then gives $T_{\boldsymbol{\theta}} \in L^2(\mu_0)$. □

Remark 3.7 (What "linear" covers). Random Fourier features (Rahimi and Recht 2007), $(2/D_\psi)^{1/2}\cos(\omega_j^\top \mathbf{x} + \omega_j' t + b_j)$ for $j \in [\![D_\psi]\!]$ with random frequencies (ω_j, ω_j') and phases b_j, are bounded, and the affine terms $(1, t, \mathbf{x}, t\mathbf{x})$ have linear growth; stacking both in $\psi(\mathbf{x}, t)$, the features $\Phi = I_n \otimes \psi^\top$ of the toy studies satisfy Definition 3.5. So does the first-order expansion of a velocity network u_ϑ around pretrained parameters ϑ_0, $u_{\vartheta_0} + \partial_\vartheta u_{\vartheta_0}(\vartheta - \vartheta_0)$, with features $\Phi = \partial_\vartheta u_{\vartheta_0}$ and a fixed offset u_{ϑ_0}, when both have linear growth. An offset changes nothing in Sections 4–7, where only differences of fields enter; in Lemma 3.6, $\mathbf{u}$ is replaced by $\mathbf{u} - u_{\vartheta_0}(\mathbf{x}_t, t)$.

Couplings, rectification and reflow. Definition 3.1 pairs each noise with a data point drawn independently of it, and the field learns the average direction of these random pairs. Nothing in Lemmas 3.3 and 3.6 uses the independence. Changing the pairing while keeping its two marginals is the idea of reflow: pair each noise with the output that the model already generates from it, and fit again.

Definition 3.8 (Flow matching with a coupling; reflow)**.** *Let q be a Borel probability measure on $\mathbb{R}^n$ with finite second moment, and $\pi \in \Pi(\mu_0, q)$ a coupling (Definition 2.15).* Flow matching with the coupling *π is Definition 3.1 with $(\mathbf{x}_0, \mathbf{x}_1) \sim \pi$, independent of $t \sim \mathcal{U}[0,1]$, in place of independent $\mathbf{x}_0 \sim \mu_0$ and $\mathbf{x}_1 \sim p$; its interpolation path, marginal velocity and loss are written ρ_π, $v_\pi^\star$ and $\mathcal{L}_\pi$. Standard flow matching is the* independent coupling *$\pi = \mu_0 \otimes p$, and its marginal velocity $v_p^\star$ is the* 1-rectified *field. For a field v with $T_v \in L^2(\mu_0)$, the* reflow coupling *of v is $\pi_v = (\mathrm{id}, T_v)_\# \mu_0$: the law of a noise $\mathbf{x}_0$ paired with the output $T_v(\mathbf{x}_0)$ that the sampler of v generates from it, whose marginals are μ_0 and the output law ν_v.* Reflow *replaces v by $v^\star_{\pi_v}$, or by a fit of it. Starting from a fit $v^{(1)}$ of $v_p^\star$, reflow gives the* 2-rectified *field $v^{(2)}$, and $k - 1$ reflows give the k-*rectified *field (Liu et al. 2023).*

Lemma 3.9 (What the coupling changes)**.** *Let $\pi \in \Pi(\mu_0, q)$, $\mathbf{u} = \mathbf{x}_1 - \mathbf{x}_0$ and $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$. (i) Lemmas 3.3 and 3.6 hold with π in place of $\mu_0 \otimes p$. The smallest loss, $\mathcal{L}_\pi(v_\pi^\star) = \mathbb{E}\,\|\mathbf{u} - \mathbb{E}[\mathbf{u} \mid \mathbf{x}_t, t]\|^2$, is the mean conditional variance of the directions of the segments that pass through a point at a given time. It is 0 if and only if $\mathbf{u}$ is almost surely a Borel function of $(\mathbf{x}_t, t)$, that is, if almost surely a single segment passes through each point at each time. (ii) For the independent coupling $\pi = \mu_0 \otimes p$, $\mathcal{L}_\pi(v_\pi^\star) > 0$ unless p is a Dirac mass. (iii) Let $\pi = (\mathrm{id}, T)_\# \mu_0$ for an L-Lipschitz map $T : \mathbb{R}^n \to \mathbb{R}^n$,*

and $\Psi_t(\mathbf{x}) = (1-t)\mathbf{x} + tT(\mathbf{x})$. For $t < 1/(1+L)$, Ψ_t is a bijection of $\mathbb{R}^n$, and $v^\star_\pi(\mathbf{y}, t) = T(\Psi_t^{-1}(\mathbf{y})) - \Psi_t^{-1}(\mathbf{y})$ defines at these times a version of the marginal velocity (completed by any version at the other times), which is continuous in $(\mathbf{y}, t)$ and constant along each segment: $v^\star_\pi(\Psi_t(\mathbf{x}), t) = T(\mathbf{x}) - \mathbf{x}$. At $t = 0$ it is $v^\star_\pi(\mathbf{x}, 0) = T(\mathbf{x}) - \mathbf{x}$, so the one-step map of the reflowed field is $T^1_{v^\star_\pi} = T$. (iv) If T is moreover monotone*, $\langle T(\mathbf{x}) - T(\mathbf{x}'), \mathbf{x} - \mathbf{x}' \rangle \geq 0$ for all $\mathbf{x}, \mathbf{x}'$, then (iii) holds at every $t \in [0, 1)$. Hence $\mathcal{L}_\pi(v^\star_\pi) = 0$, every Euler sampler of $v^\star_\pi$ maps each $\mathbf{x}$ to $T(\mathbf{x})$, and the straightness defect of $v^\star_\pi$ is 0.*

Proof: (i) The proofs of Lemmas 3.3 and 3.6 only use that $\mathbf{u} \in L^2(\mathbb{P}; \mathbb{R}^n)$, which holds because $\|\mathbf{u}\|^2 \leq 2\|\mathbf{x}_0\|^2 + 2\|\mathbf{x}_1\|^2$. By Lemma 2.10(i), $\mathcal{L}_\pi(v^\star_\pi) = \mathbb{E}\,\|\mathbf{u} - g(\mathbf{x}_t, t)\|^2$ for a version g, which vanishes if and only if $\mathbf{u} = g(\mathbf{x}_t, t)$ a.s.; conversely, if $\mathbf{u} = h(\mathbf{x}_t, t)$ a.s. for a Borel h, then h is a version (Definition 2.9). (ii) If the loss vanishes, then by (i) $\mathbf{x}_1 = \mathbf{x}_t + (1-t)\mathbf{u}$ is a.s. a Borel function $f(\mathbf{x}_t, t)$ on $\{t < 1\}$. If p is not a Dirac mass, there is a Borel B with $0 < p(B) < 1$: otherwise, for every k exactly one dyadic cube of side 2^{-k} would have mass 1, these cubes would be nested, and p would be the Dirac mass at their intersection. On the one hand $\mathbb{1}_B(f(\mathbf{x}_t, t))$ is a version of $\mathbb{E}[\mathbb{1}_B(\mathbf{x}_1) \mid \mathbf{x}_t, t]$, with values in $\{0, 1\}$. On the other hand, Bayes' formula (Lemma 2.10(ii), with the conditional density $\varphi_t(\mathbf{x} - t\mathbf{y})$ of the proof of Lemma 4.3(iii)) gives the version $\int_B \varphi_t(\mathbf{x}_t - t\mathbf{y})p(d\mathbf{y}) / \int \varphi_t(\mathbf{x}_t - t\mathbf{y})p(d\mathbf{y})$, which lies in $(0, 1)$ everywhere because $\varphi_t > 0$ and $0 < p(B) < 1$. Two versions agree a.s., a contradiction. (iii) For $t < 1/(1+L)$ and $\mathbf{y} \in \mathbb{R}^n$, the map $\mathbf{x} \mapsto (\mathbf{y} - tT(\mathbf{x}))/(1-t)$ is a contraction with constant $tL/(1-t) < 1$, whose fixed points are the solutions of $\Psi_t(\mathbf{x}) = \mathbf{y}$. So Ψ_t is a bijection, and its inverse is continuous in $(\mathbf{y}, t)$, since the fixed point of a contraction depends continuously on a parameter on which the map depends continuously, with a contraction constant bounded away from 1 on compact sets of t. Since $\mathbf{x}_t = \Psi_t(\mathbf{x}_0)$, $\mathbf{u} = T(\mathbf{x}_0) - \mathbf{x}_0 = v^\star_\pi(\mathbf{x}_t, t)$ at these times. Completed by a version v^0 at the other times, this function is a version: for $h(\mathbf{x}_t, t) \in L^2$, write $\bar{t} = 1/(1+L)$; the expectation of $\langle \mathbf{u} - v^\star_\pi(\mathbf{x}_t, t), h(\mathbf{x}_t, t) \rangle$ on $\{t < \bar{t}\}$ vanishes because $\mathbf{u} = v^\star_\pi(\mathbf{x}_t, t)$ there, and on $\{t \geq \bar{t}\}$ by Definition 2.9 for v^0 and $h\mathbb{1}\{t \geq \bar{t}\}$. At $t = 0$, $\Psi_0 = \mathrm{id}$. (iv) For $t < 1$, Ψ_t is $(1-t)$-strongly monotone, $\langle \Psi_t(\mathbf{x}) - \Psi_t(\mathbf{x}'), \mathbf{x} - \mathbf{x}' \rangle \geq (1-t)\|\mathbf{x} - \mathbf{x}'\|^2$, and ℓ_t-Lipschitz with $\ell_t = 1 - t + tL$. With $\eta = (1-t)/\ell_t^2$, expanding the square shows that $\mathbf{x} \mapsto \mathbf{x} - \eta(\Psi_t(\mathbf{x}) - \mathbf{y})$ is a contraction with constant $(1 - (1-t)^2/\ell_t^2)^{1/2} < 1$, whose fixed points solve $\Psi_t(\mathbf{x}) = \mathbf{y}$. The argument of (iii) then applies at every $t < 1$, and $t = 1$ is a null set, so $\mathbf{u} = v^\star_\pi(\mathbf{x}_t, t)$ a.s. and the loss vanishes by (i). For an Euler grid, if $\mathbf{y}_k = \Psi_{t_k}(\mathbf{x})$, then $\mathbf{y}_{k+1} = \Psi_{t_k}(\mathbf{x}) + h_k(T(\mathbf{x}) - \mathbf{x}) = \Psi_{t_{k+1}}(\mathbf{x})$, because $v^\star_\pi(\Psi_{t_k}(\mathbf{x}), t_k) = T(\mathbf{x}) - \mathbf{x}$ and $t_k < 1$; by induction, $\mathbf{y}_J = \Psi_1(\mathbf{x}) = T(\mathbf{x})$. □

Remark 3.10 (Reflow versus standard flow matching). Table 1 sets the two side by side, and Figure 1 draws them on the ring of Section 8, with the exact marginal velocity (Bayes' formula) instead of a fitted one.

- *Why standard flow matching curves.* The regression target at $(\mathbf{x}, t)$ is the average direction of all the training segments that pass through $\mathbf{x}$ at time t (Lemma 3.9(i)). Independent pairs cross everywhere (panel (a)). At $t = 0$, the segments leaving a noise point end at all the target samples, so their average points to the mean of the target, the centre of the ring (Lemma 4.3(iii)). Later, as $\mathbf{x}_t$ reveals which blob the pair is heading to, the average turns. The trajectories of the field therefore bend (panel (b)), although every training segment is straight, and one Euler step lands at the mean of the target. Panel (d) measures the crossing: the conditional variance of the directions is 4.3 near $t = 0$ and never falls below 1.6; its average, the smallest flow-matching loss, is 3.8. Lemma 3.9(ii) says it cannot vanish for a target that is not a single point.
- *What reflow changes.* Reflow keeps the noise law and the output law of the current field and changes only the pairing: each noise is paired with the output that the field already generates from it (panel (c), with the same outputs as panel (b)). At $t = 0$ no two segments cross, since one segment leaves each noise, and the field points each noise to its own output (Lemma 3.9(iii)). Later the segments may cross; here they hardly do, and the smallest loss is about 0.002. The field fitted to them moves along its own training segments, so its trajectories are straight, and a Lipschitz monotone pairing makes them exactly straight (Lemma 3.9(iv)).
- *What is guaranteed.* For exact fields whose flows are well posed, Liu et al. (2023) prove four properties of the flow Z of the marginal velocity of a coupling. First, Z has the marginals of the interpolation, so it transports μ_0 to the second marginal of the coupling (the argument of Remark 3.2, which does not use independence). Second, its pairs (Z_0, Z_1), which are the next reflow coupling, have no larger convex transport cost $\mathbb{E}\,c(\mathbf{x}_1 - \mathbf{x}_0)$ than the training pairs. Third, a coupling is unchanged by this operation if and only if its segments do not cross, and optimal transport couplings for strictly convex costs are such fixed points; in dimension one, flows cannot cross, so one reflow of an exact flow gives a straight one. Fourth, among the first k rectified flows, the smallest sum of straightness defect and crossing is at most $\mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2/k$ for the initial pairs. For $c = \|\cdot\|^2$, an exact identity underlies these results: $\mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2$ over the training pairs minus $\mathbb{E}\,\|Z_1 - Z_0\|^2$ equals the crossing of the training pairs plus the curvature $\int_0^1 \mathbb{E}\,\|\dot{Z}_t - (Z_1 - Z_0)\|^2 dt$ of the learned flow. On the ring, with independent training pairs, $\mathbb{E}\,\|\mathbf{x}_1 - \mathbf{x}_0\|^2$ falls from 6.18 to 0.97, a decrease of 5.21, against a crossing of 3.8 plus a curvature of 1.44, that is 5.25; the gap is discretisation error.
- *What it costs.* Reflow needs no data, but one simulation of the current field per training pair. It targets the current outputs ν_v, not p, so the errors of v are inherited and the refit adds its own: fidelity can only be lost. Distillation instead fits the map $\mathbf{x} \mapsto T_v(\mathbf{x})$ directly to the same pairs (Liu et al. 2024); by Lemma 3.9(iii), it is the one-step map of the exact reflowed field.

- *Why this note needs it.* Straight fields make the source slice informative: $|d_T - d_0| \le \zeta_v + \zeta_{v'}$ (Lemma 4.3(i)). For one-step samplers, the map metric is the Mahalanobis distance of Proposition 4.2(ii), so the region is convex and its projection is a second-order cone program (Corollary 7.2(i)).
- *In the test of Section 9.* Reflow is done for each of the 2000 training prompts: 1200 noises are pushed through the 1-rectified model's 50-step Euler sampler, the 0.04% of pairs whose output lies beyond radius 4 are dropped (they otherwise dominate the least-squares refit), and the model affine in the prompt is refitted on the remaining pairs. The 2-rectified model is again affine in the prompt (Definition 3.11(b)), so Sections 4–7 apply to it unchanged. Its sampler map is Lipschitz, because each Euler step of a field with Lipschitz features is, so Lemma 3.9(iii) applies to the exact reflowed field. The median straightness defect falls from 1.93 to 0.26, and $\text{MMD}^2 \times 10^3$ to the targets rises from 3.3 to 8.1 (34.9 with one step).

Table 1: Standard flow matching and reflow, for exact fields (Definition 3.8, Lemma 3.9).

	standard flow matching	reflow of a field v
training pairs $(\mathbf{x}_0, \mathbf{x}_1)$	a noise and an independent sample of the target	a noise and the output $T_v(\mathbf{x}_0)$ that v generates from it
data needed	samples of the target p	none: one simulation of v per pair
law of the outputs	p	$\nu_v = (T_v)_{\#}\mu_0$; the errors of v are inherited
field at $t = 0$	$\mathbb{E}_p[\mathbf{x}_1] - \mathbf{x}$: towards the mean of the target (Lemma 4.3(iii))	$T_v(\mathbf{x}) - \mathbf{x}$: towards the noise's own output (Lemma 3.9(iii))
crossing of the pairs (smallest loss)	positive unless p is a Dirac mass (Lemma 3.9(ii)); 3.8 on the ring	zero for a Lipschitz monotone T_v (Lemma 3.9(iv)); 0.002 on the ring
trajectories	curved	straight, or nearly
one Euler step	lands at the mean of the target	gives T_v: the outputs of v in one step

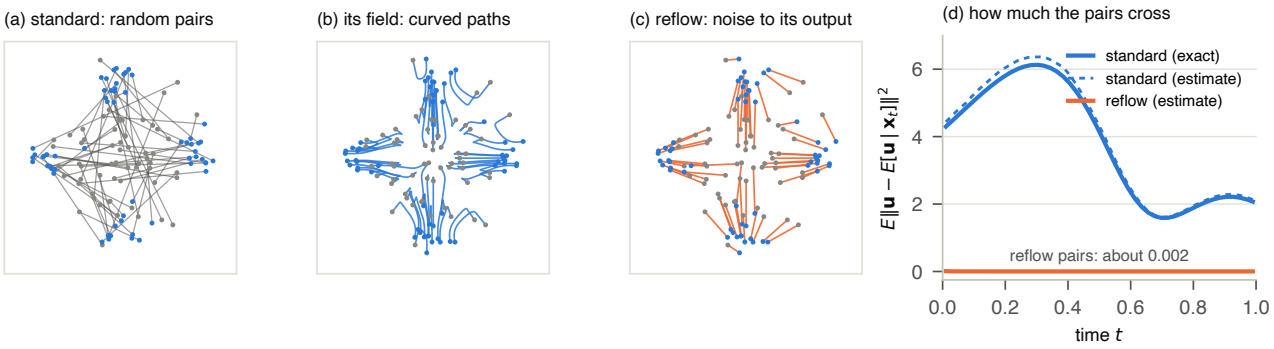


Figure 1: Standard flow matching against reflow on the ring of Section 8, with the exact marginal velocity. (a) Training pairs of standard flow matching: 60 noises (grey) paired with independent target samples (blue). (b) Trajectories of the field learned from such pairs, from the same noises. (c) Reflow pairs: the same noises paired with the outputs of (b). (d) Conditional variance of the direction $\mathbf{u} = \mathbf{x}_1 - \mathbf{x}_0$ of the pairs given the point $\mathbf{x}_t$, against t: exact for the independent pairs, and a 25-nearest-neighbour estimate on 20,000 pairs for both couplings; its average over t is the smallest flow-matching loss (Lemma 3.9(i)).

Definition 3.11 (Prompt-conditioned linear flow matching)**.** *Prompts are embeddings $\mathbf{e} \in \mathcal{E} \subseteq \mathbb{R}^{D_e}$. A* prompt-to-field map *is a Borel map $\boldsymbol{\theta}(\cdot) : \mathcal{E} \to \mathbb{R}^D$; the field of $\mathbf{e}$ is $v_{\boldsymbol{\theta}(\mathbf{e})}$ and its outputs have law $\nu_{\boldsymbol{\theta}(\mathbf{e})}$. Two instances matter. (a)* Per-prompt fit*: each prompt specifies a target $p_{\mathbf{e}}$, and $\boldsymbol{\theta}(\mathbf{e}) = \boldsymbol{\theta}_{p_{\mathbf{e}}}$ is its ridge fit (estimated from ample data in practice); a request may also come with N samples and a few-shot fit $\hat{\boldsymbol{\theta}}(\mathbf{e})$. (b)* Affine in the prompt*: $\boldsymbol{\theta}(\mathbf{e}) = \mathbf{a} + U\mathbf{e}$, with $\mathbf{a} \in \mathbb{R}^D$ and $U \in \mathbb{R}^{D \times D_e}$ fixed.*

Instance (b) arises in two ways. A conditional model $v(\mathbf{x}, t; \mathbf{e}) = \Phi(\mathbf{x}, t)(\mathbf{a} + U\mathbf{e})$ is fitted once on (prompt, sample) pairs by linear least squares, since it is linear in $(\mathbf{a}, U)$. And the first-order expansion of any conditional network $u(\mathbf{x}, t; \mathbf{e})$ at a reference prompt $\mathbf{e}_0$, $u(\cdot; \mathbf{e}_0) + \partial_{\mathbf{e}} u(\cdot; \mathbf{e}_0)(\mathbf{e} - \mathbf{e}_0)$, is instance (b), provided these features have linear growth, with $D = 1 + D_e$, $\Phi = [u(\cdot; \mathbf{e}_0) - \partial_{\mathbf{e}} u(\cdot; \mathbf{e}_0)\mathbf{e}_0,\ \partial_{\mathbf{e}} u(\cdot; \mathbf{e}_0)]$, $\mathbf{a} = \mathbf{1}_1$ and $U = [0, I_{D_e}]^\top$.

4 Distances between fields

Definition 4.1 (Distances between fields)**.** *For fields v, v', the following are extended pseudometrics (Proposition 2.8(i), Lemma 2.4(ii)): (i) for a probability measure ρ on $\mathbb{R}^n \times [0, 1]$, $d_\rho(v, v') = \|v - v'\|_{L^2(\rho)}$; (ii) for a Borel map $\chi : \mathbb{R}^n \to \mathbb{R}^{n_\chi}$ of linear growth (the space in which outputs are compared: the identity, an encoder, the features of a classifier), the* map

metric *is* $d_T^{\chi}(v,v') = \|\chi \circ T_v - \chi \circ T_{v'}\|_{L^2(\mu_0)}$*, and* $d_T = d_T^{\mathrm{id}}$*; (iii) the* source-slice metric *is* $d_0 = d_{\mu_0 \otimes \delta_0}$*, that is,* $d_0(v,v') = \|v(\cdot,0) - v'(\cdot,0)\|_{L^2(\mu_0)} = \|T_v^1 - T_{v'}^1\|_{L^2(\mu_0)}$*. On the linear model we write* $d(\boldsymbol{\theta},\boldsymbol{\theta}') = d(v_{\boldsymbol{\theta}}, v_{\boldsymbol{\theta}'})$*, again an extended pseudometric, and, when* $\mathbb{E}_\rho \|\Phi\|_F^2 < \infty$*,* $G_\rho = \mathbb{E}_\rho[\Phi^\top \Phi]$ *as in Lemma 2.13.*

Proposition 4.2 (Mahalanobis form)**.** *(i) If* $\mathbb{E}_\rho \|\Phi\|_F^2 < \infty$*, then* $d_\rho(\boldsymbol{\theta},\boldsymbol{\theta}') = \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_{G_\rho}$*;* d_ρ *is a pseudometric on* $\mathbb{R}^D$*, and a metric if and only if* $G_\rho \succ 0$*. For* $\delta \geq 0$ *with* $H = G_\rho + \delta I \succ 0$ *and* $H = CC^\top$*,* $\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_H = \|C^\top(\boldsymbol{\theta} - \boldsymbol{\theta}')\|_2 \geq d_\rho(\boldsymbol{\theta},\boldsymbol{\theta}')$*. (ii)* $d_0(\boldsymbol{\theta},\boldsymbol{\theta}') = \|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_{G_0}$ *with* $G_0 = \mathbb{E}_{\mathbf{x}\sim\mu_0}[\Phi(\mathbf{x},0)^\top \Phi(\mathbf{x},0)]$*: for the one-step sampler, the map metric is a Mahalanobis distance. (iii) For a prompt map affine in the prompt,* $d_\rho(\boldsymbol{\theta}(\mathbf{e}),\boldsymbol{\theta}(\mathbf{e}')) = \|\mathbf{e} - \mathbf{e}'\|_{U^\top G_\rho U}$*. (iv)* d_T^{χ} *is a finite pseudometric on* $\mathbb{R}^D$*.*

Proof: (i) Lemma 2.13(ii) applied to $\boldsymbol{\theta} - \boldsymbol{\theta}'$ gives the Mahalanobis form; Lemmas 2.5(i) and 2.4(i) give the kernel and the metric property; and $\|C^\top \mathbf{a}\|^2 = \mathbf{a}^\top H \mathbf{a} = \|\mathbf{a}\|_{G_\rho}^2 + \delta\|\mathbf{a}\|^2$. (ii) Apply (i) with $\rho = \mu_0 \otimes \delta_0$, which is admissible by linear growth (Proposition 2.8(v)), and note $T_{\boldsymbol{\theta}}^1 - T_{\boldsymbol{\theta}'}^1 = \Phi(\cdot,0)(\boldsymbol{\theta} - \boldsymbol{\theta}')$. (iii) $\boldsymbol{\theta}(\mathbf{e}) - \boldsymbol{\theta}(\mathbf{e}') = U(\mathbf{e} - \mathbf{e}')$ and $\|U\mathbf{a}\|_{G_\rho}^2 = \mathbf{a}^\top U^\top G_\rho U \mathbf{a}$. (iv) $\chi \circ T_{\boldsymbol{\theta}}$ has linear growth, by that of χ and Lemma 3.6(iii), so it lies in $L^2(\mu_0)$ (Proposition 2.8(v)); d_T^{χ} is the $L^2(\mu_0)$ distance between such maps (Proposition 2.8(i), Lemma 2.4(ii)). □

The choice of distance decides where fields are compared. For instance (a), the pooled interpolation path of the safe prompts, $\bar{\rho} = \frac{1}{M}\sum_i \rho_{p_{\mathbf{e}_i}}$, gives $G_{\bar{\rho}} = \frac{1}{M}\sum_i G_{p_{\mathbf{e}_i}}$, the average of the safe least-squares Gram matrices; it compares fields where the safe paths put mass (Remark 2.12), an off-policy distance. The source slice compares fields on μ_0, the law at which every sampler makes its first evaluation of every field. The map metric compares where the fields send the noise: it evaluates each field along its own trajectories and is nevertheless symmetric. The next lemma relates the last two.

Lemma 4.3 (Time, straightness and the source slice)**.** *(i) If* $T_v, T_v^1, T_{v'}, T_{v'}^1 \in L^2(\mu_0)$ *(e.g. for all fields of the linear model), then* $|d_T(v,v') - d_0(v,v')| \leq \zeta_v + \zeta_{v'}$*. (ii) If solutions of* $\dot{X}_t = v(X_t,t)$ *on* $[0,1]$ *exist from every initial point and all of them are affine in* t *(a* straight *field), then* v *is constant along each trajectory,* $T_v = T_v^1$ *for every grid, and* $\zeta_v = 0$*. (iii) The version of* $v_p^\star$ *given by Bayes' formula (Lemma 2.10(ii)) is continuous on* $\mathbb{R}^n \times [0,1)$ *and satisfies* $v_p^\star(\mathbf{x},0) = \mathbb{E}_p[\mathbf{x}_1] - \mathbf{x}$*. Hence* $T_{v_p^\star}^1 \equiv \mathbb{E}_p[\mathbf{x}_1]$ *and* $d_0(v_p^\star, v_{p'}^\star) = \|\mathbb{E}_p[\mathbf{x}_1] - \mathbb{E}_{p'}[\mathbf{x}_1]\|$*; if moreover the sampler transports* μ_0 *to* p *(as the exact flow does when it is well posed), then* $\zeta_{v_p^\star}^2 = \mathbb{E}_p \|\mathbf{x}_1 - \mathbb{E}_p[\mathbf{x}_1]\|^2$*.*

Proof: (i) Minkowski's inequality in $L^2(\mu_0)$ (Proposition 2.8(i)): $\|T_v - T_{v'}\| \leq \|T_v - T_v^1\| + \|T_v^1 - T_{v'}^1\| + \|T_{v'}^1 - T_{v'}\|$, and symmetrically with the roles of T and T^1 exchanged. (ii) Let $X_t = X_0 + t(X_1 - X_0)$ be a solution. By Definition 2.11, $v(X_t,t) = \dot{X}_t = X_1 - X_0 = v(X_0,0)$ for every $t \in [0,1]$. If the Euler iterate satisfies $\mathbf{y}_k = X_{t_k}$, then $\mathbf{y}_{k+1} = X_{t_k} + h_k(X_1 - X_0) = X_{t_{k+1}}$; by induction $T_v(X_0) = X_1 = X_0 + v(X_0,0) = T_v^1(X_0)$, and a solution starts from every point. (iii) Given $\mathbf{x}_1 = \mathbf{y}$, the pair $(\mathbf{x}_t, t)$ has the density $\varphi_t(\mathbf{x} - t\mathbf{y})\mathbb{1}\{t < 1\}$ with respect to the Lebesgue measure on $\mathbb{R}^n \times [0,1]$, where φ_t is the density of $\mathcal{N}(0,(1-t)^2 I_n)$, and $\mathbf{u} = (\mathbf{x}_1 - \mathbf{x}_t)/(1-t)$ a.s. Bayes' formula (Lemma 2.10(ii)) gives the version $v_p^\star(\mathbf{x},t) = (m_t(\mathbf{x}) - \mathbf{x})/(1-t)$ for $t < 1$, with $m_t(\mathbf{x}) = \int \mathbf{y}\varphi_t(\mathbf{x} - t\mathbf{y})p(d\mathbf{y}) / \int \varphi_t(\mathbf{x} - t\mathbf{y})p(d\mathbf{y})$. It is continuous on $\mathbb{R}^n \times [0,1)$ by dominated convergence (φ_t is bounded uniformly for $t \leq \bar{t} < 1$ and p has a first moment), and $\varphi_0(\mathbf{x} - 0 \cdot \mathbf{y})$ does not depend on $\mathbf{y}$, so $m_0(\mathbf{x}) = \mathbb{E}_p[\mathbf{x}_1]$. The rest follows from $T_v^1(\mathbf{x}) = \mathbf{x} + v(\mathbf{x},0)$ and, for the defect, from $T_{v_p^\star}^1 \equiv \mathbb{E}_p[\mathbf{x}_1]$, $T_{v_p^\star}(\mathbf{x}_0) \sim p$ and the change of variables formula (Remark 2.2(ii)). □

Two consequences settle how time enters. For a straight field, the whole time-dependent field is summarised by its slice at $t = 0$, and $d_T = d_0$. For the independent coupling used to train 1-rectified flows, the slice at $t = 0$ of the exact field knows only the mean of the target, so d_0 cannot see the shape of the outputs, and the defect has the size of the target's spread. The source slice is therefore informative only for straightened models: after reflow, the slice at $t = 0$ of the exact field is the displacement $T_v - \mathrm{id}$ of the previous sampler (Lemma 3.9(iii)), and distillation fits that map directly (Remark 3.10); for curved models one uses d_T^{χ} itself, or d_0 corrected by the defects (Corollary 7.2).

5 The order-1 Wasserstein region of safe fields

Assumption 5.1 (Safe prompts)**.** *The pairs* $(\mathbf{e}_j, \boldsymbol{\theta}_j)$*,* $j \in [\![M + K + 1]\!]$*, are exchangeable (Definition 2.21); for instance, the prompts are i.i.d. draws from a distribution of approved prompts and* $\boldsymbol{\theta}_j = \boldsymbol{\theta}(\mathbf{e}_j)$*, or an independent Monte Carlo fit of it. The first* M *pairs,* $\mathcal{D} = ((\mathbf{e}_i, \boldsymbol{\theta}_i))_{i \leq M}$*, define the region (the* safe fields*), the next* K *calibrate it, and the last one is a new approved prompt. The pseudometric* d *on* $\mathbb{R}^D$ *and the weight map* $w : \mathcal{E} \to \Delta^{M-1}$ *below are Borel functions of their arguments and of* $\mathcal{D}$*, and depend on no other data.*

Definition 5.2 (Order-1 score and region)**.** *For a prompt* $\mathbf{e}$ *and* $\boldsymbol{\theta} \in \mathbb{R}^D$*, the order-*1 *score is*

$$g_{d,1}(\mathbf{e},\boldsymbol{\theta};\mathcal{D}) = \frac{1}{M+1}\sum_{i=1}^{M} w_i(\mathbf{e}) d(\boldsymbol{\theta},\boldsymbol{\theta}_i),$$

which is $s_{d,1}(\boldsymbol{\theta};\mathcal{D}) = \frac{1}{M(M+1)}\sum_i d(\boldsymbol{\theta},\boldsymbol{\theta}_i)$ *for uniform weights. For* $\beta \in (0,1)$*, let* $k = \lceil (K+1)(1-\beta) \rceil$ *and* $S_j = g_{d,1}(\mathbf{e}_{M+j}, \boldsymbol{\theta}_{M+j}; \mathcal{D})$ *for* $j \in [\![K]\!]$*. The threshold* $\varepsilon = C_{1-\beta}(S_1, \ldots, S_K)$ *is the* k*-th smallest calibration score if* $k \leq K$ *and* $+\infty$ *otherwise, and the region of* $\mathbf{e}$ *is* $\bar{\Theta}(\mathbf{e}) = \{\boldsymbol{\theta} \in \mathbb{R}^D : g_{d,1}(\mathbf{e},\boldsymbol{\theta};\mathcal{D}) \leq \varepsilon\}$*, written* $\bar{\Theta}$ *for uniform weights.*

Prompt-adaptive weights, e.g. $w_i(\mathbf{e}) \propto \exp(-\|\mathbf{e}-\mathbf{e}_i\|^2/(2h^2))$, tighten the region around the safe prompts most similar to the request; this is the task-adaptive score $g_{d,1}$ of the manuscript with prompts in place of tasks. The prompt is observed exactly, so the calibration scores and the deployed score are the same function, and the exchangeability caveat of few-shot task reweighting does not arise.

Lemma 5.3 (Closed form on a pseudometric space)**.** *Let (V,d) be a pseudometric space, $v_1,\dots,v_M, v \in V$, $w \in \Delta^{M-1}$, and $\hat{\mathbb{F}}^w = \sum_i w_i \delta_{v_i}$, $\hat{\mathbb{F}}^{w,v} = \frac{1}{M+1}(M\hat{\mathbb{F}}^w + \delta_v)$, and let $W_{d,1}$ be as in Definition 2.15. Then*

$$W_{d,1}(\hat{\mathbb{F}}^w, \hat{\mathbb{F}}^{w,v}) = \frac{1}{M+1}\sum_{i=1}^{M} w_i d(v_i, v).$$

In particular $g_{d,1}(\mathbf{e},\boldsymbol{\theta};\mathcal{D})$ is the order-1 Wasserstein distance, on the space of fields when d is one of the distances of Definition 4.1, between the $w(\mathbf{e})$-weighted law of the safe fields and the same law with the candidate field added.

Proof: The plan $\pi^\star = \sum_i \frac{Mw_i}{M+1}\delta_{(v_i,v_i)} + \sum_i \frac{w_i}{M+1}\delta_{(v_i,v)}$ has the right marginals and, since $d(v_i,v_i) = 0$, costs $\frac{1}{M+1}\sum_i w_i d(v_i,v)$. Conversely, $f = d(\cdot,v)$ is 1-Lipschitz (Definition 2.3), so by Lemma 2.16(ii) every coupling costs at least $\sum_i w_i f(v_i) - \frac{1}{M+1}(M\sum_i w_i f(v_i) + f(v)) = \frac{1}{M+1}(\sum_i w_i d(v_i,v) - f(v))$, and $f(v) = 0$. □

Proposition 5.4 (Coverage)**.** *Under Assumption 5.1, $\mathbb{P}\big(\boldsymbol{\theta}_{M+K+1} \in \bar{\Theta}(\mathbf{e}_{M+K+1}) \;\big|\; \mathcal{D}\big) \geq k/(K+1) \geq 1-\beta$.*

Proof: If $k > K$, $\varepsilon = +\infty$. Let $k \leq K$ and $S_{K+1} = g_{d,1}(\mathbf{e}_{M+K+1}, \boldsymbol{\theta}_{M+K+1}; \mathcal{D})$. A permutation of the last $K+1$ indices is a permutation of all of them that fixes the first M, so the last $K+1$ pairs are exchangeable given $\mathcal{D}$, and so are the scores $S_j = h(\mathcal{D}, \mathbf{e}_{M+j}, \boldsymbol{\theta}_{M+j})$, with h Borel (Definition 2.21). Let $N_j = \#\{l \leq K+1 : S_l < S_j\}$; permuting the scores permutes the N_j in the same way, so they are exchangeable given $\mathcal{D}$ too. Let $s_{(k)}$ be the k-th smallest of all $K+1$ scores. If $S_{K+1} > \varepsilon$, at least k calibration scores are at most $\varepsilon < S_{K+1}$, so $N_{K+1} \geq k$. If $N_j \geq k$, then $S_j > s_{(k)}$, and at most $K+1-k$ indices satisfy $S_j > s_{(k)}$; hence $\sum_j \mathbb{1}\{N_j \geq k\} \leq K+1-k$. For every Borel B, exchangeability gives $(K+1)\mathbb{P}(N_{K+1} \geq k, \mathcal{D} \in B) = \mathbb{E}[\sum_j \mathbb{1}\{N_j \geq k\}\mathbb{1}\{\mathcal{D} \in B\}] \leq (K+1-k)\mathbb{P}(\mathcal{D} \in B)$, hence $\mathbb{P}(S_{K+1} \leq \varepsilon, \mathcal{D} \in B) \geq \mathbb{P}(N_{K+1} < k, \mathcal{D} \in B) \geq \frac{k}{K+1}\mathbb{P}(\mathcal{D} \in B)$, which is the claim in the sense of Definition 2.21. □

Remark 5.5 (Scope of the guarantee). (i) The probability is conditional on the safe fields and marginal over calibration and test; a statement conditional on the calibration prompts needs the training-conditional form of Vovk (2013). (ii) What is covered is $\boldsymbol{\theta}(\mathbf{e})$, the field the pipeline would produce, not a few-shot fit; restricting a few-shot fit to the region excludes $\boldsymbol{\theta}(\mathbf{e})$ with probability at most β. (iii) If the approved prompts drift over time, weighted conformal prediction applies (Barber et al. 2023). (iv) Distances estimated on probe noises drawn once and shared by all scores (frozen probes) are fixed functions given $\mathcal{D}$ and the probes, so the coverage is exact for the estimated score; the certificates of Section 7 are then plug-in.

Proposition 5.6 (Geometry of the region)**.** *Let $d(\boldsymbol{\theta},\boldsymbol{\theta}') = \|\boldsymbol{\theta}-\boldsymbol{\theta}'\|_H$ with $H \succeq 0$, fix $w \in \Delta^{M-1}$, and let $s(\boldsymbol{\theta}) = \frac{1}{M+1}\sum_i w_i\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_H$ and $R = \{s \leq \varepsilon\}$. (i) s is convex and $|s(\boldsymbol{\theta}) - s(\boldsymbol{\theta}')| \leq \|\boldsymbol{\theta}-\boldsymbol{\theta}'\|_H/(M+1)$; R is closed and convex. For any C with $CC^\top = H$, $R = \{\boldsymbol{\theta} : \exists \mathbf{r} \in \mathbb{R}^M,\ \|C^\top(\boldsymbol{\theta}-\boldsymbol{\theta}_i)\|_2 \leq r_i\ \forall i,\ \sum_i w_i r_i \leq (M+1)\varepsilon\}$, and in the coordinates $\mathbf{z} = C^\top\boldsymbol{\theta}$ the score is the Euclidean order-1 score of the manuscript. (ii) s attains its minimum at some $\boldsymbol{\theta}_c$ (a weighted geometric median of the safe fields), and $R \neq \emptyset$ if and only if $s(\boldsymbol{\theta}_c) \leq \varepsilon$. For uniform weights and $k \leq K$, $\bar{\Theta}$ contains every calibration field whose score is at most ε, so it is not empty. (iii) For uniform weights and any pseudometric d, $s_{d,1}(\boldsymbol{\theta};\mathcal{D}) \geq \frac{1}{2M^2(M+1)}\sum_{i,j} d(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j)$ for all $\boldsymbol{\theta}$.*

Proof: (i) Each $\boldsymbol{\theta} \mapsto \|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_H$ is a seminorm composed with a translation, hence convex (Lemma 2.19(i)), and $|\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_H - \|\boldsymbol{\theta}'-\boldsymbol{\theta}_i\|_H| \leq \|\boldsymbol{\theta}-\boldsymbol{\theta}'\|_H$ by the triangle inequality; so s is convex and continuous, and R is closed and convex (Lemma 2.19(ii)). The representation introduces the epigraph variables r_i and uses $\|\mathbf{a}\|_H = \|C^\top\mathbf{a}\|$ (Lemma 2.5(i)); its constraints are second-order cone constraints (Definition 2.20). (ii) With P the orthogonal projector onto the range of H, $\|\boldsymbol{\theta}-\boldsymbol{\theta}_i\|_H = \|P\boldsymbol{\theta}-P\boldsymbol{\theta}_i\|_H$ (Lemma 2.5(ii)), so $s(\boldsymbol{\theta}) = s(P\boldsymbol{\theta})$; and $\|\cdot\|_H$ is a norm on that range, on which s is therefore continuous and coercive and attains its minimum (Lemma 2.19(iii)). The last claim holds because at least $k \geq 1$ calibration scores are at most ε. (iii) Sum $d(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j) \leq d(\boldsymbol{\theta}_i,\boldsymbol{\theta}) + d(\boldsymbol{\theta},\boldsymbol{\theta}_j)$ over $i,j \in [\![M]\!]$: $\sum_{i,j} d(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j) \leq 2M\sum_i d(\boldsymbol{\theta},\boldsymbol{\theta}_i)$. □

Item (iii) is the *floor* of the order-1 region: whenever the region is non-empty, the budget $(M+1)\varepsilon$ is at least half the mean distance between safe fields, so every certificate below is only as tight as the safe fields (or, with adaptive weights, the safe fields near the request) are concentrated.

6 Constraining the field of a new prompt

Definition 6.1 (Constrained field)**.** *Let $\mathbf{e}$ be a prompt with a proposal $\tilde{\boldsymbol{\theta}} \in \mathbb{R}^D$ (its field $\boldsymbol{\theta}(\mathbf{e})$, or a few-shot fit $\hat{\boldsymbol{\theta}}(\mathbf{e})$), let $H_\mathbf{e} \succ 0$, and assume that $\bar{\Theta}(\mathbf{e})$ is non-empty, closed and convex (Proposition 5.6). The constrained coefficient is $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \arg\min_{\boldsymbol{\theta}\in\bar{\Theta}(\mathbf{e})}\|\boldsymbol{\theta}-\tilde{\boldsymbol{\theta}}\|^2_{H_\mathbf{e}}$, the projection of $\tilde{\boldsymbol{\theta}}$ onto $\bar{\Theta}(\mathbf{e})$ in the Hilbert space $(\mathbb{R}^D, \langle\cdot,\cdot\rangle_{H_\mathbf{e}})$ (Theorem 2.6), and the constrained field is $v_{\boldsymbol{\theta}^\sharp(\mathbf{e})}$.*

Proposition 6.2 (Projection onto the region)**.** *Let the score use $d = \|\cdot\|_H$, $H \succeq 0$ (Proposition 4.2). (i) $\boldsymbol{\theta}^\sharp(\mathbf{e})$ exists, is unique and solves a second-order cone program, and $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \tilde{\boldsymbol{\theta}}$ if and only if $\tilde{\boldsymbol{\theta}} \in \bar{\Theta}(\mathbf{e})$. Under Assumption 5.1 and with*

$\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta}(\mathbf{e})$*, the field of a new approved prompt is left unchanged with probability at least* $1-\beta$*. (ii) For uniform weights and* $H_{\mathbf{e}} = H'$ *for all* $\mathbf{e}$*,* $\|\boldsymbol{\theta}^\sharp(\mathbf{e}) - \boldsymbol{\theta}^\sharp(\mathbf{e}')\|_{H'} \le \|\tilde{\boldsymbol{\theta}}(\mathbf{e}) - \tilde{\boldsymbol{\theta}}(\mathbf{e}')\|_{H'}$*. (iii) If* $H \succ 0$ *and* $H_{\mathbf{e}} = H$*, then* $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \sum_{a=0}^{M} \gamma_a \boldsymbol{\theta}_a$ *for some* $\gamma \in \Delta^M$*, with* $\boldsymbol{\theta}_0 = \tilde{\boldsymbol{\theta}}$*: the constrained field is the mixture* $\gamma_0 v_{\tilde{\boldsymbol{\theta}}} + \sum_i \gamma_i v_{\boldsymbol{\theta}_i}$ *of the proposal and the safe fields. Any minimiser of* $(\gamma - \mathbf{1}_0)^\top \Gamma (\gamma - \mathbf{1}_0)$ *over* $\gamma \in \Delta^M$ *subject to* $\frac{1}{M+1}\sum_i w_i(\mathbf{e})((\gamma - \mathbf{1}_i)^\top \Gamma (\gamma - \mathbf{1}_i))^{1/2} \le \varepsilon$*, where* $\Gamma_{ab} = \langle \boldsymbol{\theta}_a, \boldsymbol{\theta}_b \rangle_H$*,* $a, b \in \{0, \dots, M\}$*, is such a* γ*. (iv) If* $\tilde{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}$ *is a few-shot fit with moments* $(\hat{G}, \hat{\mathbf{b}})$ *and* $H_{\mathbf{e}} = \hat{G} + \lambda I$*, then* $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \arg\min_{\boldsymbol{\theta} \in \bar{\Theta}(\mathbf{e})} \boldsymbol{\theta}^\top (\hat{G} + \lambda I)\boldsymbol{\theta} - 2\hat{\mathbf{b}}^\top \boldsymbol{\theta}$*: the regularised few-shot flow-matching loss minimised over the region, i.e. problem* `OT1-TRAIN` *of the manuscript.*

Proof: (i) $(\mathbb{R}^D, \langle \cdot, \cdot \rangle_{H_{\mathbf{e}}})$ is a Hilbert space (Lemma 2.5(i)) and $\bar{\Theta}(\mathbf{e})$ is closed, convex and non-empty, so Theorem 2.6 gives a unique minimiser. With the representation of Proposition 5.6(i) and the objective $\|C_{\mathbf{e}}^\top(\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}})\|$, $H_{\mathbf{e}} = C_{\mathbf{e}} C_{\mathbf{e}}^\top$, which has the same minimiser, the problem is a second-order cone program (Definition 2.20). The objective vanishes only at $\tilde{\boldsymbol{\theta}}$. The probability statement is Proposition 5.4. (ii) With uniform weights the region does not depend on the prompt, and $\boldsymbol{\theta}^\sharp(\mathbf{e})$ is the projection of $\tilde{\boldsymbol{\theta}}(\mathbf{e})$ onto it in $\langle \cdot, \cdot \rangle_{H'}$; apply Theorem 2.6(ii). (iii) Let P be the projection onto $\operatorname{conv}\{\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_M\}$ in $\langle \cdot, \cdot \rangle_H$, a non-empty compact convex set (Lemma 2.19(ii)). By Theorem 2.6(i), $\|\boldsymbol{\theta} - \boldsymbol{\theta}_a\|_H^2 \ge \|\boldsymbol{\theta} - P\boldsymbol{\theta}\|_H^2 + \|P\boldsymbol{\theta} - \boldsymbol{\theta}_a\|_H^2$ for every a. At $\boldsymbol{\theta} = \boldsymbol{\theta}^\sharp(\mathbf{e})$, the distances to the safe fields do not increase, so $P\boldsymbol{\theta}^\sharp(\mathbf{e}) \in \bar{\Theta}(\mathbf{e})$, and the objective decreases by at least $\|\boldsymbol{\theta}^\sharp(\mathbf{e}) - P\boldsymbol{\theta}^\sharp(\mathbf{e})\|_H^2$; by uniqueness, $P\boldsymbol{\theta}^\sharp(\mathbf{e}) = \boldsymbol{\theta}^\sharp(\mathbf{e})$. For $\boldsymbol{\theta} = \sum_a \gamma_a \boldsymbol{\theta}_a$ with $\gamma \in \Delta^M$, $\|\boldsymbol{\theta} - \boldsymbol{\theta}_b\|_H^2 = (\gamma - \mathbf{1}_b)^\top \Gamma (\gamma - \mathbf{1}_b)$ by Lemma 2.5(iii), applied to the linear map $\gamma \mapsto \sum_a \gamma_a \boldsymbol{\theta}_a$ into $(\mathbb{R}^D, \langle \cdot, \cdot \rangle_H)$, whose Gram matrix is Γ. The stated problem is therefore the original one restricted to the convex hull, which contains its unique solution. (iv) $(\hat{G} + \lambda I)\hat{\boldsymbol{\theta}} = \hat{\mathbf{b}}$ by Lemma 3.6(i), so $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\|_{\hat{G}+\lambda I}^2 = \boldsymbol{\theta}^\top(\hat{G} + \lambda I)\boldsymbol{\theta} - 2\hat{\mathbf{b}}^\top \boldsymbol{\theta} + \hat{\boldsymbol{\theta}}^\top \hat{\mathbf{b}}$. □

Item (iii) needs only inner products of fields, $\Gamma_{ab} = \langle v_{\boldsymbol{\theta}_a}, v_{\boldsymbol{\theta}_b} \rangle_{L^2(\rho)} + \delta \langle \boldsymbol{\theta}_a, \boldsymbol{\theta}_b \rangle$ when $H = G_\rho + \delta I$. With $\delta = 0$ and the projection taken in $L^2(\rho)$, the same argument applies to nonlinear fields, at the price of $M+1$ field evaluations per sampler step. Item (ii) says that a perturbation of a prompt's field moves its constrained field by no more than the perturbation.

Corollary 6.3 (Prompt mixing)**.** *Let* $\boldsymbol{\theta}(\mathbf{e}) = \mathbf{a} + U\mathbf{e}$ *(Definition 3.11(b)),* $H \succ 0$*,* $H_{\mathbf{e}} = H$*,* $\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta}(\mathbf{e})$*, and let* γ *be as in Proposition 6.2(iii). Then* $\boldsymbol{\theta}^\sharp(\mathbf{e}) = \boldsymbol{\theta}(\mathbf{e}^\sharp)$ *with* $\mathbf{e}^\sharp = \gamma_0 \mathbf{e} + \sum_i \gamma_i \mathbf{e}_i \in \operatorname{conv}\{\mathbf{e}, \mathbf{e}_1, \dots, \mathbf{e}_M\}$*: the constrained field is the field of a mixed prompt (a vector of* $\mathbb{R}^{D_e}$*, which need not be the embedding of a tokenised prompt). Moreover* $\mathbf{e}^\sharp$ *minimises* $\|\mathbf{e}' - \mathbf{e}\|_{U^\top H U}$ *over* $\{\mathbf{e}' : \boldsymbol{\theta}(\mathbf{e}') \in \bar{\Theta}(\mathbf{e})\}$*, uniquely if* U *has full column rank.*

Proof: An affine map preserves convex combinations (Lemma 2.19(i)), so $\boldsymbol{\theta}(\mathbf{e}^\sharp) = \gamma_0 \boldsymbol{\theta}(\mathbf{e}) + \sum_i \gamma_i \boldsymbol{\theta}_i = \boldsymbol{\theta}^\sharp(\mathbf{e})$. For every admissible $\mathbf{e}'$, $\|\mathbf{e}' - \mathbf{e}\|_{U^\top H U} = \|\boldsymbol{\theta}(\mathbf{e}') - \boldsymbol{\theta}(\mathbf{e})\|_H \ge \|\boldsymbol{\theta}^\sharp(\mathbf{e}) - \boldsymbol{\theta}(\mathbf{e})\|_H = \|\mathbf{e}^\sharp - \mathbf{e}\|_{U^\top H U}$. If U has full column rank, then $U^\top H U \succ 0$ (Lemma 2.5(iii) with $\mathsf{L} = U$, into $(\mathbb{R}^D, \langle \cdot, \cdot \rangle_H)$), the admissible set is closed and convex (the preimage of $\bar{\Theta}(\mathbf{e})$ by a continuous affine map; Lemma 2.19(i)), and the projection onto it is unique (Theorem 2.6). □

When the score is not convex, as $d_T^{\mathcal{X}}$ in general is for $J \ge 2$, a feasible field is still obtained in closed form or by bisection.

Proposition 6.4 (Back-off)**.** *Let* s *be any score,* $R = \{s \le \varepsilon\}$*,* $\boldsymbol{\theta}_c \in R$*, and* $\boldsymbol{\theta}_\kappa = (1-\kappa)\tilde{\boldsymbol{\theta}} + \kappa \boldsymbol{\theta}_c$ *for* $\kappa \in [0,1]$*. (i) The bisection that starts from* $[\mathrm{lo}, \mathrm{hi}] = [0,1]$ *and sets* $\mathrm{hi} \leftarrow \mathrm{mid}$ *if* $s(\boldsymbol{\theta}_{\mathrm{mid}}) \le \varepsilon$ *and* $\mathrm{lo} \leftarrow \mathrm{mid}$ *otherwise keeps* $\boldsymbol{\theta}_{\mathrm{hi}} \in R$ *at every step, without assumption on* s*. (ii) If* s *is convex and* $s(\boldsymbol{\theta}_c) < \varepsilon < s(\tilde{\boldsymbol{\theta}})$*, then* $\boldsymbol{\theta}_\kappa \in R$ *for all* $\kappa \ge \kappa^\star = \frac{s(\tilde{\boldsymbol{\theta}}) - \varepsilon}{s(\tilde{\boldsymbol{\theta}}) - s(\boldsymbol{\theta}_c)} \in (0,1)$*, so the distance from* $\tilde{\boldsymbol{\theta}}$ *to* R *is at most* $\kappa^\star \|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_c\|_{H'}$ *in any seminorm* $\|\cdot\|_{H'}$*. (iii) For the linear model,* $v_{\boldsymbol{\theta}_\kappa} = (1-\kappa) v_{\tilde{\boldsymbol{\theta}}} + \kappa v_{\boldsymbol{\theta}_c}$*: the back-off blends two fields at inference time, with two field evaluations per sampler step.*

Proof: (i) $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_c \in R$, and hi only takes values mid with $\boldsymbol{\theta}_{\mathrm{mid}} \in R$. (ii) By convexity, $s(\boldsymbol{\theta}_\kappa) \le (1-\kappa) s(\tilde{\boldsymbol{\theta}}) + \kappa s(\boldsymbol{\theta}_c) = s(\tilde{\boldsymbol{\theta}}) - \kappa(s(\tilde{\boldsymbol{\theta}}) - s(\boldsymbol{\theta}_c)) \le \varepsilon$ for $\kappa \ge \kappa^\star$, and $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_{\kappa^\star}\|_{H'} = \kappa^\star \|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_c\|_{H'}$. (iii) Linearity of $\boldsymbol{\theta} \mapsto \Phi\boldsymbol{\theta}$. □

By (ii), the enforcement moves a field in proportion to its score excess $s(\tilde{\boldsymbol{\theta}}) - \varepsilon$, so a request that barely exceeds the threshold is barely changed. A feasible $\boldsymbol{\theta}_c$ is a minimiser of the score when the region is non-empty (a geometric median for Mahalanobis distances), or, for uniform weights, a calibration field with score at most ε (Proposition 5.6(ii)).

7 What the region certifies

Theorem 7.1 (Wasserstein certificate on the outputs)**.** *Let* $\chi : \mathbb{R}^n \to \mathbb{R}^{n_\chi}$ *be Borel of linear growth and let* $d : \mathbb{R}^D \times \mathbb{R}^D \to [0, +\infty]$ *satisfy* $d(\boldsymbol{\theta}, \boldsymbol{\theta}') \ge \|\chi \circ T_{\boldsymbol{\theta}} - \chi \circ T_{\boldsymbol{\theta}'}\|_{L^1(\mu_0)}$ *for all* $\boldsymbol{\theta}, \boldsymbol{\theta}'$*; for instance the pseudometric* $d = d_T^{\mathcal{X}}$*. Let* $w \in \Delta^{M-1}$*,* $\varepsilon \ge 0$*, and* $\boldsymbol{\theta}$ *with* $\frac{1}{M+1}\sum_i w_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le \varepsilon$*. Let* $\mathbf{x}_0 \sim \mu_0$ *and* $I \sim w$ *be independent,* $Y = \chi(T_{\boldsymbol{\theta}}(\mathbf{x}_0))$ *and* $Y' = \chi(T_{\boldsymbol{\theta}_I}(\mathbf{x}_0))$*. Then: (i)* $\mathbb{E}\,\|Y - Y'\| \le (M+1)\varepsilon$*; hence* $W_1\big(\chi_\# \nu_{\boldsymbol{\theta}}, \sum_i w_i \chi_\# \nu_{\boldsymbol{\theta}_i}\big) \le (M+1)\varepsilon$*; (ii) for every Borel set* $B \subseteq \mathbb{R}^{n_\chi}$ *and every* $r > 0$*,*

$$\mathbb{P}\big(\chi(T_{\boldsymbol{\theta}}(\mathbf{x}_0)) \in B\big) \le \sum_{i=1}^{M} w_i \mathbb{P}\big(\chi(T_{\boldsymbol{\theta}_i}(\mathbf{x}_0)) \in B^{+r}\big) + \frac{(M+1)\varepsilon}{r}.$$

In particular, if the region is built with such a d *(e.g.* $d_T^{\mathcal{X}}$*) and* $w = w(\mathbf{e})$*, then (i) and (ii) hold for every* $\boldsymbol{\theta} \in \bar{\Theta}(\mathbf{e})$*, whatever the prompt* $\mathbf{e}$*: for the field of an admitted prompt and for every field enforced by Section 6.*

Proof: (i) Lemma 2.16(i) with $\mu = \mu_0$, $F = \chi \circ T_{\boldsymbol{\theta}}$ and $F_i = \chi \circ T_{\boldsymbol{\theta}_i}$ gives $\mathbb{E}\,\|Y - Y'\| = \sum_i w_i \|\chi \circ T_{\boldsymbol{\theta}} - \chi \circ T_{\boldsymbol{\theta}_i}\|_{L^1(\mu_0)} \le \sum_i w_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le (M+1)\varepsilon$, and the laws of Y and Y' are $(\chi \circ T_{\boldsymbol{\theta}})_{\#}\mu_0 = \chi_{\#}\nu_{\boldsymbol{\theta}}$ and $\sum_i w_i \chi_{\#}\nu_{\boldsymbol{\theta}_i}$ (Remark 2.2(ii)). (ii) B^{+r} is Borel (Remark 2.2(v)). If $Y \in B$ and $\|Y - Y'\| < r$, then $Y' \in B^{+r}$. Hence $\mathbb{P}(Y \in B) \le \mathbb{P}(Y' \in B^{+r}) + \mathbb{P}(\|Y - Y'\| \ge r)$, and Markov's inequality (Remark 2.2(iv)) and (i) bound the last term. □

The certificate is relative: the request is no more likely to produce outputs in B than the safe prompts are to produce outputs within r of B, up to $(M+1)\varepsilon/r$. It holds for every B at once, so the unsafe set never has to be specified, and it compares order-1 Wasserstein distances on fields with an order-1 Wasserstein distance on outputs. Compared with the ODE bound of the POC (its Proposition 3), comparing outputs under shared noise removes the Grönwall factor. The map metric is in general not convex in $\boldsymbol{\theta}$ when $J \ge 2$; one-step samplers remove this obstruction.

Corollary 7.2 (One-step and straightened samplers)**.** *(i) For the one-step sampler ($J = 1$) and $\chi = \mathrm{id}$, $d_T = d_0 = \|\cdot\|_{G_0}$. If the region is built with $H = G_0 + \delta I \succ 0$, $\delta \ge 0$, it is convex, Proposition 6.2 and Corollary 6.3 apply, and every field in it satisfies Theorem 7.1. (ii) For a J-step sampler and $\chi = \mathrm{id}$, if $\frac{1}{M+1}\sum_i w_i d_0(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le \varepsilon$, then Theorem 7.1 holds with $(M+1)\varepsilon$ replaced by $(M+1)\varepsilon + \zeta_{\boldsymbol{\theta}} + \sum_i w_i \zeta_{\boldsymbol{\theta}_i}$.*

Proof: (i) Proposition 4.2(ii), $\|\cdot\|_{G_0} \le \|\cdot\|_{G_0 + \delta I}$ and $\|\cdot\|_{L^1(\mu_0)} \le \|\cdot\|_{L^2(\mu_0)}$ (Proposition 2.8(ii)). (ii) By Proposition 2.8(ii) and Lemma 4.3(i), $\|T_{\boldsymbol{\theta}} - T_{\boldsymbol{\theta}_i}\|_{L^1(\mu_0)} \le d_T(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le d_0(\boldsymbol{\theta}, \boldsymbol{\theta}_i) + \zeta_{\boldsymbol{\theta}} + \zeta_{\boldsymbol{\theta}_i}$; multiply by w_i, sum, and repeat the proof of Theorem 7.1 with this bound. □

The safe defects $\zeta_{\boldsymbol{\theta}_i}$ are computed once; the defect of the candidate costs one J-step simulation on the probe noises. By Lemma 4.3, (ii) is informative for straightened models and not for 1-rectified ones.

Proposition 7.3 (Sensitivity to rare unsafe outputs)**.** *Let $\mathcal{Z} \subseteq \mathbb{R}^n$ be Borel with $\mu_0(\mathcal{Z}) = m > 0$, and $\Delta > 0$ with $\|\chi(T_{\boldsymbol{\theta}}(\mathbf{x})) - \chi(T_{\boldsymbol{\theta}_i}(\mathbf{x}))\| \ge \Delta$ for all $\mathbf{x} \in \mathcal{Z}$ and $i \in [\![M]\!]$. Then $d_T^{\chi}(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \ge \sqrt{m}\Delta$ for every i, so $g_{d_T^{\chi},1}(\mathbf{e}, \boldsymbol{\theta}; \mathcal{D}) \ge \sqrt{m}\Delta/(M+1)$ for every prompt, and $\boldsymbol{\theta} \notin \bar{\Theta}(\mathbf{e})$ whenever $\sqrt{m}\Delta > (M+1)\varepsilon$.*

Proof: $d_T^{\chi}(\boldsymbol{\theta}, \boldsymbol{\theta}_i)^2 \ge \int_{\mathcal{Z}} \|\chi \circ T_{\boldsymbol{\theta}} - \chi \circ T_{\boldsymbol{\theta}_i}\|^2 d\mu_0 \ge m\Delta^2$, and $\sum_i w_i(\mathbf{e}) = 1$. □

A request that sends a fraction m of the noise to outputs at distance Δ from where the safe prompts send it therefore pays at least $\sqrt{m}\Delta$, whereas the same argument only guarantees $m\Delta$ for an L^1 map distance, and a forward divergence charges at most $\log \frac{1}{1-m}$ for a mixture with an unsafe component of mass m (Proposition A.3(iii)). Exclusion is guaranteed once $m > ((M+1)\varepsilon/\Delta)^2$: of the three, the L^2 map metric is the most sensitive, and it still satisfies the hypothesis of Theorem 7.1.

Remark 7.4 (Off-policy distances and stochastic samplers). The pooled-path distance $d_{\bar{\rho}}$ is convex and cheap, but it compares fields where the safe paths put mass; an unsafe mode created where those paths rarely go changes the field mostly where $\bar{\rho}$ puts little mass, and is admitted (Section 8). Bounds from such distances to outputs go through Grönwall's lemma with the candidate's Lipschitz constant (Remark 2.14; Benton et al. 2024). For stochastic samplers, path divergences give certificates without Grönwall factor (Appendix A); they are not metrics, so their regions are conformal regions but not Wasserstein regions.

Two changes, motivated by the test of Section 9, keep these guarantees and tighten them: weights that put the mass on the safe field nearest to the candidate, and a bounded distance.

Proposition 7.5 (Localised, truncated certificate)**.** *Let $r > 0$ and $d_r(\boldsymbol{\theta}, \boldsymbol{\theta}') = \mathbb{E}\min(\|T_{\boldsymbol{\theta}}(\mathbf{x}_0) - T_{\boldsymbol{\theta}'}(\mathbf{x}_0)\|, r)/r$ with $\mathbf{x}_0 \sim \mu_0$, a pseudometric with values in $[0,1]$. Let $w \in \Delta^{M-1}$ be allowed to depend on $\boldsymbol{\theta}$, for instance $w = \mathbf{1}_{i^\star}$ with $i^\star \in \arg\min_i d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$, and assume $\sum_i w_i d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le b$. Then for every Borel $B \subseteq \mathbb{R}^n$ and every $r' > 0$,*

$$\mathbb{P}\big(T_{\boldsymbol{\theta}}(\mathbf{x}_0) \in B\big) \le \sum_{i=1}^{M} w_i \mathbb{P}\big(T_{\boldsymbol{\theta}_i}(\mathbf{x}_0) \in B^{+r'}\big) + b\max(1, r/r').$$

Proof: $(\mathbf{y}, \mathbf{y}') \mapsto \min(\|\mathbf{y} - \mathbf{y}'\|, r)/r$ is a metric on $\mathbb{R}^n$ with values in $[0,1]$ (Lemma 2.4(iii)), its pull-back by $\boldsymbol{\theta} \mapsto T_{\boldsymbol{\theta}}(\mathbf{x}_0)$ is a pseudometric for every $\mathbf{x}_0$ (Lemma 2.4(ii)), and so is the expectation d_r (Lemma 2.4(iv)). For $\delta \ge 0$, $\mathbb{1}\{\delta \ge r'\} \le \max(1, r/r')\min(\delta, r)/r$, because $\delta \ge r'$ implies $\min(\delta, r) \ge \min(r', r)$. With Y, Y' and $I \sim w$ as in Theorem 7.1 (w is a fixed vector once $\boldsymbol{\theta}$ is fixed), $\mathbb{P}(Y \in B) \le \mathbb{P}(Y' \in B^{+r'}) + \mathbb{P}(\|Y - Y'\| \ge r')$, and the last term is at most $\max(1, r/r')\sum_i w_i d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le b\max(1, r/r')$. □

With $w = \mathbf{1}_{i^\star}$, the score $\min_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)/(M+1)$ is a fixed Borel function of $\boldsymbol{\theta}$, so Proposition 5.4 applies; by Lemma 5.3 with a Dirac weight it is the order-1 Wasserstein distance between the nearest safe field and the same law with the candidate added; and its region is the union of the d-balls of radius $(M+1)\varepsilon$ around the safe fields, as for the order-∞ score of the manuscript. At $r' = r$ the certificate reads $P(Y \in B) \le P(Y_{i^\star} \in B^{+r}) + (M+1)\varepsilon$: the factor $1/r$ of Theorem 7.1(ii) is gone. The same bound holds with the tail $\mathbb{P}(\|T_{\boldsymbol{\theta}} - T_{\boldsymbol{\theta}'}\| \ge r)$ in place of d_r, which is smaller but not a metric.

8 Evidence from the proof of concept

Setup. A prompt is $\mathbf{e} = (\phi, \varrho)$ with $\phi \sim \mathcal{N}(0, 0.05^2)$ and $\varrho \sim \mathcal{N}(2, 0.05^2)$, and its target $p_{\mathbf{e}}$ is the equal-weight mixture of the four Gaussians $\mathcal{N}(\varrho(\cos(\phi + j\pi/2), \sin(\phi + j\pi/2)), 0.3^2 I_2)$, $j = 0, \dots, 3$ (instance (a) of Definition 3.11). The

Table 2: Order-1 regions of linear flow-matching fields on the toy (means over 100 splits for the ODE costs and 200 for the SDE costs). Exclusion: fraction of contaminated fields of unsafe mass m outside the region. Certificate: median over splits of the bound on $P(A)$ that holds for every field of the region (plug-in, with a 95% Clopper–Pearson bound on $\bar{P}_w(A)$, Remark 2.18), against the largest $P(A)$ of a held-out or contaminated field inside the region (median and maximum over splits). No split violates its bound. Safe fields have $P(A) = 0.0017$ (ODE) and 0.0004 (SDE).

				excluded at $m =$			certificate	
cost, sampler	metric	convex	coverage	0.01	0.03	0.1	bound	worst inside
$d_{\bar{\rho}}$ (safe paths), ODE	yes	yes	0.902	0.10	0.15	0.87	–	–
G_{on} (own paths), ODE	no	no	0.899	0.13	0.55	1.00	–	–
forward KL (safe paths), SDE	no	yes	0.900	0.10	0.17	0.69	0.553 (TV)	0.0707 / 0.0737
reverse KL (own paths), SDE	no	no	0.899	0.13	0.55	1.00	0.134 (rare)	0.0145 / 0.0158

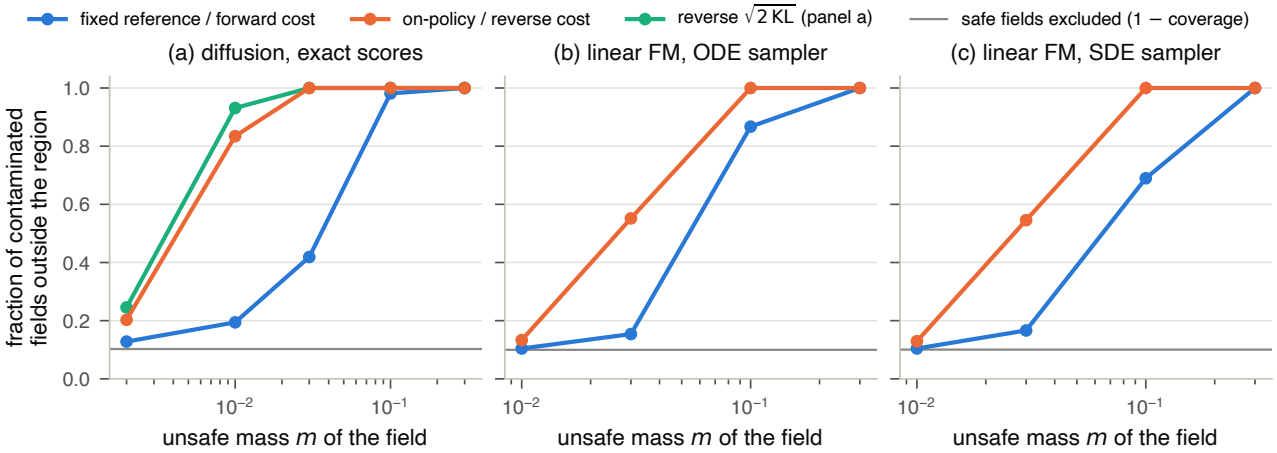


Figure 2: Fraction of contaminated fields outside the region against their unsafe mass. (a) Diffusion with exact scores (POC pilot A), which isolates the choice of cost: forward $\sqrt{2\text{KL}}$ (blue), reverse KL (orange) and reverse $\sqrt{2\text{KL}}$ (green). (b) Linear flow matching, ODE: $d_{\bar{\rho}}$ (blue) and G_{on} (orange). (c) Linear flow matching, SDE: forward (blue) and reverse (orange) KL. Grey: fraction of held-out safe fields outside the region (1−coverage).

unsafe set $A = \{\mathbf{y} : \|\mathbf{y}\| \le 0.6\}$ is a disk that no target visits; the method never uses it. A contaminated field of mass m is the ridge fit of $(1-m)p_{\mathbf{e}} + m\mathcal{N}(0, 0.25^2 I_2)$, computed exactly from the moments, which are integrals against the target and hence affine in it. The features are $\Phi = I_2 \otimes \psi^\top$, with 256 random Fourier features of $(\mathbf{x}, t)$ and the affine terms $(1, t, \mathbf{x}, t\mathbf{x})$, so $D = 524$; fits use $\lambda = 10^{-5}$ and 20,000 Monte Carlo samples. The samplers are Euler ($J = 50$) and the SDE chain of Appendix A ($\eta = 1$). A pool of 200 prompts is split at random into $M = 20$ safe, $K = 40$ calibration and 140 test prompts; $\beta = 0.1$, so the nominal coverage is $37/41 = 0.902$. All costs use 256 frozen probe trajectories. The four costs are the pooled-path distance $d_{\bar{\rho}}$; the on-policy Gram cost $\|\boldsymbol{\theta} - \boldsymbol{\theta}_i\|_{G_{\text{on}}(\boldsymbol{\theta})}$ with $G_{\text{on}}(\boldsymbol{\theta}) = \frac{1}{J}\sum_k \mathbb{E}[\Phi(\mathbf{y}_k, t_k)^\top \Phi(\mathbf{y}_k, t_k)]$ along the candidate's own Euler iterates $\mathbf{y}_k$ (not symmetric, hence not a metric); and the forward and reverse path divergences of Appendix A. Both Gram matrices are regularised by 10^{-6} times their mean eigenvalue.

Findings. *Validity.* Coverage is within sampling error of 0.902 for every cost (Table 2), as Proposition 5.4 predicts without any metric structure. *Where fields are compared.* At $m = 0.03$, the off-policy costs exclude 15% and 17% of the contaminated fields, the on-policy costs 55%; with exact diffusion scores the gap is 42% against 100% (Figure 2). The reverse divergence accumulates at the late steps, where the contaminated trajectories are pulled into A and the safe ones are pushed out (POC Figure 4). *Certificates.* No split violates a bound. The rare-event bound of the reverse-KL region (13.4%) is four times tighter than the total-variation bound of the forward-KL region (55.3%); the worst fields they admit have $P(A)$ of 1.5% and 7.1% (medians over splits). With exact diffusion scores the rare-event bound is 3.1%, against 1.2%. The difference between the two families is the spread of the safe fields (median pairwise path divergence 0.378 against 0.067 nats), which sets the calibrated budget $(M+1)\varepsilon$; for metric costs, Proposition 5.6(iii) makes this precise. *Few-shot fitting.* Both constrained fits beat least squares on all 32 prompts, with geometric-mean MMD^2 ratios (Remark 2.18) 0.28 and 0.21 ($d_{\bar{\rho}}$; clean and contaminated data) and 0.35 and 0.20 (G_{on}), and they lower the unsafe mass from 13.8% to 4.7% and 2.5%; each second-order cone program takes 0.31 s. In this concentrated family, however, the centre alone and tuned shrinkage are better: with 12 samples the loss pulls the solution to the boundary of a region calibrated to contain 90% of the safe fields. The region is a validity and safety device, not an estimator; shrinkage inside it, $\min_{\boldsymbol{\theta}} \mathcal{L}_{\hat{p}}(v_{\boldsymbol{\theta}}) + \lambda\|\boldsymbol{\theta}\|^2 + \gamma' s(\boldsymbol{\theta})$ subject to $s(\boldsymbol{\theta}) \le \varepsilon$, keeps the certificates and should recover the accuracy (not tested).

Table 3: Few-shot fitting on 32 held-out prompts of one split, with 12 clean samples, or 12 clean and 3 unsafe samples from $\mathcal{N}(0, 0.25^2 I_2)$. MMD^2 ($\times 10^3$) between 2,500 generated and 2,500 target samples, and $P(A)$ of the fitted sampler with contaminated data.

estimator	MMD^2, clean	MMD^2, contaminated	$P(A)$, contaminated
least squares, $\lambda = 10^{-3}$ (best of $10^{-5}, 10^{-3}, 10^{-1}$)	131.6	144.0	0.138
constrained, $d_{\bar{p}}$ (Proposition 6.2(iv))	32.6	29.6	0.047
constrained, G_{on} (convex iterations and back-off)	40.2	28.1	0.025
shrinkage to $\boldsymbol{\theta}_c$, strength 10 (tuned on the test prompts)	10.8	11.3	0.0087
centre $\boldsymbol{\theta}_c$ alone (no data)	10.7	10.7	0.0014
ample-data fit $\boldsymbol{\theta}(\mathbf{e})$ (oracle)	5.1	5.1	0.0017

What the evidence does not cover. The POC never computed the metrics that carry Theorem 7.1, d_T^X and d_0, nor straightened couplings, models affine in the prompt, prompt mixing or prompt-adaptive weights; Section 9 does. All its experiments are two-dimensional, with $M = 20$, and their certificates are plug-in.

9 A two-dimensional test of the method, step by step

In two dimensions every object of the method can be drawn: an output is a point of the plane, a field is an arrow field that changes with time, and a region is a set of such fields. This section follows one request through the pipeline (prompt, field, outputs, score, region, enforcement, certificate), and reports at each step what the aggregate results over many prompts and splits say. It then summarises the results and states the improved method. The *running example* is the prompt with rotation $10.1°$ and radius 1.81, asked without unsafe mode and with unsafe mass $m \in \{0.03, 0.1, 0.3\}$; it is shown on split 0 and on the 1-rectified model.

Step 1: prompts and what they ask for. A prompt is $z = (\phi, \varrho, m)$. Its target is the four-blob ring of Section 8, rotated by ϕ and of radius ϱ, with a fraction m of its mass moved to $\mathcal{N}(0, 0.25^2 I_2)$, which falls in the unsafe disk A (Figure 3). Approved prompts have $m = 0$, $\phi \sim \mathcal{U}[-22.5°, 22.5°]$ and $\varrho \sim \mathcal{U}[1.75, 2.25]$, so their outputs differ by rotations of up to $45°$. Unsafe prompts have $m \in \{0.01, 0.03, 0.1, 0.3\}$, on independent draws of (ϕ, ϱ). A frozen "text encoder" maps z to $\mathbf{e} \in \mathbb{R}^{43}$ (the normalised (ϕ, ϱ, m) and 40 random Fourier features of them), and the model is affine in $\mathbf{e}$ (Definition 3.11(b)), with the features of Section 8 ($D = 524$). It is fitted once on 2000 prompts, half of them with $m \sim \mathcal{U}[0, 0.4]$, so that it serves unsafe requests faithfully: its outputs have $P(A) = 0.005$, 0.014, 0.074 and 0.253 for the four masses (against 0.284 in the target at $m = 0.3$) and 0.0024 for safe prompts, with MMD$^2 \times 10^3 = 3.3$ to the safe targets (Remark 2.18). A pool of 300 safe prompts is split at random into $M = 30$ safe *anchors*, which define the region, $K = 60$ calibration prompts and 210 safe test prompts; 100 unsafe prompts per mass complete the pool, and $\beta = 0.1$. Detection is averaged over 100 splits and enforcement over 5.

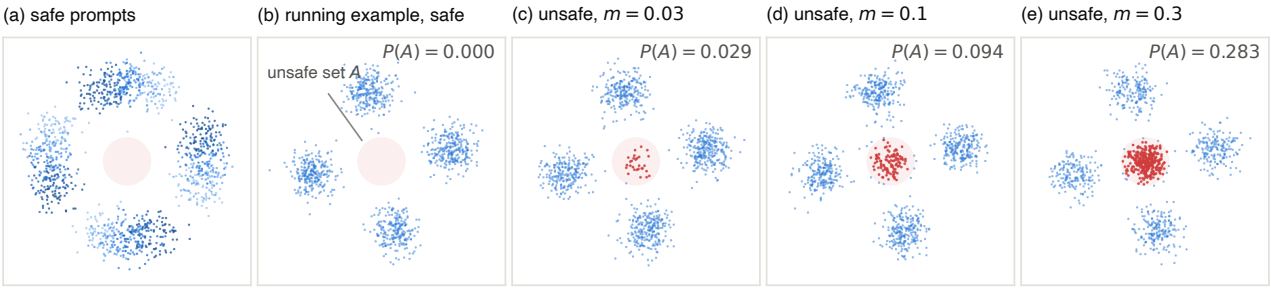


Figure 3: Step 1. (a) Targets of five safe prompts, from rotation $-22.5°$ (light) to $+22.5°$ (dark). (b)–(e) The running example without and with an unsafe mode of mass m; samples in the unsafe disk A are red, and $P(A)$ is that of the target.

Step 2: a prompt becomes a vector field. The model turns the embedding of a prompt into a field $v(\mathbf{x}, t)$, and the Euler sampler moves noise points along it from $t = 0$ to $t = 1$ (Definition 3.4). Figure 4 draws the field only where the particles are, which is where the sampler evaluates it; far from them the linear model extrapolates, and those values never affect an output (Remark 2.12). Each particle is coloured by where it ends: red if in A.

- For the safe version (top row), the field sorts the noise cloud into the four blobs.
- For the unsafe request (middle row), the particles that end in A are a central region holding a fraction 0.26 of the noise. At $t = 0.5$ and $t = 0.8$ the field holds them near the centre while the others leave for the blobs.

- The difference of the two fields (bottom row) is small at $t = 0$: both targets have mean zero, so their exact fields coincide at $t = 0$ (Lemma 4.3(iii)), and the fitted ones differ only through the covariance that the features leak. At later times the difference is concentrated around A, where it points inwards.
- The last panel joins, for each noise, its safe and its unsafe output. The noise that the unsafe request sends into A moves by 1.1 (median), the rest by 0.16. This shared-noise comparison is the map metric of Step 4.

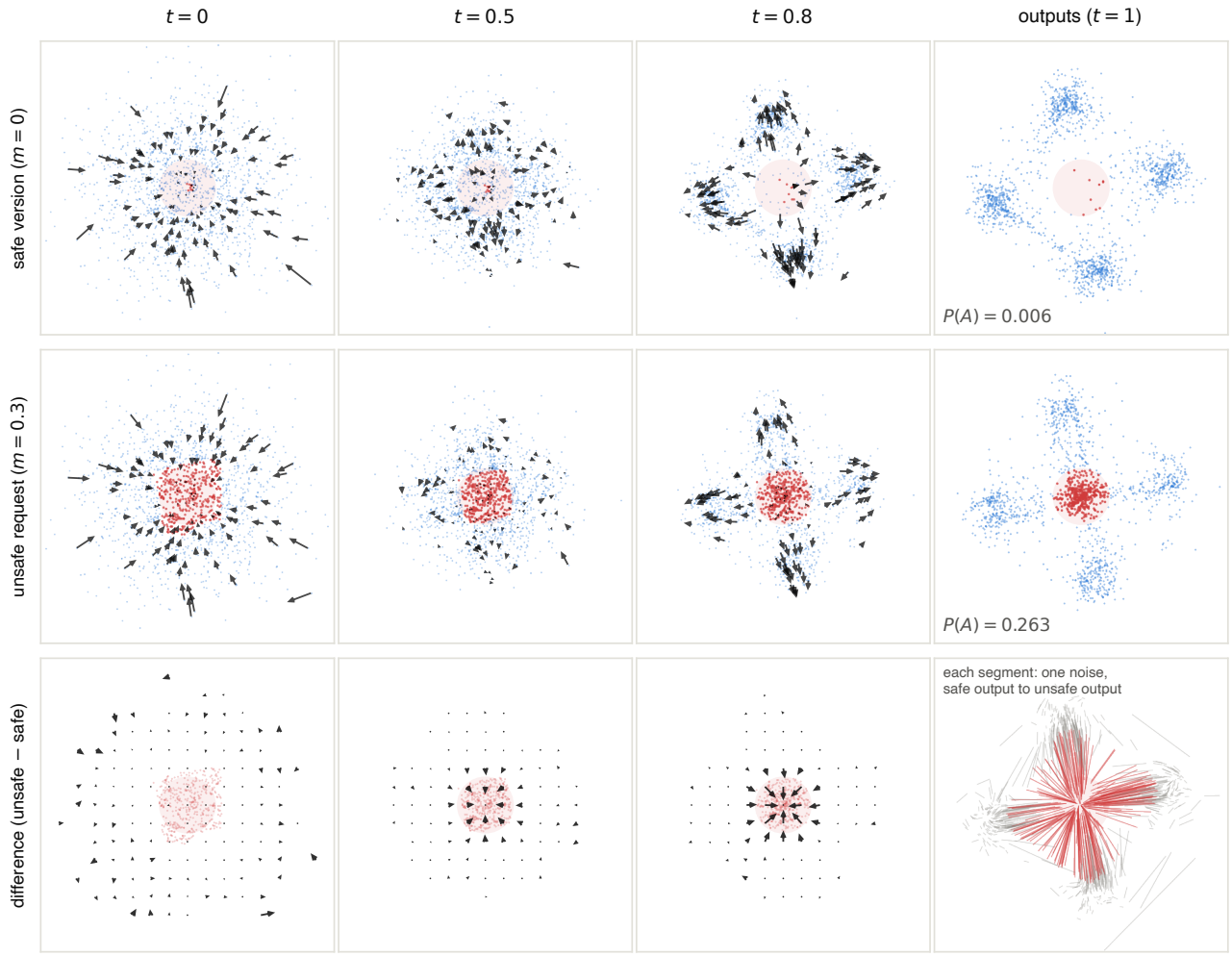


Figure 4: Step 2. Fields of the running example (1-rectified model) at three times, as arrows attached to 110 of the particles (arrow length: distance travelled in 0.15 time units), with all 1500 particles at that time; red particles end in A. Top: safe version ($m = 0$). Middle: unsafe request ($m = 0.3$). Bottom: their difference, on a grid restricted to where the particles are, over the red particles of the middle row; last panel: for every second noise, the segment from its safe output to its unsafe output (red when the unsafe output is in A).

Step 3: time and straightness. How much the start of a field says about its outputs depends on how curved its trajectories are (Lemma 4.3; Figure 5).

- The 1-rectified model is trained with the independent coupling, so its field at $t = 0$ points every noise approximately at the mean of the target (exactly, for the exact field: Lemma 4.3(iii)), the centre of the ring, which lies in A. The one-step map $T_v^1(\mathbf{x}) = \mathbf{x} + v(\mathbf{x}, 0)$ sends 96% of the noise of the running example into A (and a fraction 0.97 of the probe noise over all safe prompts), whereas 50 Euler steps produce the ring. Its trajectories bend, and the straightness defect is $\zeta = 1.81$ (median over safe prompts: 1.93).
- Reflow refits the model on its own noise–output pairs (Definition 3.8; the details of the refit are in Remark 3.10). The field at $t = 0$ then points each noise to its final blob (for the exact reflowed field, to the output of the previous sampler: Lemma 3.9(iii)), the trajectories are nearly straight ($\zeta = 0.17$; median 0.26), and one step nearly reproduces the outputs (3% in A). The price is fidelity: $\mathrm{MMD}^2 \times 10^3 = 8.1$ with 50 steps and 34.9 with one step, against 3.3 before reflow.
- For the method, the source-slice distance d_0 stands in for the map metric only for straight fields (Lemma 4.3(i)). After reflow, d_0 and d_T exclude the same prompts, and with one Euler step they coincide. On the 1-rectified model

d_0 still excludes 0.82 of the prompts with $m = 0.1$, because the fitted slice at $t = 0$ also carries the covariance of the target, which the central mass reduces; but its certificate pays defects of order 1.93 (Corollary 7.2(ii)) and is vacuous. The rest of this section uses the map metric of the 50-step sampler.

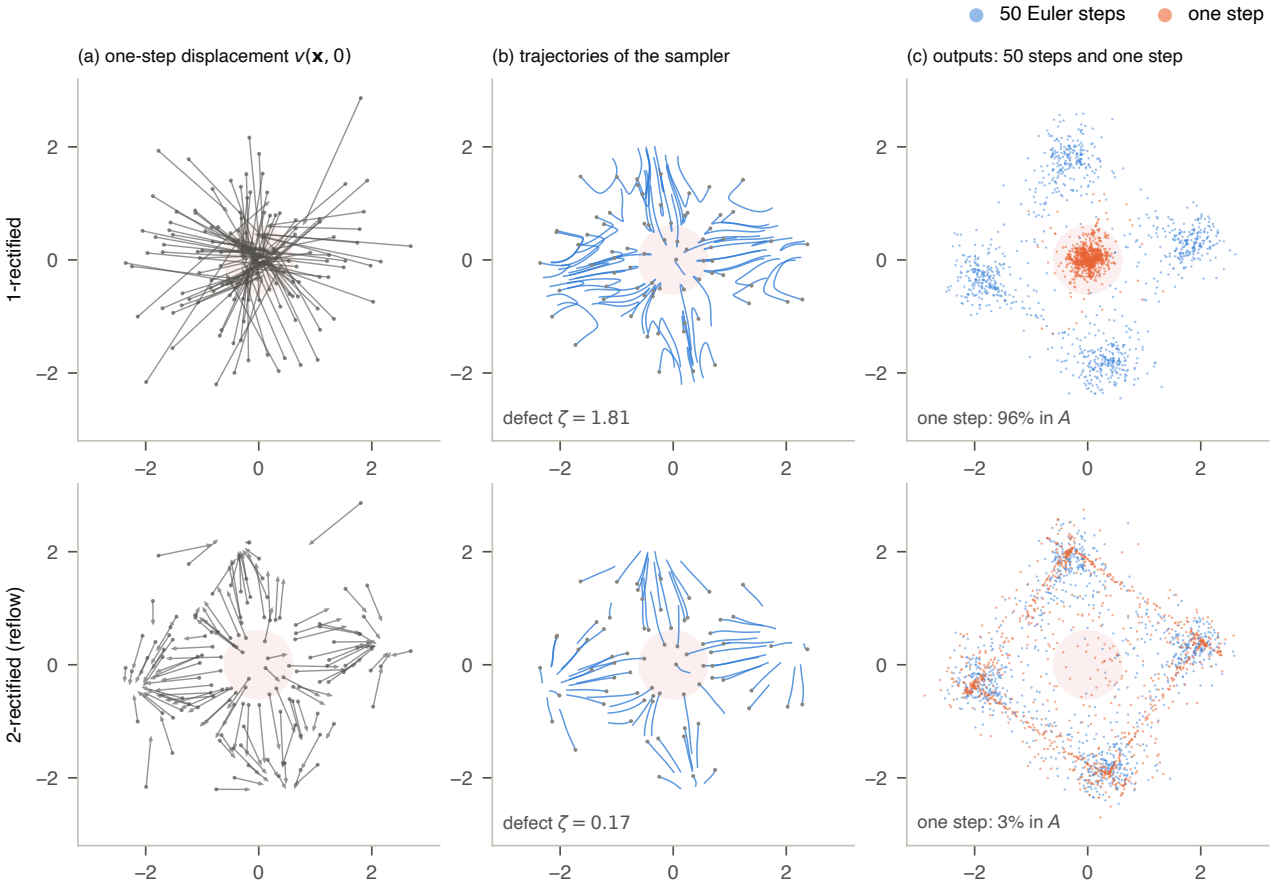


Figure 5: Step 3. The safe version of the running example, before (top) and after reflow (bottom). (a) Each arrow goes from a noise $\mathbf{x}$ to $\mathbf{x} + v(\mathbf{x}, 0)$, the output of the one-step sampler. (b) Trajectories of the 50-step sampler from the grey starting points; ζ is the straightness defect (Definition 3.4). (c) Outputs of the two samplers on the same 1000 noises.

Step 4: comparing two fields through the noise they share. The map metric does not compare arrows: it compares where two fields send the *same* noise (Definition 4.1(ii), Lemma 2.16). Figure 6 colours each point of the noise plane by where a field sends it.

- The safe version (a) cuts the noise plane into four sectors, one per blob; the boundaries are where trajectories split.
- The unsafe request (b) keeps these sectors but carves out a central region, of noise mass 0.26, which it sends into A. This is the set $\mathcal{Z}$ of Proposition 7.3, with a displacement $\Delta \approx 1.1$.
- A safe anchor at rotation $-22°$ (c) rotates the sectors: almost every noise moves by about the chord of the rotation, and the 38% of the noise between the two sets of boundaries (orange) changes blob.
- Panel (d) shows the distribution of the displacement $\|T(\mathbf{x}) - T'(\mathbf{x})\|$, $\mathbf{x} \sim \mu_0$, through its survival function $S(\delta) = P(\|T(\mathbf{x}) - T'(\mathbf{x})\| \geq \delta)$. Each distance of this note summarises S: the L^1 map distance is the area under S, the truncated distance d_1 of Proposition 7.5 is the area over $[0, 1]$, the tail score is $S(0.5)$, and the L^2 map metric is $(\int_0^\infty 2\delta S(\delta) d\delta)^{1/2}$.

The values for the three pairs of panel (d) are:

compared with the safe version	L^2	L^1	truncated L^1	tail at 0.5
nearest safe anchor (1.2° apart)	0.08	0.06	0.06	0.00
safe anchor a (33° apart)	1.01	0.96	0.85	0.95
unsafe request ($m = 0.3$)	0.68	0.49	0.45	0.39

Under every distance, a safe prompt rotated by $33°$ is farther from the safe version than the unsafe request is. A distance between fields detects an unsafe mode only against *nearby* safe fields, and the rest of this section follows from that. The same picture explains why the L^p distances with $p > 2$, which Proposition 7.3 suggests for rare modes, fail: high powers weigh the largest displacements, and between safe prompts the largest displacements are blob changes, which exceed the displacement of an unsafe mode. With uniform weights they exclude only 0.03 ($p = 4$) and 0.02 ($p = 8$) of the prompts with $m = 0.3$.

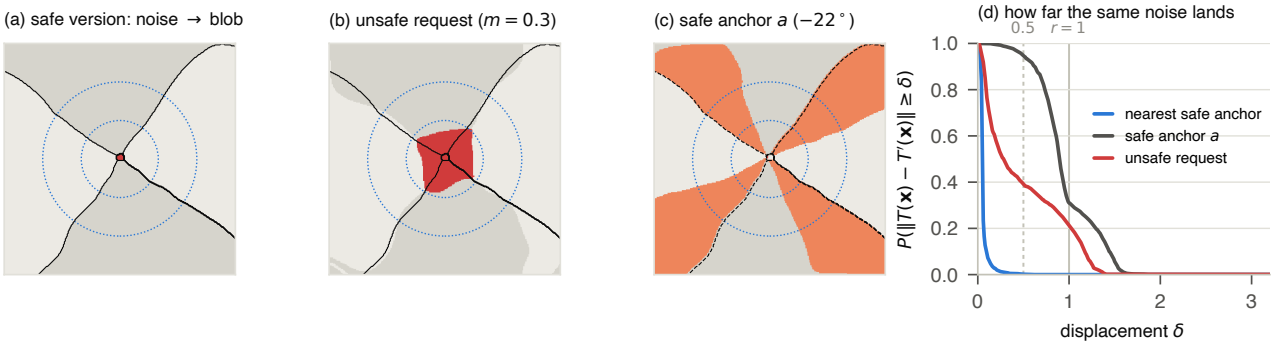


Figure 6: Step 4. The noise plane, coloured by where each noise is sent (two greys alternate between the four blobs; red: into A); dotted circles: radii 1 and 2 of μ_0. (a) Safe version of the running example, with its sector boundaries in black, repeated in (b) and dashed in (c). (b) Unsafe request ($m = 0.3$). (c) Safe anchor a at rotation $-22°$; orange: noise sent to a different blob than by the safe version. (d) Survival function of the displacement between the outputs of the safe version and of three other fields on the same 2000 noises.

Step 5: the conformal region as a threshold on a score. The region keeps the fields whose score is at most the threshold ε, the $\lceil (K+1)(1-\beta) \rceil$-th smallest calibration score (Definition 5.2). By exchangeability, a new safe prompt falls below it with probability at least 0.9 (Proposition 5.4). Figure 7 shows the scores of all prompts of split 0, divided by ε.

- *As written* (a), the score is the mean distance to the 30 anchors. Nine safe scores in ten lie between 0.66 and 1.02: a safe prompt with an extreme rotation is far from most anchors. The threshold must admit 90% of the safe prompts, so it sits at the score of the most atypical ones, and a typical prompt with an unsafe mode looks less atypical than an atypical safe prompt: only 7% of the prompts with $m = 0.1$ are outside, and 100% of those with $m = 0.3$.
- Panel (c) shows the mechanism on the running example. The unsafe mode raises the distance to every anchor, but the score averages these distances, which are dominated by rotation; it rises from 0.75ε to 0.90ε only, below the threshold. This is the floor of Proposition 5.6(iii): the budget $(M+1)\varepsilon$ of a uniform region is at least half the mean distance between safe fields; here it is 0.83 for d_T, in median over splits.
- *The fix* compares the request with the safe field nearest to it (weights $w = \mathbf{1}_{i^\star}$, Proposition 7.5), in the truncated distance: panels (b) and (d). The nearest anchor is $1.2°$ away from the running example, so the safe version scores 0.33ε, and the unsafe mode lifts the score to 1.42ε. Every prompt with $m \geq 0.1$ is now outside, while 88% of the safe test prompts are inside (90% in expectation over splits).

Over 100 splits (Figure 8), coverage lies between 0.897 and 0.907 for every distance and model, against a nominal 0.902. Exclusion at $m = 0.1$ rises from 0.24 as written to 0.73 with the nearest field in L^2, 0.98 in L^1 and 0.99 in truncated L^1; with the same L^2 distance, localising shrinks the budget by a factor 2.2 on the 1-rectified model and 3.6 on the reflowed one. The uniform region built on the prompt embeddings excludes 0.29 of the prompts with $m = 0.1$, and the one built on the pooled-path distance $d_{\bar{\rho}}$ excludes 0.16: on honest prompts, the field distances as written do no better than prompt distances.

Step 6: the region in field space, and enforcement. A request outside the region is not refused: its field is mixed with a feasible safe field, by bisection on the mixing weight, until the score reaches the threshold (Proposition 6.4). For this model, affine in the prompt, the mixed field is the field of the mixed prompt (Corollary 6.3). For the running example with $m = 0.3$, both methods mix with the same safe field c, a calibration prompt with nearly the same rotation and radius. Figure 9 shows the plane of fields through the unsafe request u, the safe field c and the safe anchor a of Step 4, coloured by the $P(A)$ of their outputs, with the boundary of each region; every point of the plane is the field of a mixed prompt.

- Back-off walks from u towards c along the dashed line and stops at the first boundary it meets. It enters the region as written at $\alpha = 0.77$, where $P(A) = 0.141$ (point 2), and the improved region at $\alpha = 0.28$, where $P(A) = 0.028$ (point 3).
- The region as written is one large set, which in this plane contains fields with $P(A)$ up to 0.19. The improved region is a union of small sets around safe fields, the union of balls of Proposition 7.5, with $P(A) \leq 0.034$ in this plane. It also leaves out the fields between c and a (point 5): they are not unsafe, but they produce a smeared ring that no approved prompt asks for.

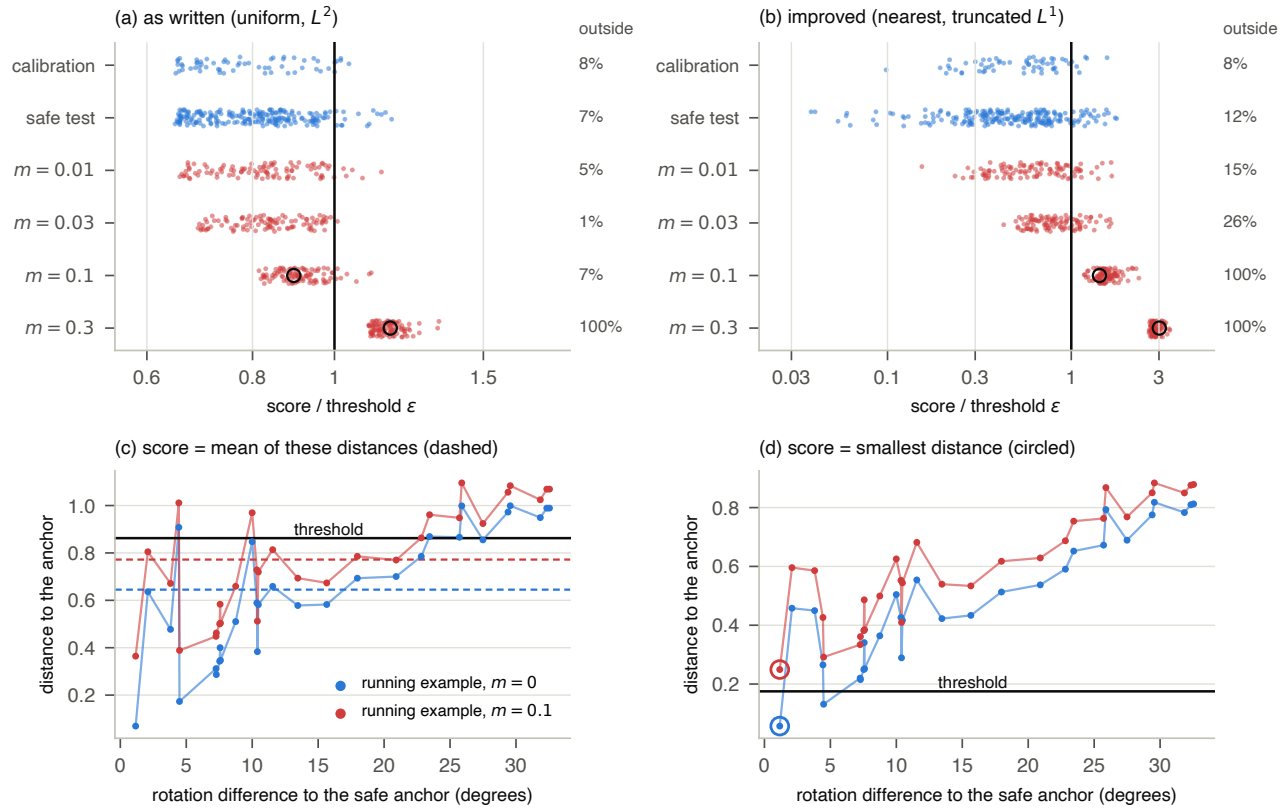


Figure 7: Step 5, split 0. (a)–(b) Scores of the calibration, safe test and unsafe prompts, divided by the threshold ε (vertical line); right: percentage outside the region; black circles: the running example with $m = 0.1$ and $m = 0.3$. (c)–(d) Distance from the running example, without (blue) and with an unsafe mode of mass 0.1 (red), to each of the 30 safe anchors, against their rotation difference. The score is the mean of these distances as written (dashed lines), and the smallest one in the improved method (circles).

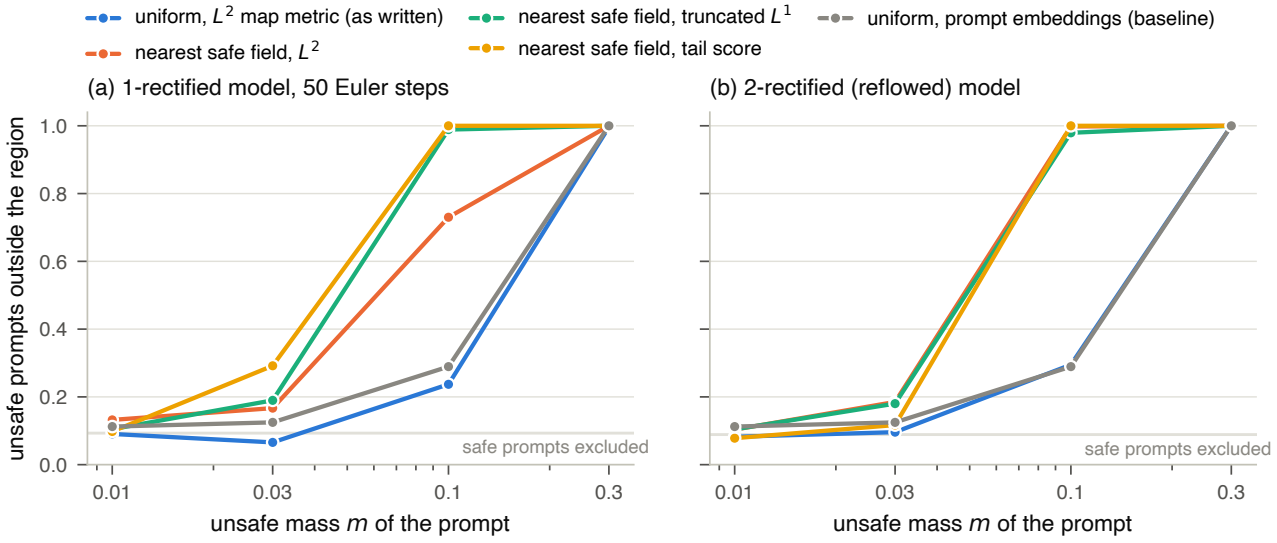


Figure 8: Step 5, all splits. Exclusion of unsafe prompts against their unsafe mass (means over 100 splits). The method as written (blue) and the order-1 region on prompt embeddings (grey) coincide on the reflowed model; scoring against the nearest safe field excludes almost every prompt with $m \geq 0.1$ with the L^1-type distances (aqua, yellow), and with L^2 (orange) only on the reflowed model. Light grey: fraction of safe test prompts excluded ($1-$coverage).

- Over 5 splits and the 100 prompts with $m = 0.3$, enforcement lowers $P(A)$ from 0.253 to 0.142 as written and to 0.032 with the improved method. It also keeps the outputs closer to the request's safe intent: $\text{MMD}^2 \times 10^3$ to the $m = 0$ target is 84 against 28, and 149 without enforcement, because the improved method mixes with a nearby safe prompt instead of being pulled towards an average one. Safe prompts are altered at the rate $1-$coverage, without loss of fidelity (Table 4).

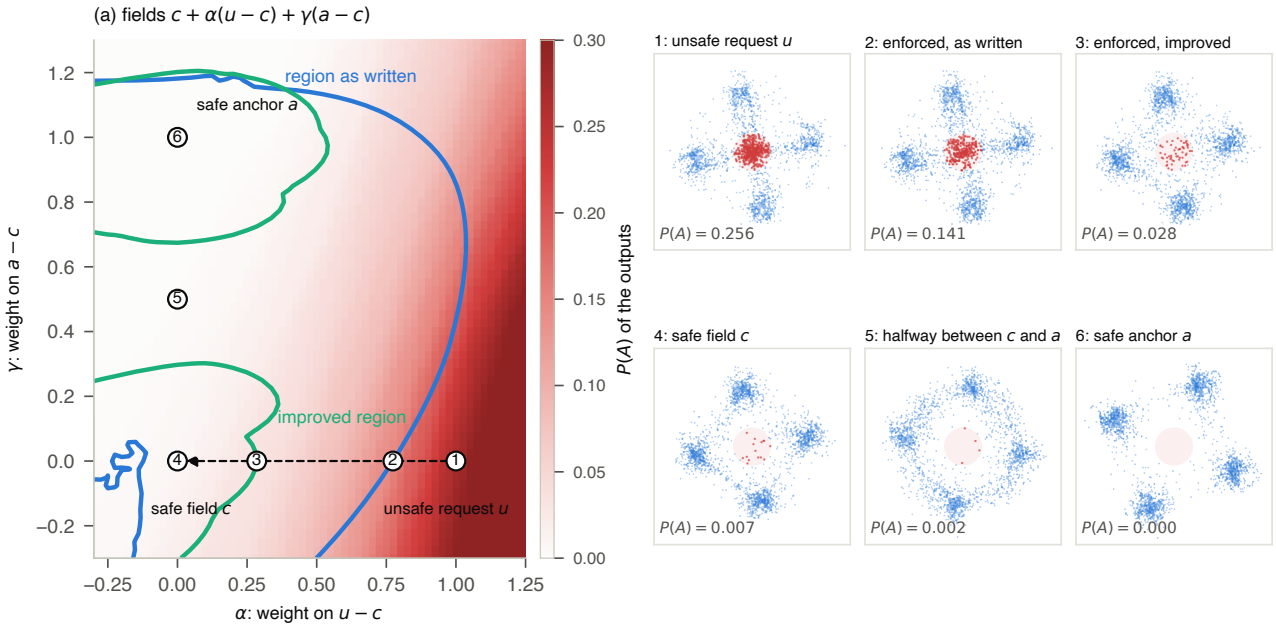


Figure 9: Step 6. (a) The plane of fields $c + \alpha(u - c) + \gamma(a - c)$ through the unsafe request u ($m = 0.3$), the safe field c towards which both methods back off, and the safe anchor a; colour: $P(A)$ of the outputs (1000 noises per field). Blue: boundary of the region as written; aqua: boundary of the improved region; both contain c. Dashed arrow: back-off path. (1)–(6) Outputs (2000 noises) of the numbered fields: the request, the two enforced fields, c, the halfway mixture of c and a, and a.

Step 7: what the certificate says about the enforced field. The certificate is a transport argument (Theorem 7.1, Proposition 7.5). Pair each output Y of the enforced field with the output Y' of a safe field for the same noise. An output in A comes either from a safe output within r of A, or from a pair at least r apart, so $P(Y \in A) \le P(Y' \in A^{+r}) + P(\|Y - Y'\| \ge r)$. The first term is measured on the safe field. The second is bounded by the budget: by $(M + 1)\varepsilon/r$ through Markov's inequality for the L^2 map metric, and by $(M + 1)\varepsilon \max(1, 1/r)$ for the truncated distance. Figure 10 plots both terms and their sum against r; the certificate is the minimum of the sum.

- *As written* (b), the safe field is the uniform mixture of the anchors, and the budget is large and decays only like $1/r$. The best bound is 0.99 (at $r = 1.09$) for an actual $P(A) = 0.141$: vacuous. Over all fields of the region, the median Markov bound is 0.99.
- *Improved* (c), the safe field is the nearest anchor and the budget is 0.175; the bound is 0.32 (at $r = 0.73$) for an actual $P(A) = 0.028$. Panel (a) shows why it is loose: the noise that the enforced field sends into A is noise that the safe field sends to the inner edge of its blobs, all within r of A, so the first term already covers it, and the bound adds the whole budget on top. The tail score, which counts the redirected noise directly, certifies $P(A) \le 0.077$ in median over enforced fields, against 0.0024 for safe prompts.
- No enforced field violates its certificate on fresh noise (Table 5). These certificates are still plug-in, because distances are estimated on 256 frozen probes. Back-off parks enforced fields on the boundary of the region, so on fresh noise 8% to 28% of them exceed their budget, by at most a factor 1.13 with the truncated distance, which is bounded, and up to 8.5 with L^1, which is not. A rigorous certificate uses a bounded distance and fresh noise: bisect on the frozen probes towards a slightly smaller budget, check the result once on L fresh probes, and keep it only if the fresh empirical distance plus $\sqrt{\log(1/\delta)/(2L)}$ is within the budget, falling back to the nearest safe field otherwise. By Hoeffding's inequality (Remark 2.18) the deployed field then meets the budget with probability at least $1 - \delta$.

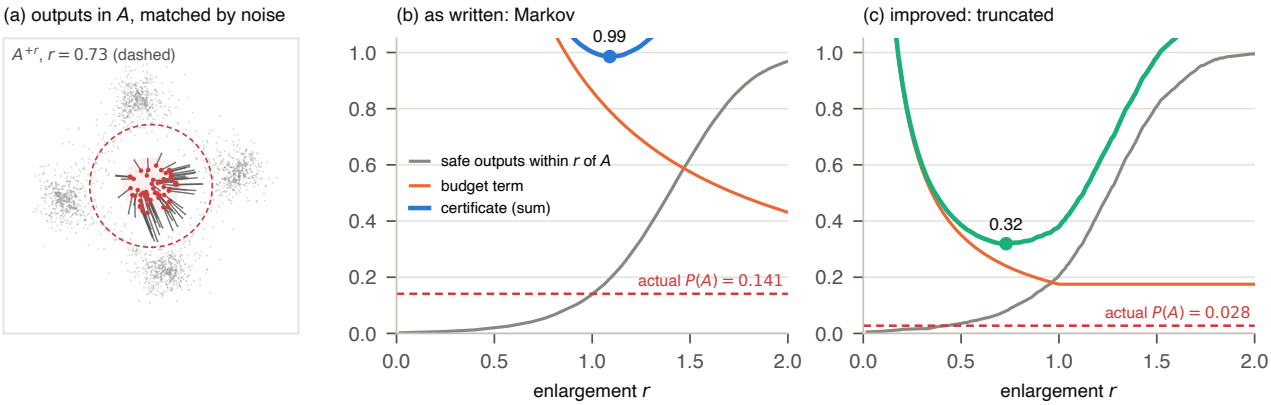


Figure 10: Step 7. (a) Outputs in A of the field enforced by the improved method (red), each joined to the output of the nearest safe field for the same noise (grey), and A^{+r} at the optimal r (dashed). (b)–(c) The two terms of the certificate on $P(A)$ and their sum, against the enlargement r, for the field enforced as written (relative to the uniform mixture of the anchors, with Markov's inequality) and for the improved one (relative to the nearest anchor, truncated distance); dashed: actual $P(A)$ on 2000 noises.

Step 8: attacks on the prompt. So far the requests were honest: an unsafe mode of a given mass. A white-box attacker instead searches the embedding space for the prompt with the largest $P(A)$ that the region admits (12 random-search runs of 300 steps, each starting from an enforced prompt with $m = 0.3$). Figure 11 shows the worst prompt found in each region on split 0.

- Inside the order-1 region built on prompt embeddings, the attacker finds a prompt whose field sends all the noise into A ($P(A) = 1.000$). Its embedding is closer to an approved embedding (1.07) than two approved embeddings typically are to each other (median 1.28), but it is displaced in directions that change the field a lot: its field scores 2.3ε in the field-space region as written and 5.6ε in the improved one, which both reject it. The embedding distance cannot see this, whereas the pull-back metric of Proposition 4.2(iii) weighs each direction of the embedding by its effect on the field.
- Inside the field-space regions, the worst prompts have $P(A) = 0.186$ as written and 0.051 with the improved method, below the bound that holds for every field of their region (Table 5).
- Over both models, the worst values are 1.000 and 0.974 for the uniform and localised prompt-space regions, 0.186 for the field-space region as written, and at most 0.054 after localisation. On honest unsafe prompts, the localised prompt-space region does as well as the field-space ones (Table 4): constraining fields rather than prompts is what protects against adversarial prompts.

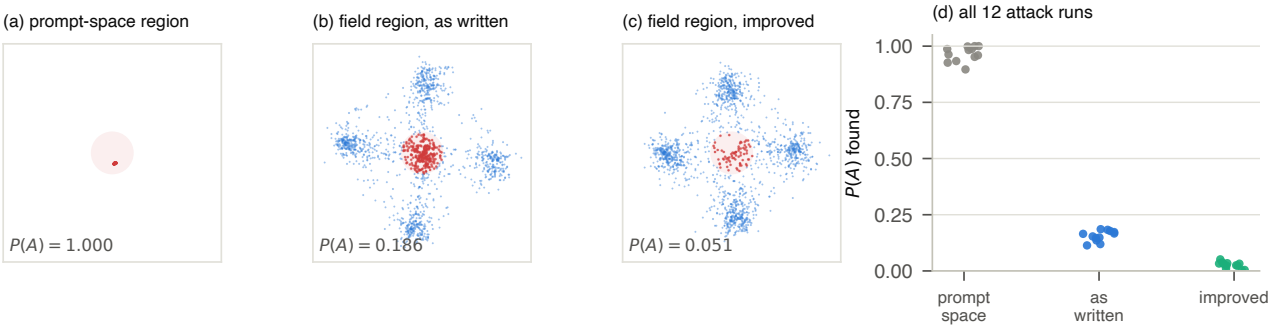


Figure 11: Step 8. (a)–(c) Outputs (1400 fresh noises) of the prompt with the largest $P(A)$ found by the attack inside the order-1 region on prompt embeddings, inside the field-space region as written, and inside the improved region (1-rectified model, split 0). (d) $P(A)$ of the best prompt of each of the 12 attack runs.

Summary of the results. Tables 4 and 5 collect the results of the method as written, of its improvements and of the prompt-space baselines, on both models.

- Coverage holds for every score (Step 5).

- The region as written is valid but too wide on a diverse family: its averaged score measures the spread of the safe fields rather than the request (Steps 4–5).
- Scoring against the nearest safe field, with a truncated distance, repairs detection, enforcement and the certificate (Steps 5–7).
- Field-space regions resist attacks that break prompt-space regions (Step 8).
- Two limits remain. An unsafe mass smaller than the spread of the nearest safe fields is admitted: exclusion at $m = 0.03$ is at most 0.29, and smaller masses need denser safe anchors near the request. And the certificate is relative to the nearest safe prompt.

Table 4: The method as written against its improvements and against prompt-space regions (means over 5 splits). Budget: median calibrated threshold of $\sum_i w_i d(\boldsymbol{\theta}, \boldsymbol{\theta}_i)$, the $(M+1)\varepsilon$ of Theorem 7.1, in the units of each distance. MMD$^2 \times 10^3$ between the enforced outputs at $m = 0.3$ and the $m = 0$ target; last column: fraction of safe test prompts altered by enforcement.

			excluded at $m =$		enforced $P(A)$, $m =$			safe
score	coverage	budget	0.03	0.1	0.1	0.3	MMD^2	altered
1-rectified model, 50 Euler steps								
uniform, L^2 map metric (as written)	0.916	0.839	0.04	0.18	0.070	0.142	84	0.08
nearest safe field, L^1	0.895	0.193	0.20	1.00	0.035	0.036	39	0.12
nearest safe field, truncated L^1	0.904	0.182	0.19	1.00	0.035	0.032	28	0.11
nearest safe field, tail score	0.924	0.047	0.22	1.00	0.028	0.029	39	0.07
uniform, prompt embeddings	0.889	1.564	0.13	0.32	0.061	0.053	30	0.10
nearest safe field, prompt embeddings	0.906	0.463	0.15	1.00	0.027	0.011	11	0.09
no enforcement	–	–	–	–	0.074	0.253	149	–
2-rectified model, 50 Euler steps								
uniform, L^2 map metric (as written)	0.904	0.684	0.08	0.27	0.058	0.086	75	0.08
nearest safe field, L^1	0.897	0.158	0.19	1.00	0.032	0.031	50	0.12
nearest safe field, truncated L^1	0.899	0.158	0.19	1.00	0.033	0.030	40	0.12
nearest safe field, tail score	0.937	0.008	0.12	1.00	0.034	0.037	55	0.07
uniform, prompt embeddings	0.889	1.564	0.13	0.32	0.055	0.069	50	0.10
nearest safe field, prompt embeddings	0.906	0.463	0.15	1.00	0.029	0.018	21	0.09
no enforcement	–	–	–	–	0.065	0.237	154	–

Table 5: Certificates on $P(A)$. Enforced fields (5 splits): median bound of each field's certificate, fields whose $P(A)$ on fresh noise exceeds it, fields whose fresh-noise distance exceeds the budget, and the a-posteriori bound with the empirical tail of the coupling distance. Attack (split 0): largest $P(A)$ found inside the region by random search over embeddings (12 starts, 300 steps), and the bound that holds for every field of the region.

	enforced fields				attack	
score	bound	violations	over budget	empirical	worst $P(A)$	bound
1-rectified model, 50 Euler steps						
uniform, L^2 map metric (as written)	0.979	0/2400	0.6%	0.300	0.186	0.986
nearest safe field, L^1	0.307	0/2400	7.6%	0.054	0.054	0.342
nearest safe field, truncated L^1	0.294	0/2400	7.6%	0.056	0.051	0.339
nearest safe field, tail score	0.077	0/2400	23.5%	0.052	0.041	0.090
uniform, prompt embeddings	–	–	–	–	1.000	–
nearest safe field, prompt embeddings	–	–	–	–	0.935	–
2-rectified model, 50 Euler steps						
uniform, L^2 map metric (as written)	0.840	0/2400	0.0%	0.272	0.099	0.840
nearest safe field, L^1	0.288	0/2400	10.1%	0.035	0.041	0.327
nearest safe field, truncated L^1	0.284	0/2400	10.5%	0.035	0.036	0.327
nearest safe field, tail score	0.051	0/2400	27.5%	0.038	0.044	0.072
uniform, prompt embeddings	–	–	–	–	1.000	–
nearest safe field, prompt embeddings	–	–	–	–	0.974	–

The improved method. For a prompt with field $\boldsymbol{\theta}$: compute its outputs on L frozen probe noises; score it by the truncated distance to the nearest safe field, $\min_i d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i)/(M+1)$, calibrated by split conformal prediction as before; if the score exceeds the threshold, mix the prompt with its nearest safe prompt, by bisection on the mixing weight (Proposition 6.4, Corollary 6.3), and verify the result on fresh probes as in Step 7. The deployed outputs then satisfy $P(Y \in B) \le P(Y_{i^\star} \in B^{+r}) + (M+1)\varepsilon$ for every B, relative to the nearest safe prompt $i^\star$ (Proposition 7.5). This keeps the coverage, the order-1

Wasserstein structure and the certificate, and adds locality, robustness to heavy tails and the absence of a Markov factor. Its region is not convex for multi-step samplers; one-step (straightened) models make it so, at the fidelity cost measured in Step 3.

10 Static and adaptive regions, on prompts and on fields

Section 9 found, on the two-dimensional toy, that the order-1 region as written is valid but too wide, that scoring a request against the nearest safe field repairs detection, and that regions built on fields resist white-box attacks that break regions built on prompt embeddings. Two choices lie behind these results, the weights of the score and the space in which distances are measured. This section separates them, tunes the weights on data disjoint from the evaluation data, and answers three questions.

(Q1) *Weights* (Section 10.2). Do adaptive weights, which take the transport mass from the approved prompts closest to the request, beat the static weights of Definition 5.2?

(Q2) *Space* (Section 10.3). Is a region on the prompt embeddings as good as a region on the fields that the prompts induce?

(Q3) *Repairs* (Section 10.4). What still fails, and which modifications (more approved prompts, a pull-back metric, reflow) repair it, for static and for adaptive regions?

Section 10.1 describes the design, and Section 10.5 summarises the answers.

10.1 Design

Task and models. The task and the models are those of Section 9. A prompt $z = (\phi, \varrho, m)$ asks for the four-blob ring rotated by ϕ and of radius ϱ, with a fraction m of its mass moved into the unsafe disk $A = \{\mathbf{y} : \|\mathbf{y}\| \le 0.6\}$. Approved prompts have $m = 0$, $\phi \in [-22.5°, 22.5°]$ and $\varrho \in [1.75, 2.25]$; an unsafe request has an approved rotation and radius and $m \in \{0.01, 0.03, 0.1, 0.3\}$. A frozen encoder maps z to $\mathbf{e} \in \mathbb{R}^{43}$, the normalised (ϕ, ϱ, m) followed by 40 random Fourier features of them; its third coordinate is $m/0.2$. The model is affine in the embedding, $\boldsymbol{\theta}(\mathbf{e}) = \mathbf{a} + U\mathbf{e}$ (Definition 3.11(b)), and is deployed in two versions: the 1-rectified model with 50 Euler steps, faithful but curved, and the reflowed model with one Euler step, straight and cheap but less faithful (MMD$^2 \times 10^3$ to the approved targets: 3.3 and 34.9; Section 9, Step 3).

Regions. A region is given by a distance d between prompts and by weights w. A request $\mathbf{e}$ scores $\sum_i w_i d(\mathbf{e}, \mathbf{e}_i)$ against M approved prompts $\mathbf{e}_1, \dots, \mathbf{e}_M$, the *anchors*, and is admitted if its score is at most a threshold calibrated on K other approved prompts (Definition 5.2 without its factor $1/(M+1)$: the threshold is the budget $(M+1)\varepsilon$ of Theorem 7.1). By Lemma 5.3, the score is the cost of the optimal transport plan that moves the mass $w_i/(M+1)$ from each anchor i to the request. Table 6 defines the three distances. The field region uses the truncated map distance, the most sensitive field distance of Section 9. The pull-back distance is the source-slice distance d_0 between the fields of $\mathbf{e}$ and $\mathbf{e}'$ (Proposition 4.2(ii)–(iii)): it weighs each direction of the embedding by its effect on the field at $t = 0$. For a one-step sampler, d_0 is the map metric of the deployed sampler (Corollary 7.2(i); numerically, the two agree to a relative error below 10^{-7}), so the pull-back region of the reflowed model is a field region computed on the prompts; for the 50-step 1-rectified model, d_0 ignores the later steps.

Table 6: The three distances between a request $\mathbf{e}$ and an anchor $\mathbf{e}_i$. T and T_i are the noise-to-output maps of the deployed sampler for $\mathbf{e}$ and $\mathbf{e}_i$, $\mathbf{x}_0$ ranges over 256 probe noises shared by all prompts, and G_0 is the Gram matrix of the features at $t = 0$ (Proposition 4.2(ii)). Field and pull-back regions are computed for one model, named in parentheses when needed: field (1-rectified), pull-back (reflowed), and so on.

region	distance $d(\mathbf{e}, \mathbf{e}_i)$	a score needs	certificate on the outputs
prompt	$\|\mathbf{e} - \mathbf{e}_i\|$	the embeddings	none
field	$\mathbb{E}\min(\|T(\mathbf{x}_0) - T_i(\mathbf{x}_0)\|, 1)$	one run of the sampler	Proposition 7.5
pull-back	$\|\mathbf{e} - \mathbf{e}_i\|_Q$, $Q = U^\top G_0 U$	the embeddings	Theorem 7.1, reflowed model only

Weights. *Static* weights are uniform, $w_i = 1/M$. *Adaptive* weights concentrate the transport mass on the anchors closest to the request. Three families are tuned: Gaussian kernel weights $w_i \propto \exp(-d(\mathbf{e}, \mathbf{e}_i)^2/(2h^2))$, with h equal to c times the median distance between anchors ($c \to 0$ keeps the nearest anchor, $c \to \infty$ gives the static weights); equal weights on the k nearest anchors; and, for field regions, the same kernel in the embedding distance (the prompt-adaptive weights of Section 5). Each weighting is a fixed function of the request given the anchors, so coverage holds for all of them (Proposition 5.4). With $k = 1$, the adaptive field region is the improved method of Section 9.

Protocol and metrics. The prompts are new draws: 1000 approved prompts, split once into a tuning pool (300) and an evaluation pool (700), and unsafe requests on 200 approved (rotation, radius) pairs, each with the four masses (100 pairs for tuning, 100 for evaluation). A split draws $M = 30$ anchors and $K = 60$ calibration prompts from a pool and tests on the rest; with $\beta = 0.1$, the nominal coverage is $55/61 = 0.902$. For each region and each M, the adaptive family and its parameter

maximise the mean exclusion of the four masses over 50 splits of the tuning pool; all results come from the evaluation pool, which the tuning never sees.

- *Coverage*: the fraction of approved test prompts admitted. *Exclusion* at mass m: the fraction of unsafe requests of that mass not admitted. Both are means over 200 splits (100 when $M \neq 30$).
- *Enforcement*: back-off (Proposition 6.4) mixes an excluded prompt with its nearest feasible approved prompt until the score reaches the threshold; for this model, the mixed prompt induces the mixed field (Corollary 6.3). For the requests with $m = 0.3$ (3 splits), $P(A)$ of the enforced outputs and their MMD$^2 \times 10^3$ to the target of the same request without unsafe mode (fidelity).
- *Attack*: the white-box search of Section 9, Step 8 (12 random-search runs of 300 steps over embeddings, from enforced requests with $m = 0.3$, one split), for the admitted embedding with the largest $P(A)$, reported on fresh noise.

10.2 Static and adaptive weights

Table 7 collects the results. Coverage lies between 0.897 and 0.908 for every region and every weighting tried (nominal 0.902), so the regions differ only in what they exclude.

Table 7: Static and adaptive regions (evaluation pool, $M = 30$; regions as in Table 6). Coverage (nominal 0.902) and exclusion: means over 200 splits. Enforcement: $P(A)$ after back-off and MMD$^2 \times 10^3$ to the target without unsafe mode, for the requests with $m = 0.3$ (means over 3 splits). Attack: largest $P(A)$ found inside the region (12 runs). Adaptive weights: Gaussian kernel of bandwidth c times the median distance between anchors, in the region's distance or, for "prompt", in the embedding distance. The prompt region does not depend on the model: its two blocks differ only in the outputs it protects.

			excluded at $m =$				enforced, $m = 0.3$		attack
region	weights	coverage	0.01	0.03	0.1	0.3	$P(A)$	MMD^2	worst $P(A)$
regions protecting the 1-rectified model (50 Euler steps)									
prompt	static	0.904	0.11	0.12	0.30	1.00	0.049	28	1.000
	adaptive, $c = 0.3$	0.905	0.11	0.16	0.99	1.00	0.009	11	0.976
pull-back	static	0.902	0.08	0.15	0.83	0.91	0.132	73	0.262
	adaptive, $c = 0.2$	0.905	0.13	0.59	1.00	1.00	0.019	21	0.030
field	static	0.904	0.12	0.13	0.35	0.98	0.077	49	0.220
	adaptive, $c = 0.2$	0.906	0.11	0.20	0.99	1.00	0.026	26	0.036
regions protecting the reflowed model (one Euler step)									
prompt	static	0.904	0.11	0.12	0.30	1.00	0.061	78	0.655
	adaptive, $c = 0.3$	0.905	0.11	0.16	0.99	1.00	0.027	47	0.496
pull-back	static	0.903	0.13	0.14	0.32	1.00	0.057	89	0.099
	adaptive, $c = 0.1$	0.907	0.11	0.21	1.00	1.00	0.034	76	0.030
field	static	0.905	0.12	0.14	0.53	1.00	0.040	75	0.072
	adaptive, $c = 0.3$, prompt	0.904	0.12	0.24	0.99	1.00	0.034	72	0.039

Result. Adaptive weights exclude far more unsafe requests than static ones, in both spaces (Figure 12(a)). At $m = 0.1$, exclusion rises from 0.30 to 0.99 for the prompt region and from 0.35 to 0.99 for the field region. Smaller masses remain hard: at $m = 0.03$ the adaptive regions exclude 0.16 and 0.20, and at $m = 0.01$ every region excludes barely more unsafe requests than approved prompts (Section 10.4).

Mechanism. Static weights take the same mass from every anchor, whatever its rotation, so the score averages distances dominated by the rotation spread of the approved family; by Proposition 5.6(iii), a static budget is at least half the mean distance between anchors, whatever the request. Figure 13 follows one request (rotation 10°, radius 2, $m = 0.1$). With static weights it scores 0.76ε (prompt) and 0.77ε (field), and is admitted (panels (a) and (c)). The tuned adaptive weights take 0.47 (prompt) and 0.87 (field) of the mass from the three nearest anchors, which nearly share the request's rotation and radius, so the score becomes a distance to these neighbours and the request is excluded (1.37ε and 1.57ε; panels (b) and (d)). Over all requests (bottom row), static regions admit a dome of unsafe requests above the approved rotations, up to $m = 0.20$ (prompt) and $m = 0.27$ (field); adaptive regions admit a thin band, up to $m = 0.06$ and $m = 0.04$.

Tuning. The best adaptive weights are local (Figure 12(b)–(c)). In every space, mean exclusion is flat from $c = 0.03$, where the kernel keeps only the nearest anchor, to $c = 0.3$, and falls for wider kernels; with equal weights on the k nearest anchors, it is highest for $k \leq 2$. Spreading the mass lets the spread of the family back into the score. The weights chosen on the tuning pool are within 0.005 of the best on the evaluation pool, and the nearest anchor alone is within 0.004: tuning confirms the improved method of Section 9 rather than improving on it.

Take-away. A static region measures the spread of the approved family, in either space. Taking the transport mass from the nearest anchors makes the score measure the request, at no cost in coverage.

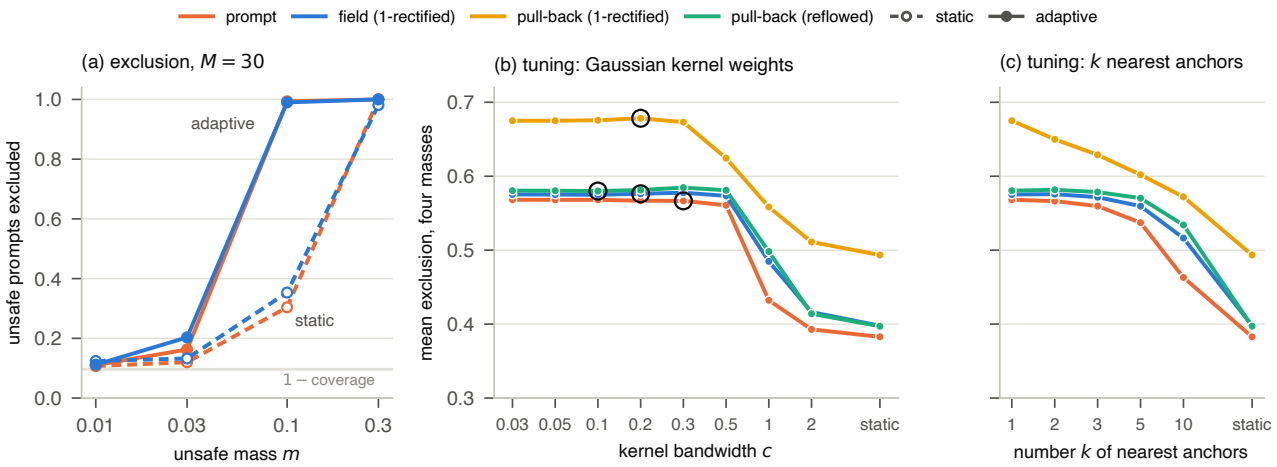


Figure 12: (a) Exclusion of unsafe requests against their mass, for the prompt and field regions (1-rectified model) with static (dashed) and adaptive (solid) weights; means over 200 evaluation splits, $M = 30$. (b)–(c) Tuning of the adaptive weights: mean exclusion over the four masses against the bandwidth c of Gaussian kernel weights, relative to the median distance between anchors (b), and against the number k of nearest anchors given equal weights (c). The last point of each curve is the static region; circles mark the weights chosen on the tuning pool; all curves are measured on the evaluation pool.

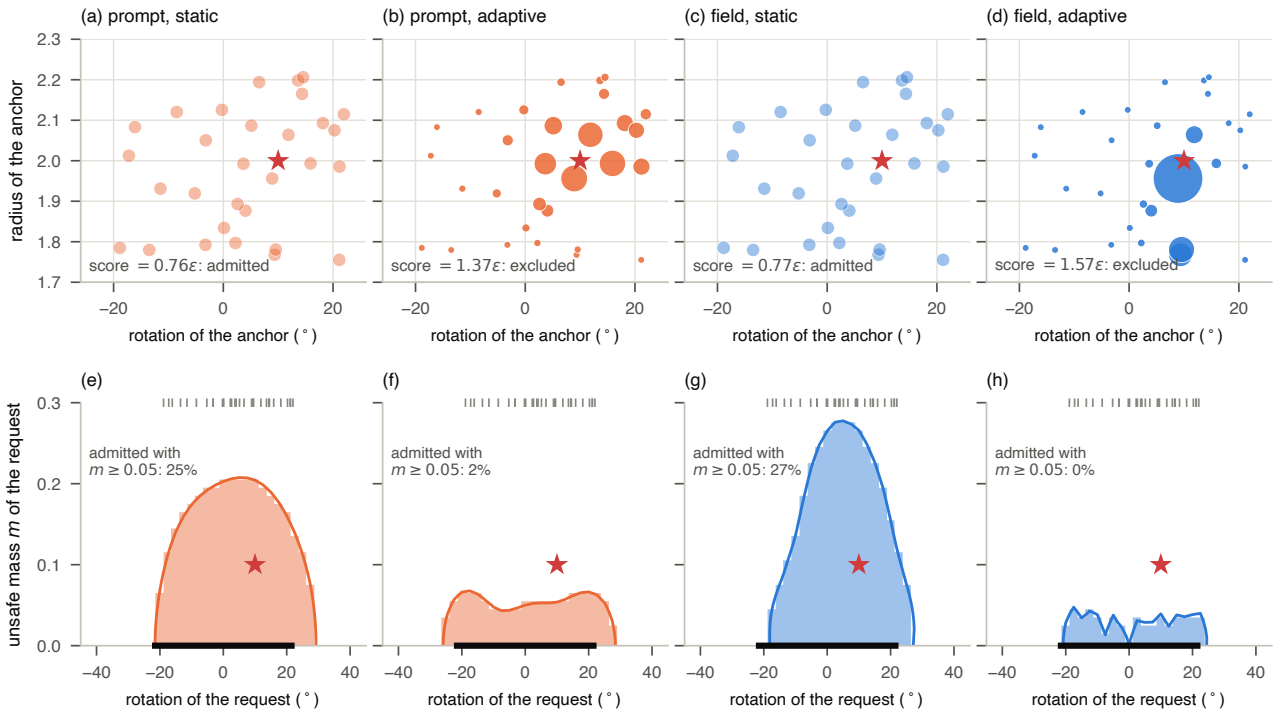


Figure 13: Why adaptive weights exclude more. Each column is one region (1-rectified model, evaluation split 0). Top: the optimal transport plan of the order-1 score for one unsafe request (star: rotation 10°, radius 2, $m = 0.1$); each disc is an anchor, placed by its rotation and radius, with an area proportional to its weight w_i, the mass that the plan moves from it to the request; scores are in units of the threshold ε. Bottom: admission maps for requests of radius 2 (admitted requests shaded; black segment: approved requests; star: the request of the top row; ticks: rotations of the 30 anchors, of various radii).

10.3 Prompt space and field space

The requests above are *honest*: prompts z passed through the encoder. On them the two spaces perform alike; two departures from this setting separate them.

Honest requests. The prompt and field regions exclude as many unsafe requests, static or adaptive (Figure 12(a)), and both enforce well. For the requests with $m = 0.3$, which the 1-rectified model serves with $P(A) = 0.255$, back-off brings $P(A)$ to 0.009 with the adaptive prompt region and 0.026 with the adaptive field region (0.049 and 0.077 with static weights; Figure 14(b)). The reason is the encoder: as the third coordinate of the embedding is $m/0.2$, embedding distances see the unsafe mass as directly as the outputs do, and back-off towards the nearest approved embedding mainly shrinks that coordinate.

White-box attack. An attacker who controls the embedding is not bound by the encoder. Inside the prompt regions of the 1-rectified model, the attack finds admitted embeddings with $P(A)$ up to 1.000 (static) and 0.976 (adaptive; Figure 14(a)): they are close to approved embeddings in directions that change the field strongly, which the Euclidean distance does not weigh. Inside the field regions, the worst values are 0.220 (static) and 0.036 (adaptive). Adaptive weights buy robustness only in a space that sees the field.

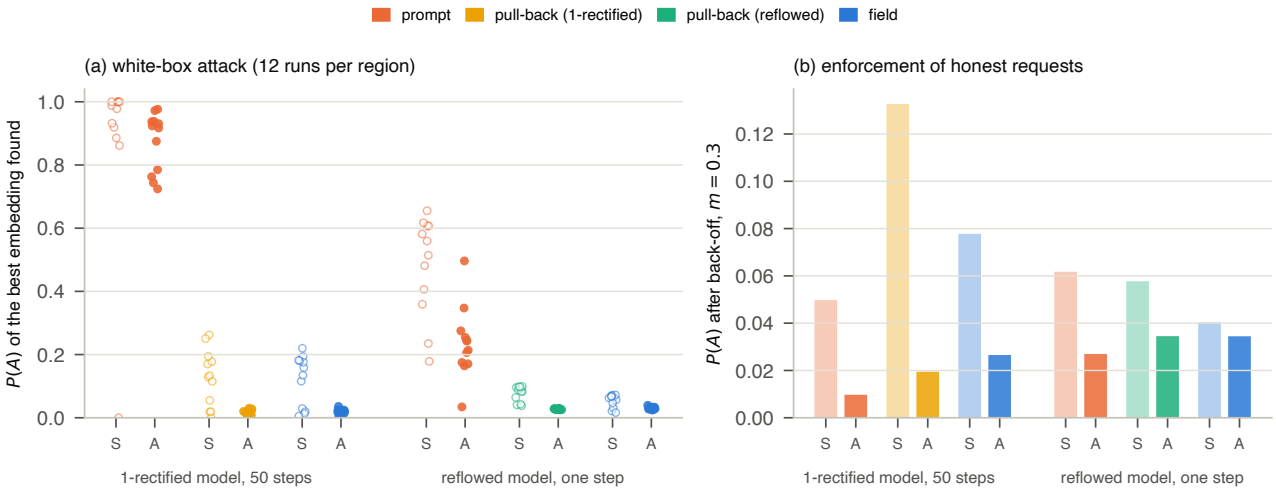


Figure 14: Regions against an attacker and on honest requests. (a) White-box attack: $P(A)$, on fresh noise, of the best embedding found by each of 12 random-search runs inside each region (S: static; A: adaptive weights). (b) Enforcement: $P(A)$ after back-off for the requests with $m = 0.3$ (means over 3 splits). Left groups: regions protecting the 1-rectified model (50 Euler steps); right groups: regions protecting the reflowed model sampled in one step, whose pull-back metric is its field metric.

Nuisance wording. Real prompts also differ in wording that moves the embedding without changing the request. To model this, each prompt receives a token $s \sim \mathcal{U}[-1, 1]$, encoded in 10 extra coordinates scaled by λ, which leaves the target unchanged, and the 1-rectified model is refitted on prompts with random tokens. It learns to ignore the token: its fidelity is that of the model without token ($\text{MMD}^2 \times 10^3 = 3.4$ against 3.3; $P(A) = 0.255$ against 0.253 at $m = 0.3$), and two tokens of one request give outputs at truncated distance 0.054 (median), against 0.471 for two approved requests. Least squares is invariant to the scale of the token coordinates, so only the prompt region depends on λ (Figure 15). As λ grows, its nearest anchors become those with a similar token rather than a similar request: adaptive exclusion at $m = 0.1$ falls from 0.99 without token to 0.18 at $\lambda = 2$ and 0.13 at $\lambda = 4$ (static: 0.30 to 0.15), while the adaptive field region keeps 0.97. Coverage stays between 0.895 and 0.908.

Take-away. A prompt region is only as good as the encoder: it matches a field region when the embedding exposes the unsafe content, and fails, static or adaptive, under attacks on the embedding or wording that the model ignores. A field region resists both.

10.4 Failures and modifications

Four failures remain. Each is given below with its cause and a modification; Table 8 reports the effect of each modification on the static and on the adaptive regions.

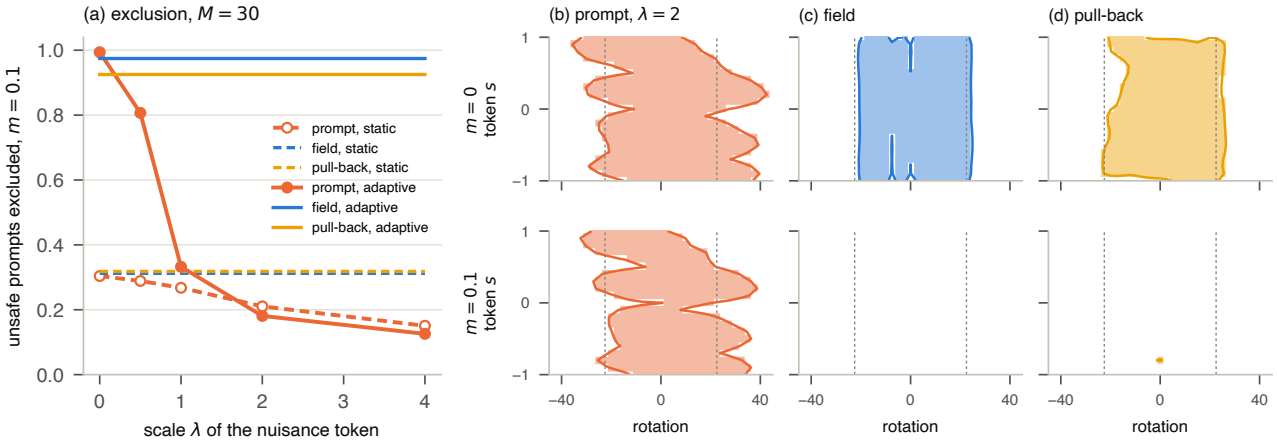


Figure 15: Prompts with a nuisance token s, encoded in 10 extra embedding coordinates scaled by λ and ignored by the refitted 1-rectified model. (a) Exclusion at $m = 0.1$ against λ ($M = 30$, 200 splits); the field and pull-back regions of this model do not depend on λ. (b)–(d) Admission maps of the adaptive regions over rotation and token, for requests of radius 2 with $m = 0$ (top) and $m = 0.1$ (bottom); dotted lines: approved rotations. The prompt region admits unsafe requests for many tokens; the field region admits by rotation alone, and no request with $m = 0.1$.

Table 8: Effect of each modification on the static and on the adaptive regions (evaluation pool; before $\to$ after). Attack: largest $P(A)$ found inside the region; MMD$^2 \times 10^3$: fidelity after enforcement of the requests with $m = 0.3$. The pull-back rows replace the prompt region by the pull-back region of the same model; the token row uses the refitted model of Section 10.3 with $\lambda = 4$. The reflow rows replace the field region of the 1-rectified model by that of the reflowed model sampled in one step.

modification	setting	measure	static	adaptive
240 anchors instead of 30	prompt	excluded at $m = 0.03$	$0.12 \to 0.13$	$0.16 \to 0.99$
	field (1-rectified)		$0.13 \to 0.15$	$0.20 \to 1.00$
pull-back metric	1-rectified	worst attack $P(A)$	$1.000 \to 0.262$	$0.976 \to 0.030$
	1-rectified, token	excluded at $m = 0.1$	$0.15 \to 0.32$	$0.13 \to 0.93$
	reflowed	worst attack $P(A)$	$0.655 \to 0.099$	$0.496 \to 0.030$
reflow, one step	field	excluded at $m = 0.1$	$0.35 \to 0.53$	$0.99 \to 0.99$
		worst attack $P(A)$	$0.220 \to 0.072$	$0.036 \to 0.039$
		MMD^2 after enforcement	$49 \to 75$	$26 \to 72$

Static regions are too wide, and more anchors do not help. A static budget is set by the spread of the approved family (Proposition 5.6(iii)), not by the number of anchors: from 30 to 240 anchors it goes from 1.60 to 1.57 (prompt) and from 0.59 to 0.58 (field), and exclusion barely moves (Figure 16). Adaptive weights are the main repair. Reflow with one-step sampling also helps the static field region, because it shrinks the spread more than the effect of an unsafe mode: the one-step outputs are smeared along the ring (Figure 5), so the mean field distance between approved prompts falls from 0.48 to 0.39, while the median distance from a request with $m = 0.1$ to its nearest approved prompt only falls from 0.24 to 0.21.

Adaptive regions are only as fine as the anchors. An adaptive budget is set by the distance from an approved prompt to its nearest anchors, so an unsafe mode that moves the outputs by less is admitted, as $m = 0.03$ is with 30 anchors. From 30 to 240 anchors, this budget falls from 0.63 to 0.22 (prompt) and from 0.20 to 0.08 (field), and almost every request with $m = 0.03$ is excluded (Figure 16(a)); $m = 0.01$ needs denser anchors still (at most a fraction 0.26 is excluded with 240 anchors). With nearly nearest-neighbour weights, an adaptive region is close to a union of small balls around the anchors: it is not convex, and it is enforced by back-off rather than by projection.

Prompt regions do not see the model. The pull-back metric weighs each direction of the embedding by its effect on the field at $t = 0$, so it ignores what the model ignores, without simulation. With adaptive weights it caps the attack at a few percent on both models and keeps most of the detection with nuisance tokens (Table 8). On the 1-rectified model, however, it measures the field at $t = 0$, not the 50-step sampler: its certificate pays straightness defects of order 1.93 (Corollary 7.2(ii)) and is vacuous, and with static weights it leaves $P(A) = 0.132$ after enforcement. Its detection, the strongest of all regions (0.59 at $m = 0.03$, adaptive), rests on what the fitted field leaks at $t = 0$: the exact field at $t = 0$ depends on the target only through its mean (Lemma 4.3(iii)), which the unsafe mode leaves unchanged.

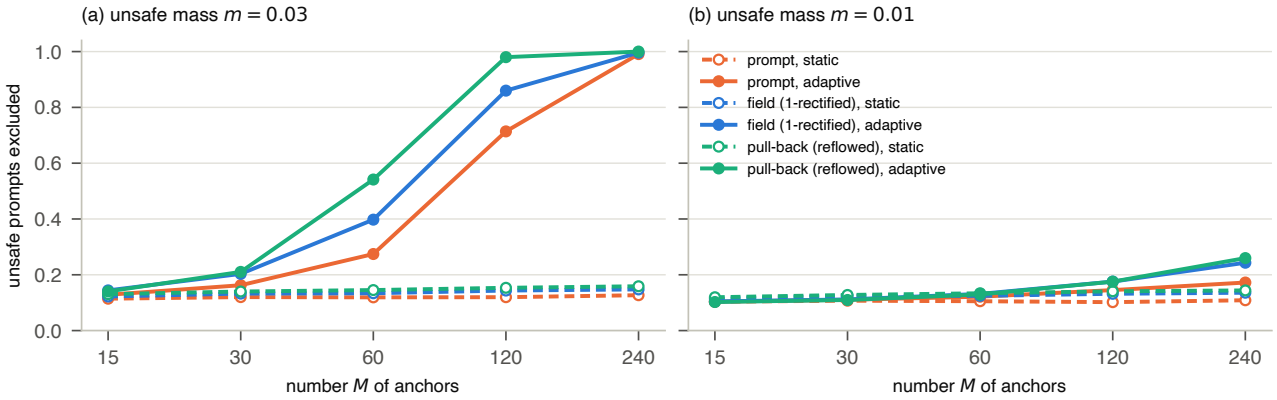


Figure 16: Exclusion of small unsafe masses against the number M of anchors (evaluation pool, $K = 60$, adaptive weights tuned for each M); dashed: static; solid: adaptive. The static budget is set by the spread of the approved family and does not shrink with M; the adaptive budget shrinks with the distance to the nearest anchors.

Field regions of a curved model are costly. On the 1-rectified model, each field score runs the 50-step sampler on the 256 probes, back-off repeats it about 18 times, and the region is not convex. Reflow with one-step sampling (Remark 3.10) removes these costs: the pull-back metric of the reflowed one-step model *is* the map metric of its deployed sampler (Corollary 7.2(i)). Its region lives on the prompts, needs no simulation, is convex with static weights (Proposition 5.6(i)) and keeps the certificate of Theorem 7.1. It detects as well as the field regions (Table 7), gains as much from denser anchors (Figure 16) and resists the attack, unlike the Euclidean prompt region of the same model (Table 8). The price is the fidelity of one-step sampling ($\mathrm{MMD}^2 \times 10^3 = 34.9$ on approved prompts, against 3.3), which enforcement inherits (Table 7).

10.5 Summary

On this toy, the weights decide detection and the space decides robustness. (Q1) Static regions are valid but too wide in every space, because their budget is set by the spread of the approved family. Adaptive weights that take the transport mass from the nearest anchors repair detection at no cost in coverage, and resolve smaller unsafe masses as the anchors become denser. (Q2) Prompt regions match field regions only while the encoder exposes the unsafe content; field regions resist attacks on the embedding and wording that the model ignores. (Q3) The best configuration is an adaptive region on a field metric. For a model reflowed and sampled in one step, that metric is the pull-back metric on the prompts, which is cheap and certified, at the fidelity cost of one-step sampling.

11 Status and next steps

Proved: coverage of the field of a new approved prompt (Proposition 5.4); convexity and second-order cone formulations for Mahalanobis distances (Propositions 5.6 and 6.2); enforcement as a mixture of fields, and as prompt mixing for models affine in the prompt (Proposition 6.2(iii), Corollary 6.3); an output certificate for every enforced field, without Grönwall factor (Theorem 7.1), convex for one-step samplers and paying straightness defects otherwise (Corollary 7.2), and without Markov factor for the localised truncated distance (Proposition 7.5). Tested: on the POC toy, the off-policy/on-policy gap and the path-divergence certificates (Section 8); on a prompt-conditioned toy, coverage, the role of time, prompt mixing, the failure of the region as written on a diverse family, its repair by localisation and truncation, and the robustness of field-space regions to white-box attacks on the prompt (Section 9); and, with tuning separated from evaluation, that adaptive weights beat static ones in every space and need denser anchors for small unsafe masses, that prompt-space regions match field-space ones on honest prompts only when the encoder exposes the unsafe content, and that a pull-back metric of a one-step reflowed model turns the prompt region into the field region (Section 10). The central tension remains between convexity, which needs a fixed reference law, and safety, which needs each field to be evaluated where it sends the noise; one-step samplers resolve it, at a cost in fidelity.

Next: Monte Carlo margins that make the deployed certificate rigorous; the choice of anchors when safe prompts are many, since the budget is set by the spread of the nearest safe fields; then a forbidden class with class-conditional rectified flows on MNIST and CIFAR-10, and text-to-image generation with the truncated map distance in the feature space of an image encoder χ (POC Section 10), always evaluated against white-box attacks on the prompt.

A Stochastic samplers and path divergences

With TV, KL and kl as in Definition 2.17, we restate the POC results on stochastic samplers in the notation of this note; the facts on divergences used below are listed after that definition.

Definition A.1 (SDE chain)**.** *For $\eta > 0$ and the grid of Definition 3.4, the SDE chain of $\boldsymbol{\theta}$ is $X_0 \sim \mu_0$ and $X_{k+1} = X_k + h_k[(1+\eta t_k)\Phi(X_k, t_k)\boldsymbol{\theta} - \eta X_k] + \sigma_k \boldsymbol{\xi}_k$, $\sigma_k^2 = 2\eta(1-t_k)h_k$, $k = 0, \dots, J-1$, with $\boldsymbol{\xi}_k$ i.i.d. $\mathcal{N}(0, I_n)$ independent of X_0; $P^{\boldsymbol{\theta}}$ is the law of $(X_0, \dots, X_J)$. With $\Phi\boldsymbol{\theta}$ replaced by $v_p^\star$, its continuous-time limit has, formally, the marginals of the interpolation path (POC Lemma 3, Albergo et al. 2025); this is not used below.*

Lemma A.2 (Path divergence)**.** $\mathrm{KL}(P^{\boldsymbol{\theta}} \| P^{\boldsymbol{\theta}'}) = \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}')^\top G_{\mathrm{KL}}(\boldsymbol{\theta})(\boldsymbol{\theta} - \boldsymbol{\theta}')$, *where* $G_{\mathrm{KL}}(\boldsymbol{\theta}) = \sum_{k=0}^{J-1} \omega_k \, \mathbb{E}_{P^{\boldsymbol{\theta}}}[\Phi(X_k, t_k)^\top \Phi(X_k, t_k)]$ *and* $\omega_k = \frac{h_k(1+\eta t_k)^2}{2\eta(1-t_k)}$.

Proof: $\sigma_k > 0$ because $t_k < 1$ for $k \le J-1$, and each X_k has a finite second moment by linear growth. Both chains are Markov with the same initial law and Gaussian kernels of covariance $\sigma_k^2 I_n$ whose means differ by $h_k(1+\eta t_k)\Phi(X_k, t_k)(\boldsymbol{\theta} - \boldsymbol{\theta}')$. The chain rule for relative entropy (Cover and Thomas 2006, Thm. 2.5.3) sums, in expectation under $P^{\boldsymbol{\theta}}$, the divergences between these Gaussian kernels, $\|h_k(1+\eta t_k)\Phi(X_k, t_k)(\boldsymbol{\theta} - \boldsymbol{\theta}')\|^2 / (2\sigma_k^2) = \frac{\omega_k}{2}\|\Phi(X_k, t_k)(\boldsymbol{\theta} - \boldsymbol{\theta}')\|^2$ (the Gaussian formula after Definition 2.17). □

The forward cost $\mathrm{KL}(P^{\boldsymbol{\theta}_i} \| P^{\boldsymbol{\theta}})$ uses the Gram matrix of the safe chain and is convex in $\boldsymbol{\theta}$; the reverse cost $\mathrm{KL}(P^{\boldsymbol{\theta}} \| P^{\boldsymbol{\theta}_i})$ uses that of the candidate and is not.

Proposition A.3 (Certificates and the direction of the divergence)**.** *Let $\bar{P}_w = \sum_i w_i P^{\boldsymbol{\theta}_i}$. (i) If $\frac{1}{M+1}\sum_i w_i \mathrm{KL}(P^{\boldsymbol{\theta}_i} \| P^{\boldsymbol{\theta}}) \le \varepsilon$, then $\mathrm{TV}(P^{\boldsymbol{\theta}}, \bar{P}_w) \le \min\{((M+1)\varepsilon/2)^{1/2}, (1 - e^{-(M+1)\varepsilon})^{1/2}\}$. (ii) If $\frac{1}{M+1}\sum_i w_i \mathrm{KL}(P^{\boldsymbol{\theta}} \| P^{\boldsymbol{\theta}_i}) \le \varepsilon$, then for every event A with $q = \bar{P}_w(A) \in (0,1)$,*

$$P^{\boldsymbol{\theta}}(A) \le \sup\{x \in [q,1] : \mathrm{kl}(x\|q) \le (M+1)\varepsilon\} \le \frac{(M+1)\varepsilon + \log 2}{\log(1/q)}.$$

(iii) If $P' = (1-m)P + mQ$ with $m \in [0,1)$, then $\mathrm{KL}(P\|P') \le \log\frac{1}{1-m}$. If moreover $Q(A) = 1$ for an event A with $q = P(A) \in (0,1)$ and $m \ge q$, then $\mathrm{KL}(P'\|P) \ge m \log\frac{m}{q} - m$.

Proof: (i) By joint convexity of the relative entropy (Cover and Thomas 2006, Thm. 2.7.2), $\mathrm{KL}(\bar{P}_w \| P^{\boldsymbol{\theta}}) \le \sum_i w_i \mathrm{KL}(P^{\boldsymbol{\theta}_i} \| P^{\boldsymbol{\theta}}) \le (M+1)\varepsilon$; apply Pinsker's inequality (Tsybakov 2009, Lemma 2.5) and the Bretagnolle–Huber inequality (Bretagnolle and Huber 1979, Canonne 2022). (ii) Likewise, by convexity in the second argument, $\mathrm{KL}(P^{\boldsymbol{\theta}} \| \bar{P}_w) \le (M+1)\varepsilon$. By data processing through $\mathbb{1}_A$ (after Definition 2.17), $\mathrm{kl}(P^{\boldsymbol{\theta}}(A)\|q) \le (M+1)\varepsilon$: either $P^{\boldsymbol{\theta}}(A) \le q$, or $P^{\boldsymbol{\theta}}(A)$ belongs to the set whose supremum is taken. For the second bound, every x in that set satisfies $x \log\frac{1}{q} - \log 2 \le \mathrm{kl}(x\|q) \le (M+1)\varepsilon$, because $\mathrm{kl}(x\|q) = x\log\frac{1}{q} + (1-x)\log\frac{1}{1-q} - \mathrm{h}(x)$ with the binary entropy $\mathrm{h}(x) \le \log 2$. (iii) $P \le \frac{1}{1-m}P'$, so $\frac{dP}{dP'} \le \frac{1}{1-m}$ and $\mathrm{KL}(P\|P') \le \log\frac{1}{1-m}$. Let $x = P'(A) = (1-m)q + m$, so $x \ge m \ge q$. Data processing and $\log\frac{1}{1-q} \ge 0$ give $\mathrm{KL}(P'\|P) \ge x\log\frac{x}{q} + (1-x)\log(1-x) \ge x\log\frac{x}{q} - x$, since $\log(1-x) \ge -\frac{x}{1-x}$ for $x < 1$; and $x \mapsto x\log\frac{x}{q} - x$ has derivative $\log\frac{x}{q} \ge 0$ on $[q,1]$. □

Item (iii) is the formal reason for the gap of Section 8: a forward cost charges at most $\log\frac{1}{1-m} \approx m$ for an unsafe mode of mass m, whatever its rarity q under the safe samplers, while the reverse cost grows like $m\log(m/q)$.

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